Pålitelig dyp læring for ekkokardiografisk bildeanalyse

Ultralydavbildning av hjertet, også kjent som ekkokardiografi, er et av de viktigste verktøyene for å diagnostisere hjertesykdom. Det er en tilgjengelig og rimelig metode som gir informasjon i sanntid uten å utsette pasienten eller personellet for ioniserende stråling. Ekkokardiografi er imidlertid krevende å utføre og tolke, og det trengs lang opplæring for å beherske både skanningen og bildeanalysen. Det er en økende mangel på klinikere i ekkokardiografi som forventes å forverres i løpet av dette tiåret.

Automatisert bildeanalyse kan avlaste klinikerne ved å gjøre diagnostikken av hjertesykdom mer effektiv og veilede mindre erfarne klinikere. I dagens ekkokardiografi utføres det en rekke tidkrevende målinger fra ulike vinkler av hjertet for å hente kvantitativ informasjon som brukes til både diagnostikk og videre oppfølging av pasienten. Her kan dyp læring og kunstig intelligens (KI) spille en avgjørende rolle ved å muliggjøre rask, reproduserbar og presis tolkning av ultralydbildene.

Til tross for betydelige fremskritt møter KI-baserte løsninger i ekkokardiografi utfordringer som begrenset tilgang til data, varierende bildekvalitet, samt mangel på entydig fasit og ekspertkonsensus. Dette er faktorer som svekker både generaliserbarhet og pålitelighet. Formålet med dette arbeidet var å undersøke teknikker som kan gjøre dyp-læringsverktøy for ekkokardiografi mer robuste. Arbeidet bidrar særlig til å håndtere tre forskjellige utfordringer.

Først introduseres en metode for å avdekke svikt i dyp-læringsbaserte segmenteringsmodeller ved hjelp av en sekundær modell som fanger opp anatomisk umulige resultater. Deretter beskrives en teknikk som automatisk anslår bildekvaliteten til forskjellige hjerteregioner, hvor resultatene viste at bedre bildekvalitet gir bedre samsvar mellom automatiserte og manuelle målinger. Det siste bidraget viser hvordan en generativ diffusjonsmodell kan brukes til å utvide datasett med få augmenteringer, noe som økte både robustheten og generaliserbarheten til segmenteringsmodellen.

Abstract

Cardiac ultrasound imaging, also known as echocardiography, is one of the main tools to diagnose heart disease, due to its availability, affordability, ability to provide real-time results, and absence of ionizing radiation. However, echocardiography is challenging and requires a lot of practice to master, both in terms of scanning and in terms of interpreting the images. There is a growing shortage of clinicians in echocardiography, which is expected to worsen in the current decade. Automated analysis of cardiac ultrasound images can help clinicians diagnose heart disease more efficiently and guide non-experts.

In echocardiography, many time-consuming measurements are performed on different views of the heart to provide quantitative information for diagnosis and follow-up of patients. Deep learning and artificial intelligence (AI) tools can play a crucial role in these echocardiography challenges by enabling fast, reproducible and accurate interpretation of cardiac ultrasound images.

Despite advances in AI, its adoption in echocardiography faces challenges such as limited data, varying image quality, and lack of ground truth and expert consensus, which affects generalisability and reliability. The overall goal of this work was to investigate techniques to make deep learning tools for echocardiography more robust. Specifically, the work provides contributions to dealing with three different challenges. The first contribution is a method for detecting failing cases of deep learning based segmentation models. The study demonstrates that failures in segmentation models can be identified using a secondary model specifically designed to prevent anatomical errors. The second contribution is a method for automatically estimating the image quality of different regions of the heart. Different methods were tested and the results showed that higher quality images correlate with tighter limits of agreement between automated and manual clinical measurements. Finally, the last contribution provides a technique to deal with limited available annotated data for segmentation models using a generative diffusion model. The work shows how generative augmentations can improve the robustness and generalisability of segmentation models.

Preface

This thesis is submitted in partial fulfillment of the requirements for the degree of *Philosophiae Doctor* (Ph.D.) at the Faculty of Medicine of the Norwegian University of Science and Technology (NTNU). The research was funded by the Research Council of Norway and carried out at the Department of Circulation and Medical Imaging (ISB), NTNU. The main supervisor has been Erik Smistad, and co-supervisors have been Lasse Løvstakken, and Håvard Dalen, all from ISB, NTNU.

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I also want to thank Bjørnar, Espen, Sindre, and John for their time to help with annotations and clinical questions I had along the way. Data is the cornerstone of machine learning, and your contributions provided this essential data.

Finally, I want to thank all of my technical colleagues. Specifically, I would like to thank Sarina and Guy for their suggestions and clarifications on their work that I used as a starting point during the first phase of my PhD. I am also grateful to my colleagues at NTNU: I want to thank David for explaining the statistics behind clinical evaluations and the help with loading medical data, Sigurd for helpful general suggestions and discussions on diffusion models, Artem for insights and discussions about neural network architectures, and Jieyu for the vital help with data management and remote setup. I thank Mia for helping with Norwegian translation of the abstract and proofreading.

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Abbreviations

AI	Artificial intelligence
A4C	Apical four chamber view
A2C	Apical two chamber view
ALAX	Apical long-axis view
CR	Contrast ratio
CNR	Contrast to noise ratio
GCN	Graph Convolutional Network
LV	Left ventricle
ROI	Region of interest
HUNT4	Helse Undersøkelsen i Nord-Trøndelag ultrasound dataset
CAMUS	Cardiac Acquisitions for Multi-structure Ultrasound Segmentation
CNN	Convolutional neural network
GPU	Graphics processing unit
Hz	Hertz
B-Mode	Brightness-mode
M-mode	Motion-mode
TTE	Transthoracic echocardiograph
TEE	Transesophageal echocardiograph
ASE	American Society of Echocardiography
EACVI	European Association of Cardiovascular Imaging
EF	Ejection fraction
EDV	End diastolic volume
ESV	End systolic volume
HFpEF	Heart failure with preserved ejection fractio

GLS	Global longitudinal strain
MAPSE	mitral annular plane systolic excursion
LoA	Limits of agreement
MSE	Mean squared error
SGD	Stochastic gradient descent
MC	Monte Carlo
BN	Batch normalization
Adam	Adaptive moment estimation
ReLU	Rectified linear unit
GAN	Generative adversarial network
IS	Inception score
KL	Kullback–Leibler
FID	Fréchet Inception Distance
LV	Left atrium
MYO	Myocardium
GE	General Electric
SOTA	state-of-the-art
VAE	variational autoencoder
GIC	Global image coherence
VLCD	Very Large Cardiac Channel Data Database
PSAX	Parasternal short axis
PLAX	Parasternal long axis
SSIM	Structural similarity index measure
PSNR	Peak signal-to-noise ratio
RPE	Relative pixel error
AO	Aorta
CVC	Cardiac view classification
DDPM	Denoising diffusion probabilistic model

Introduction

1

One out of three deaths in the United States is caused by cardiovascular diseases [1]. While the age-standardised death rate has been falling for the last thirty years, is it still the most common cause of death worldwide. Countries that invest a larger part of their gross domestic product in healthcare have a lower death rate by these diseases [2], which shows how important a good healthcare system is and why investments made to make it as effective and precise as possible are important. Finally, the current demographic shifts towards a higher percentage of elderly people who are at an elevated risk of cardiovascular diseases.

Ultrasound imaging is one of the main tools to diagnose heart disease [3]. By sending high frequency sound waves into the body and recording the reflections, accurate 2D and more recently also 3D gray-scale images of soft tissue organs inside the body can be imaged non-invasively. Ultrasound has several key advantages. It is a) real-time, b) relatively inexpensive, c) safe, and d) easy to deploy and distribute. These advantages are especially useful for developing countries, where precisely the most cases of related deaths occur [2], but also for point-of-care applications, emergency situations, and field hospitals.

The main disadvantages of ultrasound are lower image quality compared to other imaging techniques and the high level of operator skill required to obtain clear images. For image quality, ultrasound imaging is an active field of research and development and image quality is improving in each generation of new scanners. On the other hand, there is a global trend towards increasing prevalence of obesity [4] and obese patients have generally lower echocardiographic image quality [5]. This problem is particularly significant in echocardiography, as the heart's location deep within the chest makes it more challenging to image in obese patients. At the same time, these patients are more at risk for cardiovascular diseases [6]. Thus, image quality remains one of the main challenges in ultrasound imaging. While CT and MRI generally require only positioning the patient inside the machine, ultrasound imaging demands extensive training to master proper scanning techniques.

There is currently a shortage health personnel and this trend is projected to continue

in the current decade [7] [8]. This shortage is also prevalent in echocardiography. This leads to a discrepancy between the guidelines for echocardiography and actual clinical practice. The guidelines recommend performing a wide range of quantitative measurements repeated over several cardiac cycles to take into account measurement variability [3]. However, in today's clinical practice many of these quantitative measurements are either not acquired or only performed on a single loop due to time constraints.

As a final challenge, quantitative measurements in echocardiography are inherently subject to inter- and intra-observer variability. Already during the acquisition of the ultrasound images, slight changes of the probe and heart orientation can potentially have a big impact on measurements outcomes. Afterwards, the extraction of these measurements suffers from inter- and intra-observer variability due to differences in interpretation, experience and a lack of clear consensus [9].

Despite these challenges, advancements in technology are enabling the creation of novel solutions. One such promising approach is the application of deep learning in echocardiography. Deep learning, a subset of machine learning and artificial intelligence, uses neural networks with multiple layers to model and understand complex patterns in data. This method has already revolutionized various fields such as image recognition, natural language processing, and autonomous driving, and its potential in medical imaging is increasingly being recognized.

Deep learning is attractive for several reasons. First of all, it can provide instantaneous measurements [10]. This allows clinicians to perform multiple quantitative measurements as the guidelines recommend and focused their efforts on making the correct diagnosis. Secondly, as deep learning networks are deterministic, they have the potential to lower inter- and intra-observer variability significantly [11]. Another way to reduce variability is to guide operators towards standardised views during acquisition [12]. Finally, deep learning can handle large volumes of data efficiently, enabling the analysis of extensive patient datasets to uncover patterns and correlations that might be missed by traditional methods, ultimately contributing to more precise and personalized treatment plans.

Despite these advantages, most quantitative measurements today are still being performed manually or semi-automatically. One reason for the lack of large-scale deployment of deep leaning solutions is the lack of generalizability and reliability. This is especially difficult in the medical domain where human lives are at stake and responsibility is an important topic.

The challenge of generalizability and reliability in deep learning for cardiac echocardiography is increased by the limited availability of annotated data, a

consequence of strict data protection regulations and the labor-intensive process of curation and annotation. Furthermore, variations in annotation preferences and differences between imaging vendors require careful consideration when integrating datasets from multiple sources. This contrasts sharply with the field of computer vision for natural images, where large, standardized, and labeled datasets are readily accessible, enabling more robust and transferable model development.

1.1 Research questions

The goal of this thesis was to investigate the following research questions. Each research question corresponds to one of the three main contributions of this thesis.

1) Can a real-time deep learning segmentation model be created that does not produce anatomical incorrect segmentations?

The U-Net architecture is currently the most popular deep learning architecture for pixel-level segmentation tasks in the medical domain [13, 14] and a well configured U-Net can still achieve state-of-the-art performance [15]. While the U-Net shows good performance on average and can run in real-time, it can create segmentation masks that do not make sense anatomically. This is partly due to the inherent problem of translational invariance of the convolutional layers used in deep neural networks. Thus, convolutional neural networks have no strong contraints in terms of how different anatomical regions relate to each other. Graph convolutional networks are a promising new approach to segmentation which provides stronger anatomical constraints by having a model of the anatomy while still having a low computational complexity [16–18]. An open question at the start of this work was whether these stronger constraints would lead to a reduction in pixel-wise accuracy. While anatomical coherence is improved, it is crucial to evaluate whether this comes at the cost of segmentation precision at the pixel level.

2) How can we detect when segmentation fails?

Our research group has observed over many years that segmentation models based on the U-net architecture can often produce incorrect segmentations. This typically happens when the model is run on data that was not in the training distribution. Since cardiac ultrasound datasets are scarce and heterogeneous, this issue arises frequently in practice. For example, it occurs when a model trained on data from one center is applied to data from another center. It also happens when a model trained on recordings made by an experienced user is used on recordings made by an inexperienced sonographer [14]. These differences in data quality, acquisition protocols, and operator expertise lead to segmentation masks that are anatomically incorrect. One approach to detecting failing segmentation is to have

multiple different segmentation models, and measure the consensus between these models. The hypothesis is that if all the models agree on the segmentation and the models are based on fundamentally different architectures or approaches, then the segmentation is most likely correct. However, if they have a large disagreement on the segmentation, the segmentation is most likely failing, as the different models typically fail in different ways. Detecting failing segmentations is important as long as current segmentation models can't always produce perfect results. Detecting when segmentations fail can serve as a warning to users of the automatic segmentation tool in real-world use. Additionally, it could be used when crafting a dataset of annotations in an active learning setting. Annotating a case where the current model fails will add more information than a case where it already works.

3) Can a deep learning model assess regional image quality in a cardiac ultrasound image?

In current research, regional image quality is assessed using metrics like the contrast ratio (CR) [19] or the contrast to noise ratio (CNR) [20]. These methods have as drawback that they require a region of interest and noise region to be selected manually. For cardiac ultrasound, deep learning can automatically segment the regions of interest using segmentation [21], so the question naturally arises whether segmentation can be used to automatically calculate these metrics and how well they correlate with expert perception of image quality. This also raises the question whether a deep learning model can learn to predict image quality directly with better accuracy than classic metrics when compared to experts opinion. A model that automatically assesses image quality could be used in the following ways. First, it could warn operators in real-time that the image quality is too poor to extract quantitative metrics. Second, it could automatically select the best recordings, cardiac cycles, or myocardial regions to be used for quantitative analysis. Finally, it could be used in data mining projects to sort recordings based on image quality.

4) Can diffusion models improve segmentation models in terms of robustness and generalisability?

On of the main challenges in cardiac ultrasound segmentation in the limited amount of annotated data [14]. Creating high quality pixel-wise reference masks is time consuming and annotation conventions can differ between datasets. On the other hand, unannotated data is much easier to obtain in larger quantities. Generative AI learns the distribution of a dataset to generate new samples. Thus, the question arises whether a generative model such as diffusion models, that learns the distribution of an unannotated dataset can be used to enhance the diversity of a limited annotated

training dataset. This would then make the model trained on the new dataset more robust and generalizable to different datasets.

1.2 Summary of contributions

Three articles were produced as part of this thesis, targeting the research questions from the previous section.

Contribution 1: Towards Robust Cardiac Segmentation using Graph Convolutional Networks

The first contribution of this thesis answers the two first research questions of how to create a more robust segmentation model that does not produce anatomical outliers and how to detect failing segmentations. The proposed method builds upon the work of Thomas et al. [18], which proposes a method that predicts keypoints on the contour of the LV. This makes it fundamentally different from the U-Net based approaches, which predict a label for each pixel in the image. The proposed architecture is a MobileNetV2 [22] encoder with a Graph Convolutional Network (GCN) as a decoder. The work of Thomas et al. [18] was limited to single-structure segmentation (left ventricle endocardium) on a single view (apical four-chamber) and lacked a thorough evaluation. The contribution in this thesis explores ways to extend the model to multi-structure segmentation, how the network can be used in combination with U-Net, and evaluates it clinically using automatic EF estimation from segmentation masks.

To extend the model to multi-structure segmentation, the work proposed a clinically motivated design where the model predicts the keypoints of the endocardium and the thickness of the myocardium. Fig. 1.1 shows the placement of the contour keypoints the model predicts. The result show that this design leads to a model that does not create anatomical incorrect segmentations. However, the delineation details and precise placement of the segmented structures might still be inaccurate and the U-Net achieves slightly higher scores on the classic segmentation metrics like Dice score and Haussdorf distance on average. The design of the graph-based network makes it a more regularized segmentation model than the U-Net.

Combining the segmentation output of the U-Net with that of the graph-based model gives a lightweight tool that can detect failed segmentations in real-time. On the one hand, the graph based network predicts the correct shape to segment, but can be more inaccurate. On the other hand, the U-Net can generate highly accurate segmentation masks, but can sometime fail drastically. Thus, when there is little overlap between the segmentation masks of both networks, the segmentation failed in at least one of the networks. This was usually due to an out-of-distribution input

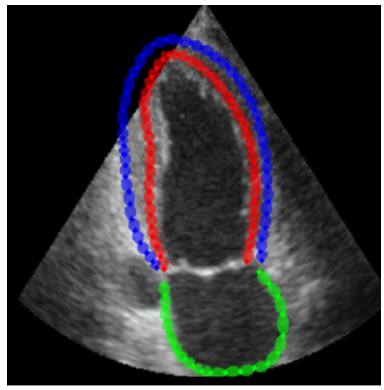


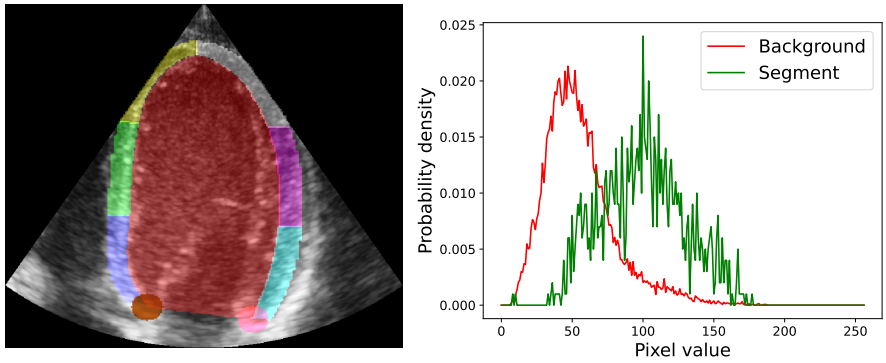
Figure 1.1: Instead of predicting a class for each pixel, the graph based network predicts the coordinates of the contour points of the structures being segmented.

or low image quality. Finally, the work also explores a cascade approach, where the graph-based model first generates a coarse segmentation mask, which is then refined by the U-Net to produce a more detailed and precise output. While this approach improves Dice and Haussdorf distance over the graph based network alone, it can also re-introduce anatomical incorrect segmentations.

Contribution 2: Regional image quality scoring for 2D echocardiography using deep learning

Ultrasound image quality in echocardiography is influenced by factors like the scanner, patient, and probe, and it impacts the accuracy of measurements such as LV volume, EF, and strain. Regional image quality assessment of the heart wall can guide operators, select optimal images, ensure quality assurance, and exclude poor-quality cases in data analysis. Existing metrics like the contrast ratio (CR) [19], contrast to noise ratio (CNR) [20], and generalized CNR (gCNR) [23] evaluate local image quality but are not fully automatic as they require a selection of a region of interest (ROI) and a background region to compare against. Previous deep-learning based work only evaluates the quality of images on the global level and does not distinguish clearly between view correctness and image quality [24–28].

Our study compares three different approaches for assessing regional image quality. The first approach uses the classic metrics CR, CNR, and gCNR, evaluated on the the regions extracted using a segmentation model. In this approach, the foreground regions are the regions of the myocardium extracted from the segmentation mask of a segmentation model. The background region is the LV lumen itself, as extracted by the same model. Fig. 1.2 shows an example of the first approach. The second approach used the same foreground regions, but uses the local image coherence as



(a) Segmentation **(b)** Probability density functions

Figure 1.2: With automatic segmentation, ROIs and background regions for the signal-no-noise-ratio calculation can be extracted automatically.

a metric instead of the classic metrics. Since channel data is required to calculate the local image coherence and this is often not available, a model was developed that estimates local image coherence from the B-Mode image. Finally, the third and last approach consists of a deep learning network that directly learns to predict the subjective scores given by three senior cardiologists.

Comparing the prediction of each of the networks to the assessments of the cardiologists, the third approach achieves the best agreement, followed by the second and first approach. However, the first two approaches are more general because they only require the selection of foreground and background. In contrast, the third approach involves training a task-specific model that is designed specifically for quality prediction, which requires annotated data to train the model.

The agreement between clinical metrics extracted using deep learning and those extracted manually was better in recordings of higher quality. This suggests that the quality of ultrasound images has a direct impact on the reliability and consistency of clinical measurements, emphasizing the importance of accurate regional image quality assessment.

Contribution 3: Generative augmentations for improved cardiac ultrasound segmentation using diffusion models

This study explores the use of diffusion models to generate synthetic images to enhance the robustness of segmentation models without requiring additional manually labeled data.

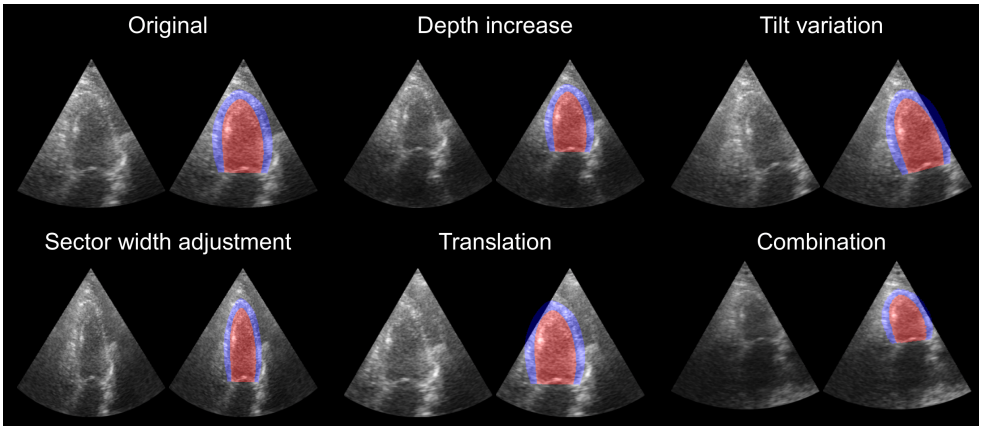


Figure 1.3: Generative augmentations create context while keeping the annotations intact.

The proposed method generates new training samples by modifying existing labeled images. First, the part belonging to the left ventricle is moved, rotated, stretched, and resized. The image is then edited using RePaint [29] to generate missing parts to create a completed sector. During this process, the method does not modify the part of the image belonging to the left ventricle, apart from geometrical transformations such as scaling and rotating. Therefore, the proposed method ensures that the precise, high-quality annotations of the clinicians remain intact. Figure 1.3 gives an example of how a single annotations can be augmented with different transformations.

The study uses two datasets: HUNT4Echo [30], a standardized dataset focused on the left ventricle, and CAMUS [31], which includes a more diverse range of acquisitions. The diffusion model of the RePaint method was trained on CAMUS. The impact of incorporating the proposed augmentations was evaluated both within each dataset individually and through cross-dataset testing between HUNT4Echo and CAMUS. Segmentation was performed using nnU-Net [32].

The results demonstrate that generative augmentations improve segmentation performance, particularly on external datasets. Dice scores increased by up to 0.08 and Hausdorff distances were reduced by up to 20 millimeters. Furthermore, the limits of agreement between the clinical reference and automatic estimation of ejection fraction using segmentation improved by 6% by reducing outliers. A visual evaluation by experts showed that clinical and technical ultrasound experts struggled to differentiate real from synthetic images, achieving only 56.4% accuracy, with senior cardiologists performing slightly better at 63.7%. The study concludes that generative augmentations provide a promising approach to enhance dataset diversity

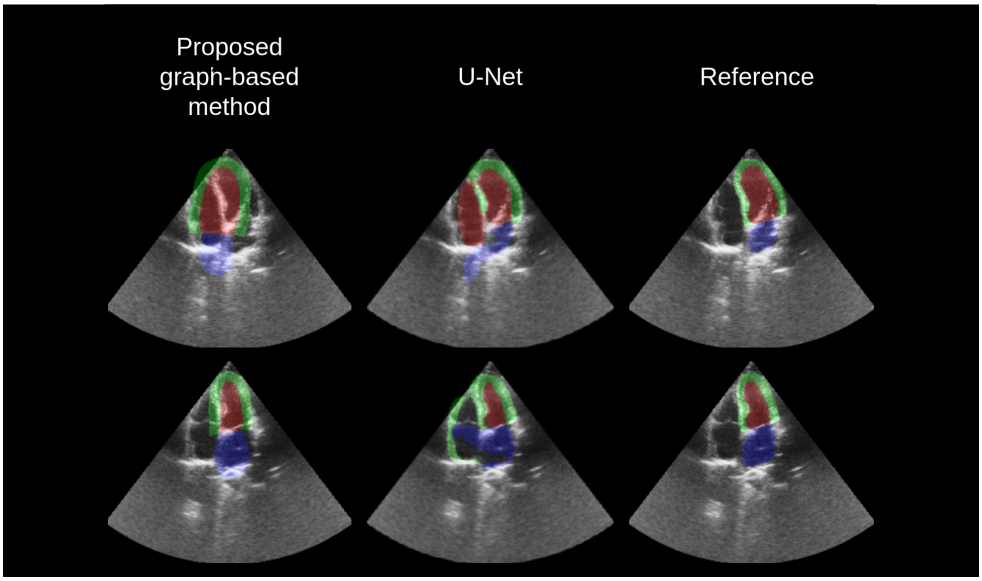


Figure 1.4: An out of distribution case for segmentation due to greatly increased depth. Although still incorrect, the proposed graph-based segmentation method does not create an anatomical incorrect shape.

and improve segmentation robustness.

1.3 Discussion of results

1.3.1 Towards Robust Cardiac Segmentation using Graph Convolutional Networks

Anatomical correctness at the cost of lack of attention to details

The design of the graph based network put constraints on what kind of shapes the network can create. This acts as a regularization effect that avoids disconnected parts and highly irregular shapes by design. When a out-of-distribution or low quality image is given as input, the network would not produce an anatomically incorrect segmentation as the U-Net often does, but the segmentation would not necessarily be placed correctly in the image. Fig. 1.4 compares the segmentation of U-Net to the proposed graph-based model on an out-of-distribution case.

The advantages and disadvantages of the U-Net and the graph based model are complementary. On the one hand, the severe anatomical incorrect segmentations of the U-Net might give rise to distrust in the deep learning model by end users. On the other hand, clinical measurements extracted with the U-Net tend to agree more

with the reference than the graph based model. An optimal solution therefore would be to only use the output of the model when both networks agree up to a certain extent.

Detecting segmentation failures

If the two methods, each with their distinct data representations, produce similar segmentations, there is a higher likelihood that the segmentations are accurate. Conversely, when one method makes a significant error, it is unlikely, though not impossible, that the other method will make the same mistake due to the inherent differences in their data representations. Therefore, the inter-model agreement between these two different segmentation architectures identifies failed segmentations. While this is not a guarantee, it is a useful heuristic that can help identify uncertain predictions. Thus, in essence, uncertainty is estimated from an ensemble with only two models. The advantage of having two is that it makes real-time application feasible.

From a theoretical point of view, the disagreement of the two models measures mostly epistemic uncertainty [33, 34]. This is the uncertainty that arises from a lack of knowledge or information about the model or the data, leading to uncertainty in the predictions. Epistemic uncertainty is typically reduced as more data becomes available or as the model architecture is improved. It reflects the limitations of the model in capturing the underlying distribution of the data it is tested on. In practice, this means the epistemic uncertainty mostly detects out-of-distribution cases. If the training data only includes high or medium quality images, this also includes low quality images.

Model design

Several variations of the neural network architecture were tested during the design phase of the network, but are not included in the publication due to space limits in the publication. The main goal of the project was to design a neural network architecture that would not create anatomical incorrect segmentation masks. The first designs were a logical expansion of the work by Thomas et al. [18] where the architecture was the same, but with more keypoints as outputs. These initial models did not perform well and the transition to an output structure with a dual ring of keypoints improved results drastically. Still, the network would sometimes confuse the inner and outer keypoints and predict the outer points inside the inner points or on top of each other. This did not happen very often, but the goal was to create a model that would always create an anatomical correct shape. Therefore, the model with the displacement method was designed to make this kind of error impossible by design.

The actual graph convolutional part of the final proposed network has only two layers. The ablation study shows that increasing the depth has a small impact on performance. Additional experiments included versions with skip connections from the convolutional neural network (CNN) encoder to the GCN decoder. This design inspired by the skip connections in the U-Net architecture would allow for a deeper decoder to be effective by providing low level information. However, the results with these networks were not statistically significantly better than the simple two layer network. This can indicate the shapes of the cardiac structures are simple and do not require a complex GCN decoder.

1.3.2 Regional image quality scoring for 2D echocardiography using deep learning

Objective image quality assessment: balancing automation and human consistency

Human perception of image quality is inherently subjective. Metrics like (g)CNR offer a more objective way to measure image quality. However, the results of these methods are strongly affected by the chosen region of interest. Automating the selection of these regions creates a more objective way of measuring image quality. However, this showed relatively low correlation with the assessments given by human annotators. The model trained to mimic these assessments directly gave significantly higher correlations. While a method trained like this is inherently biased towards the opinion of the annotator who created the labels, it can still be more objective than manual assessment since it produces the same response for the same input every time. Fig. 1.5 shows an example of the quality as predicted by the deep learning model.

Before annotating the image quality dataset for this project, each of the three clinicians evaluated the same set of ten frames selected from multiple recordings of two patients. Next, the clinicians held a meeting to discuss the differences in annotations between each other. This reduces subjectiveness, and ensures that the labels used for training the model are as consistent as possible.

Model design

Several attempts were made to improve the results of the first two approaches in the study that use segmentation masks as ROI and background regions. The segmentation mask of the myocardium regions often includes pixels that actually do not correspond to tissue. Meanwhile, the segmentation mask of the LV lumen often includes pixels belonging to tissue, originating from the valves or trabeculations.

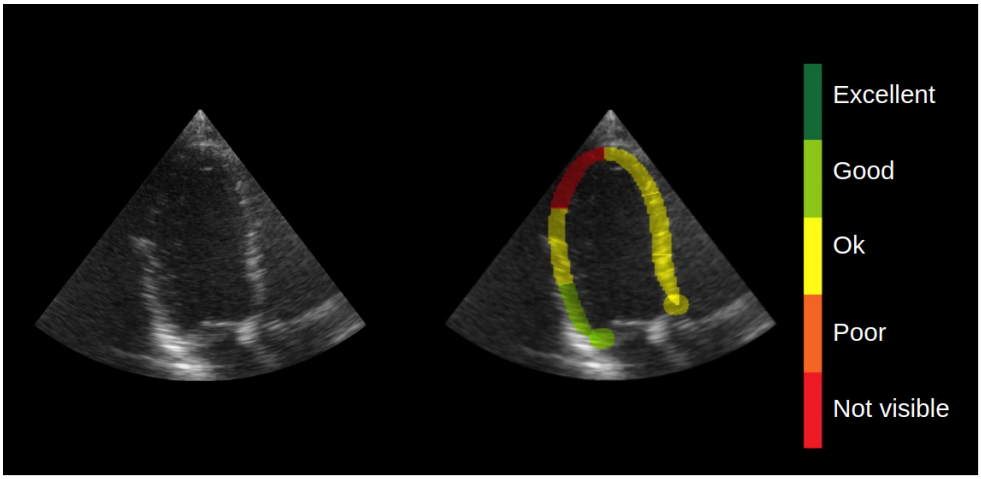


Figure 1.5: An example bmode image and the corresponding regional image quality in the myocardium, as predicted by the deep learning model.

Extensive experiments were performed using Otsu thresholding [35] and percentile filters to automatically separate tissue from background inside the masks. Although these filters were mostly successful in separating the tissue from the background, the agreement with the reference did not improve.

The final end-to-end deep learning model used was, in fact, the most simple design. This was somewhat surprising, as previous work showed the advantage of adding attention to the network [36] when the ROI of the relevant object is known. This was not investigated further, as the simple model already showed performance below inter-observer variability.

Future work

Making the project code open source and well documented facilitates its use, comparison, and further development in other projects. The methods in this work are already being used in two other research projects. The first project focuses on artificially degrading the image quality of ultrasound images using post-hoc digital beamforming. The predicted quality is then utilized to validate the image degradation network by comparing the quality of the original images to that of the degraded ones. The second project involves the analysis of strain rate curves extracted from tissue Doppler imaging. Here the code is used to segment the myocardium and divide it into regions for regional strain analysis.

The deep learning network that predicts the coherence image from the B-Mode image was developed in this work as a necessary tool for using local image

coherence when channel data is not available. However, this in itself is a useful result. Image coherence and its applications is an active field of research [37]. However, most available datasets do not have the channel data available necessary to calculate the coherence factor in the usual way.

The methods in this work were designed such that they only focus on the image quality separate from the view correctness. This allows future work to use existing methods that extract view correctness [38] and combine it with the image quality method of this work.

A direct use case for automatic regional image quality prediction is relating it to the expected accuracy of measurements. This was already demonstrated briefly in the published work were it was shown that automatic and manual measurements have more agreement with higher quality. Future work could also look into similar experiments, but also factoring in other aspects like view correctness [39].

The results of the image quality prediction network were already below inter-observer variability on apical views. Further advancements in the methodology could be to generalize the model for different views or non-standardized images. The techniques in chapter 5 could be used to make the model robust to non-standardized views.

1.3.3 Generative augmentations for improved cardiac ultrasound segmentation using diffusion models

Balancing AI-Generated Content with Data Integrity

Generative AI models, including those used in this work, are prone to hallucinations. Therefore, their output should not be trusted blindly. This consideration led to the decision to preserve the original annotated regions while generating only the surrounding areas. This makes sure the segmentation annotations are still correct. Still, just like traditional data augmentation techniques, the augmentation process can produce unrealistic samples. However, this is not a concern, as the primary goal of data augmentation is not to generate realistic images but to enhance the performance of deep learning models on real data.

Model design

Due to the diffusion model's training time of approximately 90 hours on the A100 GPU used for this project, only a few variations were tested. Specifically, attempts to add more layers and adjust the noise scheduler were made, but neither led to any improvements. However, as part of the preceding work by Lenz [40], many variations were tested on a model that uses images with reduced dimensions. The

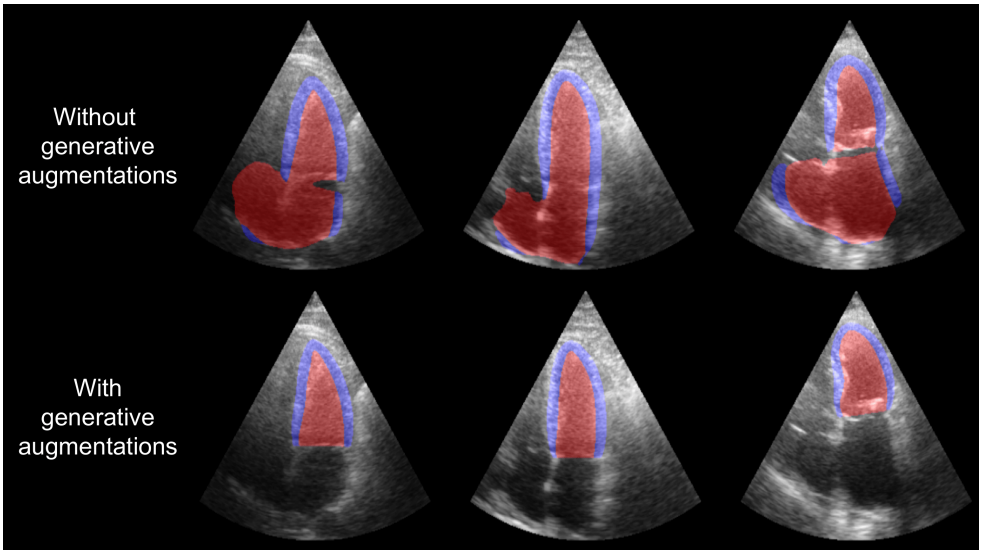


Figure 1.6: A segmentation model trained with the augmented dataset generalizes better to different acquisition setups.

design that worked best on these images was then upscaled to the images with the full 256x256 dimensions and was used as a starting point for this study.

The most important difference to the model from Lenz [40], is additional sector width augmentations. The large majority of images in the CAMUS dataset has the same sector angle of 75 degrees. As a result, a diffusion model trained directly on CAMUS overfits on this angle and only learn to produce images with this angle. When such a model is used for inpainting, it fails to reproduce the correct sector shape, especially when used on the HUNT4 data where most recordings have a sector angle of 65 degrees.

Result on out of distribution data

The results show that a model trained on HUNT4Echo with generative augmentations achieve significantly higher accuracy than a model trained without them when evaluating on the CAMUS dataset or on out-of-distribution cases in the HUNT4Echo data. At the same time, performance in the HUNT4Echo in-distribution cases does not deteriorate. Fig. 1.6 shows how a model trained with generative augmentations generalizes better to images with large depth.

Future work

The capabilities of generative AI have increased rapidly over the last decade thanks to technical innovations [41–46] and it is still an active field of research. Future advancements in generative AI models could further improve the realism and diversity of augmented cardiac ultrasound images.

This work only looked into generating single frames, but future work could look into generating full recordings. However, this brings challenges of scale, as these models currently require vast amounts of data and hardware to train [47, 48]. For example, the Open-Sora project [49] that attempts to recreate and open-source the Sora model from OpenAI [50], used 35000 hours of training on H100 GPUs [51]. This would translate into around $87 500 USD when using cloud computing. However, a generative model for video generation in electrocardiography could be much more specialized, thus generally not requiring as large models and datasets as on natural images.

Another potential direction is the integration of physics-based constraints into generative models to ensure anatomical and structural consistency. By incorporating domain knowledge about ultrasound imaging physics and cardiac anatomy, future models could generate synthetic images that better reflect real-world variations while maintaining medical plausibility. Existing simulation methods that do not use deep learning are computationally expensive and do not generate fully realistic images [52–55].

1.4 Concluding remarks

This thesis explored deep learning methods applied to echocardiography with a special focus on improving the robustness and reliability of these methods. Each of the three main contributions tackles one of the challenges of deep learning in echocardiography that hinder its use in the clinic.

The first contribution offers a safety metric in the form of a warning for when a deep learning model, a segmentation model in this case, fails to provide a sensible solution. Clinicians are understandingly reluctant to rely on the output of automated methods. In the health industry, a wrong decision based on a failed deep learning model can potentially cost lives. However, this does not mean these tools should not be used altogether. In contrast, the usage of these tools might improve the accuracy of clinical decisions and save lives in the long run. The proposed approach in this contribution draws a parallel to the aviation industry, where redundant systems and cross-checks are integral to safety. Just as aircraft rely on multiple independent sensors and flight control systems to detect and mitigate

failures, the proposed method uses a pair of networks to solve the segmentation problem in a fundamentally different way and validate each other's outputs. In aviation, discrepancies between systems trigger alerts for pilots to intervene. Could a similar framework of redundant, cross-validating systems be used in the future to enhance reliability of deep learning in echocardiography?

The second contribution offers a way to quantify image quality regionally. When talking about the limitations or challenges of ultrasound, image quality is consistently one of the main points mentioned. However, while image quality impacts the performance and precision of virtually any method or clinical ultrasound measurement, there are currently no satisfactory ways to measure local image quality automatically and objectively. Thus, the methods proposed in this thesis is a new approach to this challenge and opens the door for new applications.

The third and final contribution deals with another typical problem in deep learning for echocardiography: limited available labeled data. In deep learning for natural images, public datasets with millions of labeled images are available [56], while for echocardiography, the amount of images in public datasets is in the order of thousands [31]. Simply annotating more data is not easily done since these annotations should only be performed by clinical experts and it is time consuming to create annotations of high quality. Additionally, there can be large variation in how even experts annotate echocardiography images. This highlights the need to have multiple experts discuss to reach a consensus on how to annotate images, which is even more time consuming. As the common saying in deep learning goes, a deep learning model is only as good as the data it is trained on. Thus, the proposed generative augmentations provide an interesting way to leverage unannotated data, thereby enhancing model generalisability without the need for extensive manual labeling.

While some deep learning based models are already available in current scanners [57, 58] or as a separate tool [59], its widespread use remains limited due partly to concerns about reliability, generalisability, and interpretability. The three contributions presented in this thesis address these barriers by enhancing the trustworthiness of automated analyses. Future adoption will depend on continued collaboration between clinicians and engineers to refine these tools, validate their impact on patient outcomes, and ensure they enhance, rather than replace, clinical expertise. Only then can deep learning fulfill its potential to democratize access to precise and efficient cardiac diagnostics while upholding the highest standards of patient care.

1.5 Publication list

Below follows a complete list of all publications and presentations made as part of this thesis.

Publications included in the thesis

1. **Gilles Van De Vyver**, Sarina Thomas, Guy Ben-Yosef, Sindre Hellum Olaisen, Håvard Dalen, Lasse Løvstakken, Erik Smistad. *Toward Robust Cardiac Segmentation using Graph Convolutional Networks*, Published in IEEE Access, March 2024. Volume 12: 2024, Pages:33876 - 33888

2. **Gilles Van De Vyver**, Svein-Erik Måsøy, Håvard Dalen, Bjørnar Leangen Grenne, Espen Holte, Sindre Hellum Olaisen, John Nyberg, Andreas Østvik, Lasse Løvstakken, Erik Smistad, *Regional Image Quality Scoring for 2-D Echocardiography Using Deep Learning*, Published in Ultrasound in Medicine & Biology, Volume 51, Issue 4, April 2025, Pages 638-649.

3. **Gilles Van De Vyver**, Aksel Try Lenz, Erik Smistad, Sindre Hellum Olaisen, Bjørnar Leangen Grenne, Espen Holte, Håvard Dalen, Lasse Løvstakken, *Generative augmentations for cardiac ultrasound segmentation*, Submitted for review to Scientific Reports on March 6, 2025.

Presentations and abstracts

1. **Gilles Van De Vyver**, Sarina Thomas, Guy Ben-Yosef, Sindre Hellum Olaisen, Håvard Dalen, Lasse Løvstakken, Erik Smistad, Towards Robust Cardiac Segmentation using Graph Convolutional Networks, Poster, Nordic Probabilistic AI School (ProbAI), June 2023.

2. **Gilles Van De Vyver**, Sarina Thomas, Guy Ben-Yosef, Sindre Hellum Olaisen, Håvard Dalen, Lasse Løvstakken, Erik Smistad, Towards Robust Cardiac Segmentation using Graph Convolutional Networks, Lecture, IEEE International ultrasonics symposium (IUS), September 2023.

3. Artem Chernyshov, Jahn Frederik Grue, John Nyberg, Andreas Østvik, **Gilles Van De Vyver**, Erik Smistad, Lasse Løvstakken, Automated segmentation and quantification of the right ventricle in 2D echocardiography, Poster, IEEE International ultrasonics symposium (IUS), September 2023.

4. **Gilles Van De Vyver**, Svein-Erik Måsøy, Håvard Dalen, Bjørnar Leangen Grenne, Espen Holte, Sindre Hellum Olaisen, John Nyberg, Andreas Østvik,

Lasse Løvstakken, Erik Smistad, Regional quality estimation for echocardiography using deep learning, Lecture, IEEE International ultrasonics symposium (IUS), September 2024.

5. **Gilles Van De Vyver**, Svein-Erik Måsøy, Håvard Dalen, Bjørnar Leangen Grenne, Espen Holte, Sindre Hellum Olaisen, John Nyberg, Andreas Østvik, Lasse Løvstakken, Erik Smistad, Regional quality estimation for echocardiography using deep learning, Poster, 27th. International Conference On Medical Image Computing and computer Assisted Intervention (MICCAI), Ocober 2024.

Thesis outline

Chapter 2 presents technical and clinical background related to this work. The contributions are presented in chapters 3, 4, and 5.

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Background

2

2.1 Ultrasound imaging

This section gives a brief introduction to ultrasound imaging. This is a technical thesis about applied deep learning in cardiac ultrasound imaging. To make successful models, it is important to understand the domain in which the deep learning models are applied. This section starts with a brief introduction to ultrasound imaging in general. It then describes image coherence and it's relation to image quality, a concept that is studied in chapter 4.

Ultrasound imaging

Ultrasound imaging is based on the principle of sound wave reflection. An ultrasound probe (transducer) emits ultrahigh-frequency sound waves that travel through the body, usually in the range of 2 to 18 MHz. Just like any type of physical wave, a portion of the wave is reflected whenever there is a change in impedance in the material encountered and the remainder of the wave continues to travel forward. The complex internal composition of human tissue creates numerous small-scale impedance variations, resulting in a multitude of reflections while an ultrasound wave traverses human tissue. On the other hand, blood is relatively uniform and does not reflect as much. Finally, the impedance difference between soft tissue and bones is large, resulting in most of the wave being reflected, creating in a shadow region behind the bone. A receiving transducer receives the reflected ultrasound waves and an image is formed through a signal processing pipeline called beamforming.

Delay-and-sum beamforming is a signal processing technique used in ultrasound imaging to focus on specific points within the body. The main idea is to align and sum up signals from multiple transducer elements so that the reflected waves from a particular location add up constructively, while reflections from other locations cancel out. When an ultrasound pulse is sent into the body, it reflects off structures like tissues and organs. These reflected waves travel back to the transducer elements at different times, depending on the distance of the reflecting point. Since the speed

of sound in tissue is relatively constant, we can estimate how long it takes for the waves to return. To form an image of a specific point, the signals received by each transducer element are adjusted by introducing delays. These delays compensate for the differences in travel time, ensuring that the waves from the desired point arrive in phase (aligned). When these delayed signals are summed together, their amplitudes reinforce each other, enhancing the signal from that point. In contrast, signals from other locations arrive out of phase and tend to cancel out [1].

Transmit beamforming involves focusing transmitted waves to the physical region of interest, the focal region. Delay-and-sum transmit beamforming works in a similar way as the receive beamforming described above. The time delay for each transducer element is calculated such that the transmitted waves from all elements arrive simultaneously, and thus with equal phase, at the focal region. This requires each element to fire at slightly different times, creating a synchronized wavefront that converges at the target location. The result is a concentration of energy in the focal region. In practice, transmit and receive beamforming are combined. Transmit beamforming can focus the ultrasound waves on a single specific region at a time, while receive beamforming allows for focusing on multiple regions throughout the entire image simultaneously [1]. In practice, many different beamforming techniques exist, but they all fundamentally build upon the concept of combining coherent signals [2].

The description above describes brightness-mode (B-Mode) imaging. Other ultrasound modalities like motion-mode (M-mode) and Doppler imaging are not used in this work.

Image coherence and image quality

The **coherence factor** is a measure of the phase consistency or coherence of the received ultrasound signals across different elements of the transducer array. There are several ways to calculate the coherence factor in the literature [3–7]. In this work, we use the coherence factor defined by Rigby [8]

$$\mathrm{CF} = \frac{|\sum_{i=1}^{N} S_i|}{\sum_{i=1}^{N} |S_i|} \tag{2.1}$$

where S_i is the delayed signal for the i-th transducer element. This is equivalent to dividing the coherent sum of the signals by the sum of their absolute values, which represents the incoherent sum. Fig. 2.1 shows the coherent and incoherent sum schematically. In traditional beamforming, the sound of speed is assumed to be constant inside the body. This assumptions is mostly valid for human tissue, but is violated when there are for example layers of fat between the probe and the

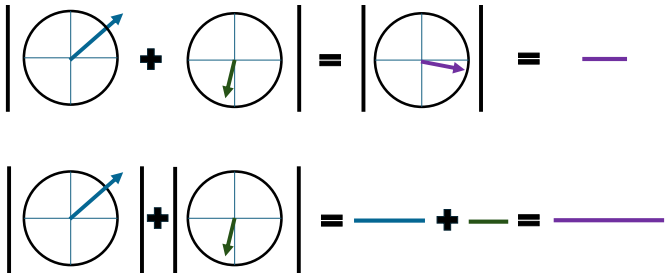


Figure 2.1: Schematic illustration of the coherent and incoherent sum of the transducer signals in the coherence factor calculation. The top part of the image shows the coherent sum that takes into account the phase. The sum of the signals cancels out to some extent because of the difference in phase. The bottom part of the image shows the incoherent sum where the phase is not taken into account.

structure of interest. The resulting distortions are aberrations and will result in a lower coherence value. Clinicians perceive ultrasound images with higher coherence as having higher image quality [9]. Image coherence plays a central role in one of the three proposed image quality estimation methods in chapter 4.

Ultrasound image quality is influenced by a lot of factors and the perceived image quality is highly subjective. There is an inherent trade-off in the frequency used by a transducer with respect to theoretical image quality. A higher frequency gives better axial resolution, but also worse depth penetration, resulting in poor performance for deeper structures. Another trade-off lies in spatial versus temporal resolution. Forming many images with short time intervals will give more accurate time resolution, but will generally deteriorate image quality [1]. Finally, acoustic artifacts such as reverberations, haze, and shadows also influence the image quality [10].

2.2 Echocardiography

This section describes how ultrasound is used to image the heart. It starts by briefly describing basics of circulatory physiology relevant for this work. It then continues with describing the current clinical workflow and common measurements performed. The field of echocardiography is large and the focus of this thesis is on machine learning. Thus, only relevant topics and measurements are discussed here.

Relevant basics of circulatory physiology

The circulatory system is responsible for transporting oxygen, nutrients and waste products to and from tissues through blood. It consists of the systemic, the pulmonary, and the coronary circuit.

The systemic circulation is responsible for transporting oxygenated blood from the heart to the rest of the body and returning deoxygenated blood back to the heart. This circuit begins in the left ventricle, where oxygen-rich blood is pumped into the aorta and distributed to tissues and organs via arteries and capillaries. After delivering oxygen and nutrients, blood collects waste products and carbon dioxide, returning to the right atrium through progressively larger veins.

The pulmonary circuit carries deoxygenated blood from the heart to the lungs, where it is oxygenated, and then returns the oxygen-rich blood to the heart. This process begins in the right ventricle, which pumps blood into the pulmonary arteries, leading to the lungs. Once oxygenated in the lungs, the blood travels back to the left atrium of the heart via the pulmonary veins. Since this circuit only involves pumping blood between the lungs and the heart, the pressures in the right ventricle are significantly lower than in the left ventricle.

The coronary circuit specifically supplies blood to the heart muscle (myocardium) itself. The coronary arteries branch off from the aorta, providing oxygen and nutrients to the heart tissue. Deoxygenated blood from the heart muscle is then collected by coronary veins and returned to the right atrium through the coronary sinus.

The contraction and relaxation phase of the heart muscle are called systole and diastole respectively. During systole, the ventricles contract to pump blood out of the heart. During diastole, the heart muscle relaxes, and the ventricles fill with blood from the atria in preparation for the next contraction. In a healthy heart, the left and right ventricles contract simultaneously. End systole is the point in the cardiac cycle just after contraction when the ventricles have the smallest volume of blood, and end diastole is the point just before contraction begins when the ventricles hold their largest volume of blood. These timepoints mark the extremes of blood volume in the heart during each beat.

Clinical workflow of routine examinations

Transthoracic echocardiography (TTE) is a non-invasive acquisition technique where the transducer is placed on the chest wall (thorax) to obtain images of the heart. In addition to TTE, transesophageal echocardiography (TEE) is another commonly used ultrasound technique, which involves placing the transducer within the food pipe (esophagus) to obtain high-resolution images of the heart. Due to the

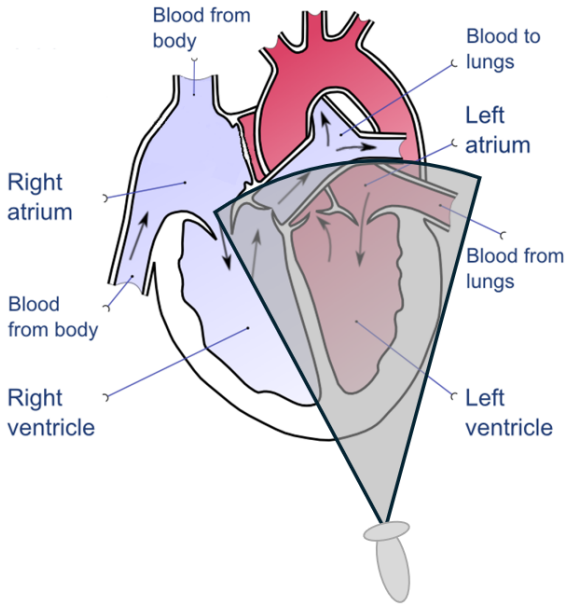
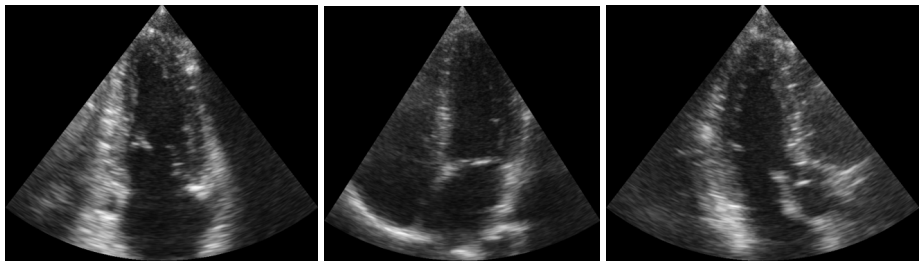


Figure 2.2: Alignment of the ultrasound probe relative to the heart for apical views. Figure adapted from Manco Capac[1]

proximity of the esophagus to the heart, TEE provides superior image quality. While TTE is non-invasive and typically the initial diagnostic tool, TEE is often used in specific scenarios. This thesis focuses on TTE.

During a routine echocardiographic recording, the operator obtains a set of standard imaging views of the heart, typically following guidelines established by the American Society of Echocardiography (ASE) or the European Association of Cardiovascular Imaging (EACVI). The standard imaging planes include parasternal, apical, subcostal, and suprasternal views. This work focuses on apical views, for which the ultrasound probe should be placed in the extension of the apex, as shown schematically in Figure 2.2. By rotating the probe, the operator can get different apical views. Figure 2.2 shows the most common apical views. Standardization of these views allows measurements to be performed in a repeatable way so they can be compared to normal values [11].

[1]Manco Capac, Atrial septal defect, retrieved from https://upload.wikimedia.org/wikipedia/commons/9/92/Atrial_septal_defect-en.png, CC BY-SA 3.0



(a) Apical two chamber (A2C) **(b)** Apical four chamber (A4C) **(c)** Apical long axis (ALAX)

Figure 2.3: The most common apical views

Relevant Measurements

Using the standard imaging planes, several clinical measurements can be performed. In this section, the clinical measurements studied in this thesis are described.

Ejection fraction (EF) is the percentage of blood ejected from the left or right ventricle during each heartbeat. It is calculated as

$$EF = \frac{EDV - ESV}{EDV}$$

where EDV and ESV are the end diastolic and end systolic volumes respectively. Currently, the recommended way to measure volume in ultrasound is Simpson's biplane method, also called the disk summation method [11]. The method takes as input the traces of the endocardium (inner heart wall) in both the A2C and A4C view and gives the volume as output. It consists of the following steps:

1. In each view, draw a line from the middle of the base to the apex. The base is defined as a straight line between the annulus points in the 2D frame.

2. In each view, divide the line from step 1 into an equidistant number of segments. Typically, 20 segments are used. These remaining steps calculate and sum the volume of each disk corresponding to these segments.

3. In each view, calculate the diameter from the 2D view.

4. Calculate the volume of each disk as

$$\text{Volume}_{\text{disk}} = \pi h \frac{d_{A4C}}{2} \frac{d_{A2C}}{2}$$

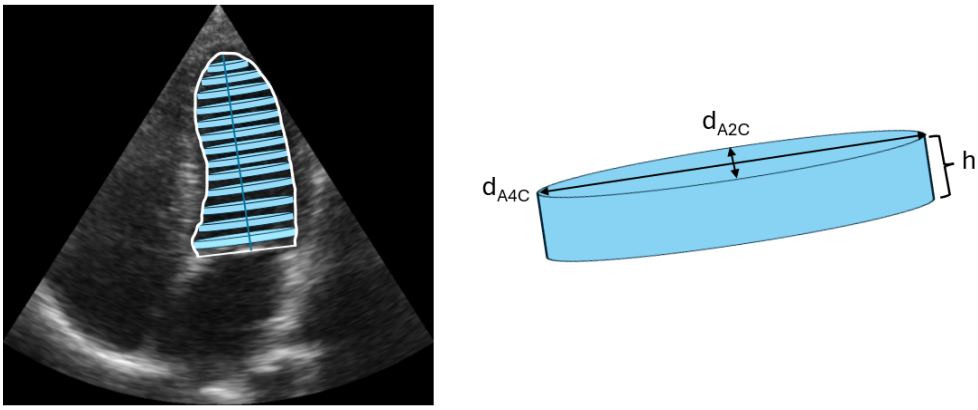


Figure 2.4: Schematic illustration of Simpon's biplane method for calculating LV volumes. For illustrative purposes, the figure shows only a portion of the disks. In reality, there are no gaps between the disks and the next disks starts where the previous one end.

where d_{A2C} and d_{A4C} are the diameters in A2C and A4C and h is the height of each disk from step 2. Thus, the method assumes the volume is ellipsoid.

5. Sum the volume of each disk.

Figure 2.4 schematically shows the disks of the Simpson's method for the LV. Normal LV EF values are in the range 53-73% and EF values below that range are suggestive of abnormal LV systolic function [11]. However, this is not a necessary condition, and heart failure with preserved ejection fraction (HFpEF) is common [12].

Strain measures how much the heart muscle (myocardium) stretches or contracts during the cardiac cycle. It is calculated by measuring the relative change in length of the myocardium. Longitudinal strain is the most commonly evaluated type of strain and measures the relative change in length of the myocardium along the direction that runs from the base to the apex. Global longitudinal strain (GLS) evaluates the average longitudinal strain across the entire myocardium and is typically a negative value, reflecting the shortening of the myocardium during systole. Normal values for LV peak GLS are vendor and software specific [11, 13–16]. Strain measures are obtained by tracking the myocardium, which is currently done with speckle tracking or doppler tissue imaging [11]. It is also possible to measure strain regionally, typically in the six segments shown in Figure 2.5. Another way to asses longitudinal strain is to measure **mitral annular plane systolic excursion (MAPSE)**, which measures the longitudinal motion of the mitral annulus points during systole. This method is recommended when three or more

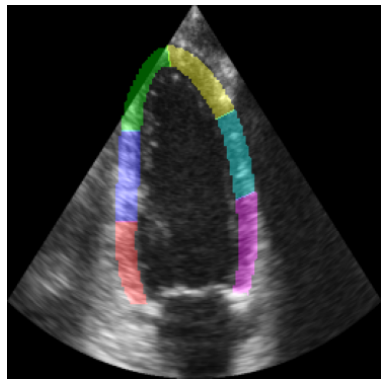


Figure 2.5: Myocardial regions typically used to calculate regional strain.

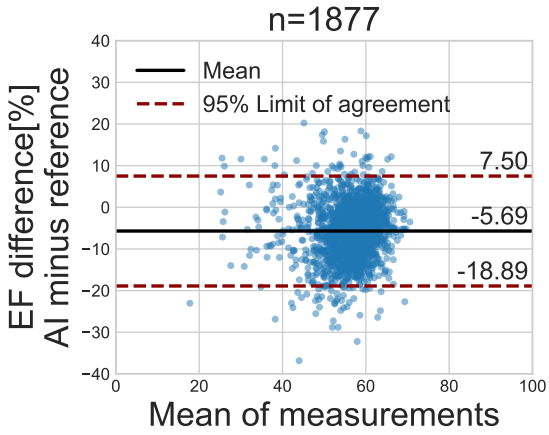


Figure 2.6: Example of a Bland-Altman plot [17]. In this case, the clinical measurement is EF and the two methods are automatic EF using segmentation and reference EF from GE EchoPAC software.

myocardial segments can not be tracked accurately [11].

Bland-Altman analysis

Bland-Altman analysis is a statistical method that assesses the agreement between two methods measuring the same clinical measurement [18]. Typically, a new alternative method is tested against an existing, gold standard method. The Bland-

Altman plots in this work compare measurements extracted automatically using deep learning tools with the gold standard reference values obtained using clinical software. Figure 2.6 shows an example of such a Bland-Altman plot. The Bland-Altman plot quantifies bias and limits of agreement (LoA) and enables visual detection of systematic differences between the two methods. For example, it could be that one method is underestimating large values or vice versa. In that case, the bias is not valid over the full range of possible measurement values. Otherwise, if the bias remains consistent across all values, it can be adjusted for by subtracting the mean difference between the methods from the output of the new method. It could also be that the spread of the differences is not consistent across the range of averages. In that case, the LoA are not valid over the full range. Finally, outliers can easily be spotted in the Bland-Altman plot. The publications in chapters 3 and 4 use Bland-Altman plots when evaluating the proposed methods used to extract clinical measurements.

2.3 Theoretical machine learning background

This section gives a brief theoretical background of machine learning. The section begins with a broad, abstract perspective, gradually focusing on topics more closely related to subjects of the thesis. Section 2.4 describes more concrete and practical tools used in this work.

Foundations of Machine Learning and Neural Networks

The core concept of machine learning involves creating algorithms that can learn to recognize patterns and make decisions from data without being explicitly programmed. The most popular branch of machine learning at the moment is artificial neural networks. A neural network can be abstractly described as a collection of processing units, known as nodes or neurons, arranged in two or more layers. The neural networks include an input layer, zero or more hidden layers, and an output layer. Each node in a neural network receives multiple inputs, where each input is multiplied by a corresponding learned weight. Additionally, each node has a learned bias value that is added to the weighted sum of inputs. The result is then passed through an activation function, which is typically non-linear. These non-linear activation functions have been shown to be crucial for the expressive power of neural networks [19].

For neural networks, learning is achieved by training, which involves repeated iterations of forward propagation, where input data is processed through the network to produce predictions, and backpropagation, where error gradients are calculated and weights are updated according to how much they contributed to the error [20].

This iterative adjustment minimizes the difference between the predicted output and the actual target, thereby improving the network's ability to learn and generalize from data. Training is often performed in batches, where instead of processing the entire dataset at once, the data is divided into smaller subsets (batches). This approach improves computational efficiency and stabilizes learning by enabling more frequent weight updates. The randomness introduced by batches helps reduce overfitting and can help in escaping local minima. Central to backpropagation is the use of a loss function, which quantifies the error between predictions and actual outcomes [19].

In theory, a neural network with only a single hidden layer can learn highly complex models. In fact, the universal approximation theorem asserts that such a network can approximate any continuous function on a compact set with arbitrary precision. However, this theorem does not specify the number of neurons or the amount of data required for this approximation [21]. In practice, shallow neural networks are less effective compared to deep neural networks, which build upon the same principle but utilize multiple hidden layers. Deep learning methods consistently outperform other machine learning approaches given sufficient data. Classic methods, such as statistical techniques or shallow neural networks, tend to reach performance plateaus with increasing amounts of data, unlike deep learning methods.

Machine learning as a complex curve fitting problem

The goal of a machine learning model is to model the relationships that are present in the data. For supervised learning, we can abstractly represent the relationships as a function f that maps an input x to an output y, potentially with some additional noise ϵ. For example, we can estimate tomorrow's temperature (y), by looking at given meteorological observations (x). The goal is to construct a model $\hat{f}(x)$ that matches the true relationship f(x) as closely as possible. However, even if we had the perfect model, $\hat{f}(x) = f(x)$, there would still be an error due to random effects or lack of observations. For example, if x only contains the time of the year, there will be more random error than when it also includes data from weather stations, even if the model is optimal. We denote this irreducible randomness as $\epsilon \sim \mathcal{N}(0, \sigma^2)$, and $y = f(x) + \epsilon$. If all possible combinations of x and y were known, it would be possible to reconstruct f(x) exactly and the error would correspond to ϵ. In practice, a model is fitted on a dataset D containing a limited amount of samples of x and y pairs [22].

Machine learning problems are often applied in high dimensional spaces. For example, in a computer vision application with images of size 256×256, x will have 256^2 dimensions. For segmentation problems, y also has the same dimensions

and the problem can be seen as fitting a plane in this high dimensional space. The curse of dimensionality [23] makes that the datapoints in D are actually far from each other and D is in fact a sparse sample. The challenge is then to fit the plane in between these points [22].

The bias-variance trade-off and overfitting

To evaluate the performance of a machine learning model, we need to consider both the bias and variance contributions to the overall error. It can be shown that the expected mean squared error (MSE) of the fitted model is [24]

$$\text{MSE} = \mathbb{E}_{D,\epsilon}(y - \hat{f}(x;D))^2 = \text{Var}_D[\hat{f}(x;D)] + (\text{Bias}_D[\hat{f}(x;D)])^2 + \sigma^2 \tag{2.2}$$

$\text{Var}_D[\hat{f}(x;D)]$, the **variance**, quantifies how sensitive the model is to fluctuations in the training data D. A model with high variance is overly complex, capturing noise in the data rather than just the underlying pattern. This results in significant changes in predictions when training on different datasets, leading to poor generalization to new data. $\text{Bias}_D[\hat{f}(x;D)]$, the **bias**, measures the inability of the model to learn the relationships present even with perfect data available. It typically measures systematic errors caused by incorrect, implicit assumptions built into the model [25–27].

The trade-off arises when developing the model to minimize the expected error. It is typically not possible to minimize both the bias and the variance at the same time. Models that are not complex enough to capture the true relationships present in the data **underfit** the data. This introduces bias that stems from limited model capacity, which can be seen as wrong assumptions being made. The classic example of this is modeling noisy data with a high order polynomial relationship with a linear model. On the other hand, high-variance models will **overfit**, where the model learns to memorize the training data. This can lead to poor performance when using the model on new, unseen data. Here, the classic example is modeling noisy data with a linear relationship with a high order polynomial [27]. While too small models have low variance and underfit the training data, the reverse is not always true: complex models do not necessarily have high variance. While large, over-parameterized models learn to memorize the training data, they can still generalise well to new data. This concept is known as double descent [28, 29].

As an example of the bias-variance trade-off, regularization typically reduces the variance but increases the bias of a model. Adding regularization constraints makes the model less flexible by adding implicit assumptions of smoothness, thus potentially increasing the bias. At the same time, these same constraints will potentially prevent the model to create solutions to the fitting problem highly

sensitive to small changes in the training data, thus reducing the variance. The classic example of regularization is L2 regularization for polynomial fitting, where the magnitude of the model's weights are prevented from becoming very large, creating an implicit smoothness assumption. Intuitively, the increase in bias due to regularization is a 'bias towards smoothness'.

More data typically helps to reduce the error due to variance. With a larger training set, the model has more examples to learn from, which helps it generalize better and become less sensitive to the noise in any single training example. Therefore, if there is a large amount of data available, a model with lower bias but higher variance will generally perform better. As an example, we can compare classical, statistical models against deep learning models. Statistical models typically have low variance but have implicit modeling constraints based on human-designed features, resulting in relatively high bias. On the other hand, deep learning models typically have fewer model constraints resulting in lower bias, but higher variance. This explains why classical, statistical methods tend to perform well when the available data is limited, while deep learning is more successful with large amounts of available data.

In this thesis, the bias-variance trade-off is particularly noticeable in 3, where a graph convolutional model is proposed that introduces explicit and implicit constraints on the output of the model. This increases the bias of the model and decreases the variance.

Supervised learning

This work uses mostly supervised learning, one of the most fundamental paradigms in machine learning, where a model is trained on a labeled dataset. In this approach, the algorithm learns a mapping function from an input to corresponding output labels by minimizing the error between the predicted and actual labels. This error is quantified by a task specific loss function and minimized by iteratively adjusting the model parameters, also known as weights, to reduce the loss. This adjustment is done using backpropagation, an algorithm that efficiently computes the gradients of the loss function with respect to the model's parameters. Backpropagation works by propagating the error (the difference between the predicted and actual output) backward through the network, updating the parameters at each layer. This is done by applying the chain rule of calculus to compute how each parameter in the network contributes to the overall error [24].

Once the gradients are calculated, an optimizer updates the parameters in a way that minimizes the loss function. Most optimization methods are based on stochastic gradient descent (SGD). In SGD, instead of computing the gradient over the entire dataset, the model parameters are updated using gradients computed from a single

or a small batch of training examples. This is not only a practical solution, but can also help the algorithm escape local minima and find a more optimal solution [24].

In supervised learning, the dataset is typically split into three subsets: the training set, the validation set, and the test set. The training set is used to train the model by iteratively optimizing its parameters to minimize the loss function. However, optimizing solely for minimizing the training loss risks overfitting, where the model memorizes noise, outliers, or irrelevant patterns in the training data, leading to poor performance on unseen examples. This is why the validation set is used. It acts as a proxy for unseen data during training. By evaluating performance on the validation set after each epoch or training iteration, one can detect overfitting early, signaled by a growing gap between declining training loss and stagnant or rising validation loss. Hyperparameters, such as learning rate, regularization strength, or model architecture choices, are tuned based on validation performance to find a balance in the trade-off between underfitting (high bias) and overfitting (high variance). The test set, held out entirely until the final evaluation, provides an unbiased assessment of the model's real-world performance after all adjustments are complete. Crucially, the training, validation and test sets must remain distinct: repeatedly tuning hyperparameters on the test set would leak information, artificially inflating performance and defeating its purpose.

Cross-validation is often used to enhance the reliability of validation and mitigate the risk of overfitting, especially when working with limited data. This technique partitions the training data into k subsets (or "folds"), iteratively training the model on $k-1$ folds and validating on the remaining fold. By rotating the validation fold across all k iterations and averaging the results, cross-validation provides a more robust estimate of model performance and reduces dependency on a single train-validation split. A common variant is stratified cross-validation, which preserve class distribution in each fold. The test set should remain strictly reserved for final evaluation, ensuring an unbiased assessment of generalization [24].

Domain shift and generalizability

Robustness of machine learning models is tightly linked to generalizability. Domain shift occurs when the statistical properties of the training data differ from those of the data encountered during deployment, often resulting in degraded model performance. Machine learning models show impressive results when trained and tested on the same kind of data with sufficient data available, but tend to perform badly when there is domain shift present in the data. From the theoretical point of view of machine learning as a complex curve fitting problem, this makes sense, as a machine learning model interpolates between data points. The problem

of generalization is a general problem in machine learning not only specific to echocardiography. This generalization problem is intrinsically connected with the problem of a lack of general AI [30].

Uncertainty in deep learning

Uncertainty in machine learning falls into two theoretic categories: aleatoric and epistemic [31]. Aleatoric uncertainty arises from inherent variability or noise in the data itself, such as sensor errors or ambiguous labels, and is irreducible even with additional data. In contrast, epistemic uncertainty stems from a lack of knowledge about the model or the data, such as insufficient training data or limitations in the model's representational capacity. This type of uncertainty is reducible and can be mitigated by incorporating more data or improving the model architecture. Aleatoric uncertainty can be quantified by directly modeling the noise in the data, such as using probabilistic models [32] or learning a variance parameter alongside predictions [33]. On the other hand, epistemic uncertainty is often estimated through techniques like Monte Carlo (MC) dropout [34], Bayesian inference [32, 35], or ensemble methods [36, 37], which explore model uncertainty by considering multiple possible representations or predictions. An ensemble method is used in Chapter 3.

2.4 Machine learning components and architectures

This section starts by giving an overview of the neural network components used in this thesis. It continues by describing how these components are combined to form the neural network architectures used in this work.

Neural network components used in this thesis

The fully connected layer is the fundamental building block of neural networks. In this layer, an input of size n is transformed into an output of size m. Each output unit is computed as a weighted sum of all input units plus a bias term, resulting in a total of $m(n+1)$ trainable parameters. Due to the dense connectivity, fully connected layers can model complex relationships within the data. However, this also leads to a large number of parameters, making the layer computationally expensive, harder to train, and more prone to overfitting [24].

The **fully connected layer** is the most elementary neural network layer. At each layer, an input of size n is transformed into an output of size m. For each output, all the inputs and a bias term are combined with different weights, resulting in a total of $m * (n+1)$ parameters for the layer. Compared to other types of layers, this is typically a large number of parameters. This gives the layer the potential

to learn complex relationships, but also makes them harder to train and at risk of overfitting [24].

The **convolutional layer** is the most commonly used type of neural network layer, and networks that utilize it are known as convolutional neural networks (CNNs). The central concept in a convolutional layer is the reuse of weights in a kernel at multiple locations in the input via a sliding window fashion, which is biologically inspired by the organization of the visual cortex in animals. Fig. 2.7 shows this principle schematically. At each layer of a 2D CNN, an input of size

$$\text{input size} = C_{\text{in}} \times W_{\text{in}} \times H_{\text{in}} \tag{2.3}$$

is transformed to an output of size

$$\text{output size} = C_{\text{out}} \times W_{\text{out}} \times H_{\text{out}} \tag{2.4}$$

where C_{in} and C_{out} are the number of channels, W_{in} and W_{out} the width, and H_{in} and H_{out} the height of the in- and output respectively. For other dimensions, the width and height is replaced by the corresponding dimensions, such as depth in the case of 3D convolutions. The kernel only uses a subset of all elements in the input that are within the receptive field and combines these in a fully connected way. Therefore, the number of parameters is

$$\text{number of parameters} = (K_W * K_H * C_{\text{in}} + 1) * C_{\text{out}} \tag{2.5}$$

where $K_w \times K_H$ is the kernel size[2]. This is typically a lot lower than the number of parameters in a fully connected layer. This efficient reuse of parameters can be seen as a type of regularization that lowers variance but adds constraints and thus some bias. Because the same kernel is applied, a convolutional neural network (CNN) can learn to extract visual features like shapes and edges wherever they occur in the input image [38]. However, this property also means that CNNs can struggle with understanding spatial relationships between objects. This limitation can be relevant in tasks like anatomical segmentation, where relative positioning between different structures can be important.

The **self-attention layer** computes weighted representations of a sequence of input

[2]The derivation assumes a 2D input. The same logic applies to other dimensionalities

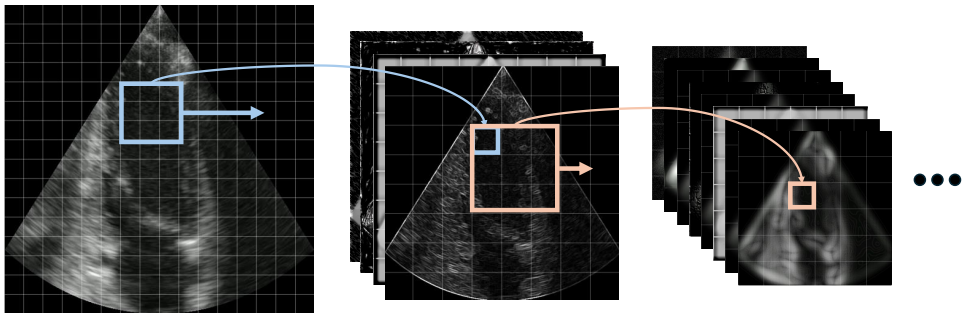


Figure 2.7: Schematic illustration of the sliding window mechanism of a convolutional neural network. The convolutional window slides over the input extracting local features. Each output channel uses its own set of weights and thus extracts different kind of features as the other channels. Typically, the width and height decreases and number of channels increases deeper into the network, resulting in the layers extracting more global features. The visualization of these features are only for illustrative purposes and not actual feature maps from a model.

data by assigning attention weights to each element [39]

$$\begin{aligned} \text{Attention}(X) &= \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \\ Q &= XW_Q \\ K &= XW_K \\ V &= XW_V \end{aligned}$$

where $X \in \mathbb{R}^{n \times d}$ is the input sequence, consisting of n elements, each represented by a d-dimensional feature vector, $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$ are learned weight matrices that convert each element to a d_k-dimensional feature vector. The output of the softmax combining the K and Q vectors gives higher values for the most relevant parts of the input [39], an is therefore referred to as attention weights. The layer is called self-attention because the same input vector is used to calculate the Q, K, and V vectors. Multi-head attention [40] computes multiple sets of attention weights in parallel, each with a different set of learned weight matrices. The results of these multiple attention heads are usually concatenated and linearly transformed into the final output representation

$$\text{MultiHead}(X) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h)W_O \tag{2.6}$$

where $W_O \in \mathbb{R}^{hd_k \times d}$ is a learned weight matrix. The attention mechanism was

first proposed for natural language processing tasks, where the sequence of input vectors is the sequence of word embeddings [39]. This was later expanded to image processing, where the sequence of input vectors is a set of image patches. The Vision Transformer (ViT) [41] applies the same self-attention mechanism to sequences of image patches rather than words. In this approach, an image is divided into non-overlapping patches, which are then flattened and linearly embedded into a sequence of vectors. These patch embeddings serve as the input to the self-attention layer, and the model applies multi-head attention over these patch embeddings in the same way as in the original transformer architecture.

As **upsampling** operator, the models in this work use a deconvolutional layer or nearest neighbour upsampling. A deconvolutional layer is the opposite of a convolutional layer. Instead of combining $K_W \times K_H$ neighbouring input elements into a single output elements, a single input element is transformed into $K_W \times K_H$ neighbouring output elements. An alternative to a deconvolutionoal layer is nearest neighbour upsampling, where each input value is directly copied over to the output multiple times in the vicinity of the input, thus increasing the resolution.

For **donwsampling**, the models in this work use max or average pooling. Downsampling in convolutional neural networks (CNNs) serves two main purposes: reducing computational complexity and increasing the receptive field without enlarging the filter size. A Max Pooling operator decreases resolution of an input by converting neighboring input values to a single output value by taking the maximum input value. An alternative is Average Pooling, which uses the average instead of the maximum.

The models in this thesis use ReLU, leaky ReLU, Mish [42], SiLU [43], and Softmax as **activation functions**. The ReLU funciton is defined as

$$\text{ReLU}(x) = max(0, x) \tag{2.7}$$

The ReLU6 function is a variant of the ReLU that is constant for large values of x. This helps to avoid some outputs becoming too big, flooding the other outputs.

$$\text{ReLU6}(x) = min(max(0, x), 6) \tag{2.8}$$

The leaky ReLU is similar, but avoids dying neurons from signals being zeroed when they are negative

$$\text{Leaky} \quad \text{ReLU}(x) = max(0, x) \tag{2.9}$$

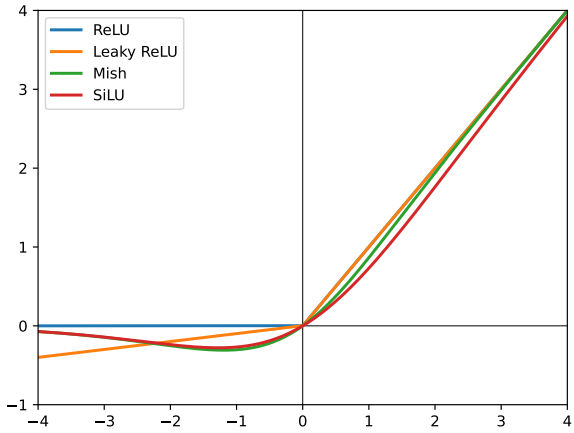


Figure 2.8: Activation functions used in this thesis. ReLU6 is the same as ReLU until $x = 6$, and is then constant after that.

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ \alpha x & \text{if } x < 0 \end{cases} \tag{2.10}$$

The Mish function is defined as

$$y = x \tanh\left(\log\left(1 + e^{x}\right)\right) \tag{2.11}$$

The SiLU funcion is defined as

$$\text{SiLU}(x) = x \cdot \frac{1}{1 + e^{-x}} \tag{2.12}$$

Fig. 2.8 shows these functions. Finally, the softmax activation function is a little different to the functions above as it is usually only applied to the final output of a neural network and is a multivariate function:

$$Softmax(x) = \frac{e^{z_i}}{\sum_{j=1}^{N} e^{z_j}} \tag{2.13}$$

where z_i is the i-th element of the output vector z. It is generally difficult to say why one activation function works better than another and the choice is often based on empirical results [42, 44].

A **skip connection** simply copies the input or latent features to a deeper part of

the network. This is then combined together with an output coming from another branch. Usually this combination is done by adding the features element-wise or concatenating the features channel-wise [24].

As **normalization schemes**, the models in this work use batch normalization (BN), Z-score normalization or GroupNorm. First, BN adjusts the distribution of each batch of input. For an input x,

$$\mathrm{BN}(x) = \gamma\left(\frac{x-\mu}{\sqrt{\sigma^2+\epsilon}}\right)+\beta \tag{2.14}$$

where μ is the mean value of of the batch, σ^2 the variance of the batch, ϵ a small constant added for numerical stability, and γ and β are learnable parameters. BN helps deep neural networks tackle the problem of internal covariate shift, which means a shift in distribution to the input of layers due to updates in preceeding layers. By normalizing the inputs of each layer, BN helps to stabilize and accelerate the training process. The parameters γ (scale) and β (shift) allow the network to learn the optimal scale and shift for the normalized activations, ensuring that the model has the flexibility to adapt to the specific requirements of the task while maintaining stable training dynamics. Batch normalization is typically applied after each layer [45]. Second, Z-score normalization is the classic statistical normalization that changes the distribution of the input to have zero mean and unit variance

$$Z(x) = \frac{x-\mu}{\sigma} \tag{2.15}$$

It is typically applied to the input of the neural network. Finally, GroupNorm is similar to BN, but divides the channels into groups and calculates the mean and variance for each group instead of for each channel separately [46]. GroupNorm is particurly effective when training with small batch sizes. Normalization is applied in neural networks to improve training stability, accelerate convergence, and mitigate issues related to different input data scales. It is typically applied before or after each layer.

As **loss functions**, this work uses the mean squared error (MSE), cross-entropy loss and Dice loss. The MSE is the most widely used loss function for regression tasks. It penalizes larger errors more heavily due to the squaring of differences, making it sensitive to outliers. The Cross-entropy loss is the most widely used loss function for classification tasks.

$$L = -\sum_{i=1}^{C} y_{ref}\log(p_{out}) \tag{2.16}$$

where C is the number of classes, y_{ref} the reference vector, and p_{out} the output distribution over the classes. The reference vector is usually a one-hot encoded vector equal to one for the single correct class and zero elsewhere. Finally, the Dice loss is the is a measure often used in segmentation tasks, particularly in medical imaging.

$$L_{\text{Dice}} = 1 - \frac{2\sum_i y_{i,\text{ref}}\, y_{i,\text{pred}}}{\sum_i y_{i,\text{ref}} + \sum_i y_{i,\text{pred}} + \epsilon} \tag{2.17}$$

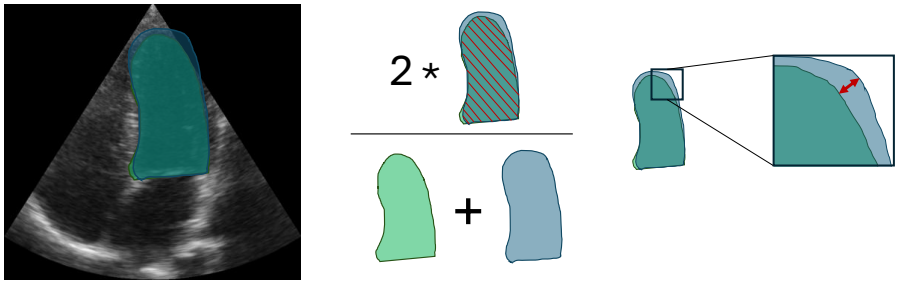
where $y_{i,\text{ref}}$ and $y_{i,\text{pred}}$ are the reference and predicted vectors for the i-th pixel and ϵ a small value to avoid dividing by zero. As before, $y_{i,\text{ref}}$ is usually a one-hot encoded vector equal to one for the single correct class the pixel belongs to and zero elsewhere. For segmentation, the Dice loss can be seen as a measure of overlap between the prediction and reference segmentation masks. Figure 2.9 shows this schematically.

As **evaluation metrics** on a test set, this work uses the loss functions above and the Hausdorff distance. The Hausdorff distance is a metric used to evaluate the similarity between two sets of points, often used in segmentation tasks to measure the spatial accuracy of predicted segmentation masks compared to reference masks. The Hausdorff distance takes into account the maximum distance between any point in one set to its closest point in the other set, in both directions. The Hausdorff distance is formally defined as:

$$H(A,B) = \max\left(\max_{a\in A}\min_{b\in B}\|a-b\|, \max_{b\in B}\min_{a\in A}\|a-b\|\right) \tag{2.18}$$

where A and B are two sets of points, $\|a-b\|$ the Euclidean distance between a and b. In the context of segmentation, A represents the boundary points of the predicted mask, and B represents the boundary points of the ground truth mask. A smaller Hausdorff distance indicates better alignment between the predicted and reference masks. Figure 2.9 shows how the Dice and Hausdorff distance are calculated for a cardiac segmentation task. The Hausdorff distance is more sensitive to outliers than the Dice score. For example, small disconnected parts far away from the main segmented blob will produce a large Hausdorff distance, while the Dice score will not be affected as much.

The **Adam optimizer** [47] is one of the most popular methods for training neural networks. Its main goal is to update the weights of the network based on the gradients of the loss function. Instead of directly using the raw gradients, Adam keeps track of two things: an average of past gradients and an average of the squared



(a) Example segmentations. **(b)** Dice score **(c)** Hausdorff distance

Figure 2.9: Dice score and Hausdorff distance for an example segmentation.

gradients. The averaging makes the updates more stable and less sensitive to noise in the data, while the second term slows down updates when the gradients vary a lot.

Dropout is a regularization technique used to prevent overfitting in neural networks by randomly deactivating a subset of neurons during training. By forcing the network to learn more robust and distributed representations, dropout helps reduce reliance on specific neurons and prevents co-adaptation. It has been shown to improve generalization across various deep learning tasks [**?**].

Data augmentation is a widely used technique in machine learning to artificially increase the size and diversity of training datasets. During training, each image is transformed with one or more randomized augmentation transformations before being used as input to the network. In image processing, typical transformations are rotations, translations, scaling, flipping, and color and brightness adjustments to the existing data. This process is particularly crucial in tasks with limited labeled data, where augmentation can mitigate overfitting [48].

Neural network architectures used in the thesis

The **U-Net** [49] architecture is a type of fully convolutional encoder-decoder neural network designed for image segmentation. It features a symmetric, U-shaped structure consisting of a downsampling/encoder and an upsampling/decoder path. The downsampling path reduces the spatial dimensions of the features while increasing the number of channels and extracts global features. The upsampling path gradually restores the original spatial dimensions while reducing the number of channels. In order to capture local, high-resolution details in the input, the architecture includes skip connections that copy the features in the downsampling path to the equally sized layers in the upsampling path. Fig. 2.10 shows a schematic diagram of the U-Net 1 architecture from Leclerc et al. [50]. With around two

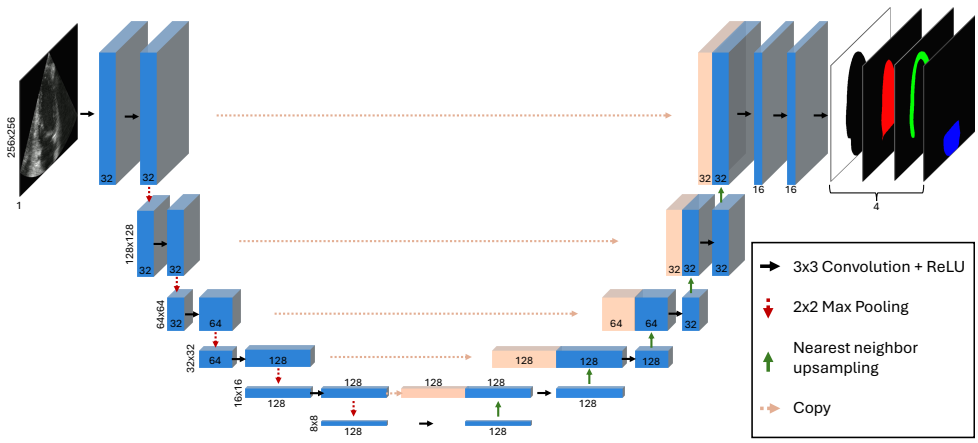


Figure 2.10: Schematic diagram of the U-Net 1 architecture from Leclerc et al. [50].

million parameters, this is a relatively small realization of the U-Net architecture with a low runtime. The nnU-Net [51], described in more detail below, is a bigger realization of the U-Net architecture, but with typically higher accuracy than U-Net 1.

The **MobileNet-v2** architecture [52] is a CNN encoder developed to be small and computationally lightweight. Tables 2.1 and 2.2 give an overview of the architecture. Fig. 2.11 gives an schematic overview of a bottleneck block, which is the main component of the architecture. The bottleneck block is characterized by a depth-wise convolutional layer followed by a 1×1 convolution without activation function. The depth-wise convolution convolves each channel independently with a separate kernel and concatenates the result. The result is then fed to the 1×1 convolution to produce the final output. These two layers combined perform a similar operation to a normal convolution over all channels, but have less parameters and are computationally less demanding. The lack of activation function in the 1×1 convolution prevents ReLU from destroying too many negative signals when compressing the features back to fewer channels. The resulting architecture is a small and computationally efficient network that can be used in real-time on low-end mobile devices. The limited size also gives it less variance and more bias compared to similar deep learning networks. This work uses a modified MobileNet-v2 architecture in chapter 3.

A **graph neural network (GNN)** represents data as nodes in a graph. Each node has a neighbourhood of nodes to which it is connected via edges in the graph. During a forward pass through a graph network, the contents of its nodes are updated based

Table 2.1: Overview of the MobileNet-v2 architecture. Each row represents one level of the architecture consisting of a sequence of blocks of one or more layers. The paramter t is the expansion factor of the bottleneck layer. The parameter n is the number of times the block is repeated. Finally, the parameter s is the stride of the first convolutional layer in the block. Whenever the stride is equal to 2, the dimensions of the features are halved in that first layer. All remaining convolutions have stride 1. Table 2.2 gives the specifications of the bottlneck block. The table is adapted from Sandler et al. [52].

Input	Operator	t	n	s
$224^2 \times 3$	1×1 convolution	-	1	2
$112^2 \times 32$	bottleneck	1	1	1
$112^2 \times 16$	bottleneck	6	2	2
$56^2 \times 24$	bottleneck	6	3	2
$28^2 \times 32$	bottleneck	6	4	2
$14^2 \times 64$	bottleneck	6	3	1
$14^2 \times 96$	bottleneck	6	3	2
$7^2 \times 160$	bottleneck	6	1	1
$7^2 \times 320$	1×1 convolution	-	1	1
$7^2 \times 1280$	1×1 average pool	-	1	-
$1 \times 1 \times 1280$	1×1 convolution	-	-	-

on their neighbors. The updated content of node u is

$$\mathbf{h}_u = \phi\left(\mathbf{x}_u, \bigoplus_{v \in N_u} \psi(\mathbf{x}_u, \mathbf{x}_v)\right)$$

where ϕ and ψ are functions, x_u the node that is being updated, N_u the neighbourhood of node u, and $\bigoplus$ an aggregation function that combines the signals from each node. The functions ϕ, ψ, and $\bigoplus$ contain the learnable parameters. After traversing the network, the final output can be extracted as the content of some or all of the nodes [53]. This is a very broad definition, and in fact, all common architectures can be seen as special cases of graph neural networks. For example, a CNN is a graph neural network where each pixel or feature map value is a node and the neighbourhood of each node is determined by the convolutional kernel.

Similar to a CNN, a **graph convolutional neural network (GCN)** [54] reuses learned weights at several locations. Typically, one layer reuses the same weights to update all nodes once. This allows the GCN to generalize across different nodes and their neighborhoods, irrespective of their specific positions in the graph. Similar to CNNs, this weight sharing reduces variance and increases bias. The decoder of the proposed segmentation network in chapter 3 is a GCN.

Table 2.2: Specifications of the bottleneck block of MobileNet-v2. The table is adapted from Sandler et al. [52].

Input	Layer	Output
$H \times W \times C_{in}$	1×1 convolution + ReLU6	$H \times W \times tC_{in}$
$H \times W \times tC_{in}$	3×3 depthwise convolution with stride s + ReLU6	$\frac{H}{s} \times \frac{W}{s} \times tC_{in}$
$\frac{H}{s} \times \frac{W}{s} \times tC_{in}$	1×1 convolution	$\frac{H}{s} \times \frac{W}{s} \times tC_{out}$

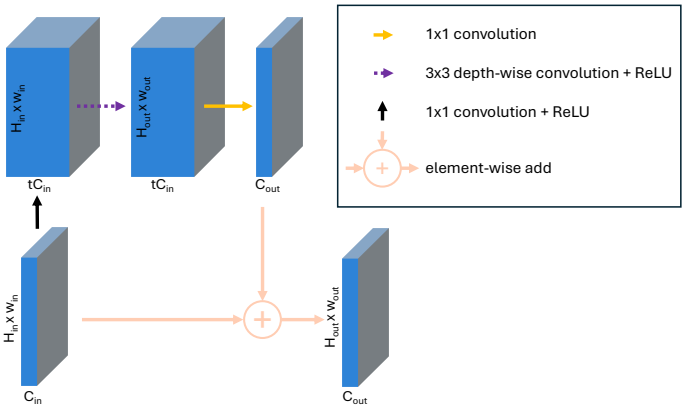


Figure 2.11: Schematic diagram of the bottleneck block in the MobileNet-v2 architecture. The first time the block is applied at a certain level in the architecture, $C_{in} \neq C_{out}$, and the element-wise add with the input is not performed. Additionally, if the stride is equal to 2, the dimensions will be halved in the first block, $W_{out} = \frac{W_{in}}{2}$ and $H_{out} = \frac{H_{in}}{2}$. For the remaining blocks at the same level, $C_{in} = C_{out}$, $H_{in} = H_{out}$, and $W_{in} = W_{out}$, and the element-wise add can be applied.

2.5 Echocardiographic image analysis using deep learning

This section starts with the main challenges and opportunities of automated image analysis for echocardiography. It continues by focusing more specifically on annotation uncertainty as it is also studied more extensively in this work.

Challenges of automated image analysis for echocardiography

Domain shift and generalization are an important challenge in machine learning, and machine learning for echocardiography is no exception. For example, a segmentation model trained on only apical two chamber views will perform badly on apical long axis views. In such cases, machine learning models in echocardiography can create solutions that do not make any sense from an anatomical point of view. This is shown in chapter 3.

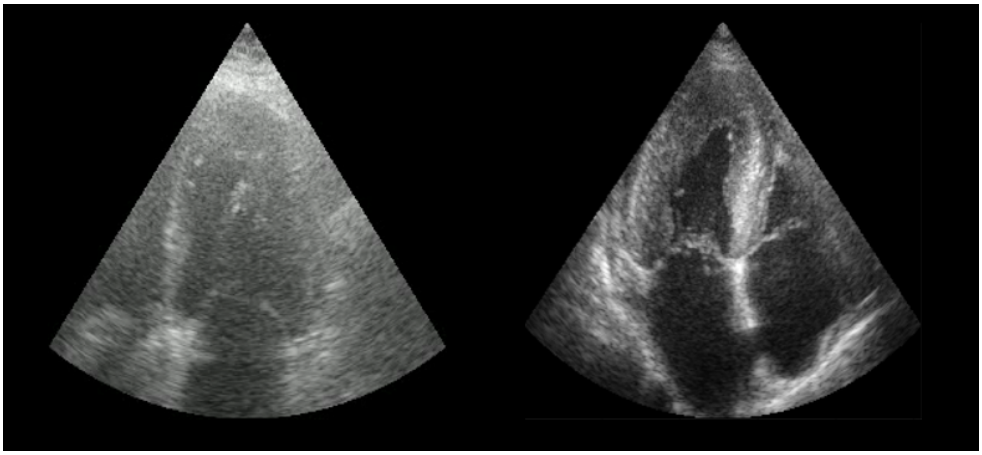


Figure 2.12: These two echoes where obtained using the same scanner, yet the quality is vastly different.

Another potential challenge of automated image analysis is the lack of explainability. This is especially important in the medical domain, where the health and well-being of patients can be at stake. A partial workaround for this problem is to divide the problem into smaller tasks that can be verified easily instead of solving a complex problem directly. For example, measuring ejection fraction (EF) with deep learning can either be done end-to-end [55], or with a pipeline that involves timing, segmentation, and the biplane Simpson's method [56].

Ultrasound images can have low image quality, especially in low-end devices. Image quality is affected by noise and artifacts, and varies by patient. In some recordings, all structures are clearly visible and analysis is relatively straightforward, while for others it is a considerable challenge to find the contours of the structures being imaged. Fig. 2.12 shows how the quality of images can vary considerably, even when they are recorded with the same ultrasound scanner. In 2D ultrasound imaging,out-of-plane motions are an additional challenge. Cardiac structures move in and out of the imaging plane during the cardiac cycle, which complicate tracking and segmentation [57].

Image quality has been shown to directly impact the reliability of ultrasound-based measurements. Studies report that low-quality images contribute to higher inter-observer variability [58], as clinicians and automated systems struggle to precisely delineate anatomical boundaries.

The lack of a universally accepted ground truth or expert-consensus labels also poses a significant challenge, as different cardiologists may provide varying interpretations

of the same images. Furthermore, echocardiographic image acquisition is highly operator-dependent, leading to variability in probe positioning, gain settings, and angle selection, which can impact model performance due to distribution shift [57].

Finally, machine learning tools in echocardiography are often required to run in or near real-time. This poses limitations on the complexity of the machine learning methods being used.

Opportunities of automated image analysis for echocardiography

Deep learning has the possibility of being able to run in real-time while an operator is scanning. In the current workflow, the operator first acquires the recordings, and only starts analyzing the acquired images later on, often when the patient is already gone. Thus, it can happen that the recorded images are not ideal to perform the desired measurements. It can also happen that the operator wants to perform extra measurements based on the new information from image analysis that requires a specific way of recording, for example a specific cardiac view or focus on a specific part of the heart. Real-time, automated analysis eliminates these problems. The exponential advances in computing power using graphic processing units (GPUs) have made it possible to run bigger, more powerful models in real-time.

Automated image analysis has the possibility to create new applications that improve the quality of medical image analysis in general. One example is guiding operators to produce standardised views [59], which helps avoiding errors due to inaccurate acquisitions [60]. Another example is image enhancement and artefact removal [61]. A final example that, to the best of my knowledge, has not yet been implemented, is a quality assurance model that can give an idea to the operator how much variability is to be expected from a given measurement and what is the cause of such variability. An important factor of this variability is image quality, which is studied extensively in chapter 4.

Eliminating inter- and intra-observer variability of clinical measurements used for diagnosis and follow-up is a major opportunity for automated image analysis and is described in more detail in the following subsection.

Inter- and intra-observer variability

Measurements obtained manually or semi-automatically from ultrasound images involves several steps that each introduce a source of variability. For example, to obtain the EF , the operator first needs to 1) record an apical two chamber (A2C) and an apical four chamber (A4C) recording, then 2) needs to select the frames of end diastole (ED) and end systole (ES) and finally 3) needs to place contour points in each of these frames. Each of these steps introduce variability into the measurement.

Thus, unlike natural image analysis, there is no fixed ground truth.

Two important benchmarks for variability are the **inter-observer variability** and **intra-observer variability**. The intra-observer variability is the variability between repeated measurements of the same operator. Each operator has its own intra-observer variability. The inter-observer variability is the variability between different operators performing the same measurements. Finally, there can also be systematic differences between annotators [62]. Typically operators from the same institution or work group will have smaller systematic differences than between different groups. The inter-observer variability is typically larger than the intra-observer variability.

Automated image analysis has the opportunity to eliminate the variability completely since they are deterministic. This consistency can greatly enhance the reliability and reproducibility of measurements from ultrasound images. By removing the human element from the process, both the inter-observer and intra-observer variability are eliminated, leading to more stable and potentially more accurate results.

Ultrasound image segmentation

Automatic image segmentation is a computer vision task that entails dividing an image into different regions or segments with little or no human intervention. There exist several techniques for segmentation that do not rely on deep learning, including active contours [63], level sets [64] and Kalman filters [65]. However, deep learning has revolutionized the field of image segmentation with models that can segment more complex regions and has been proven to be more accurate also for cardiac ultrasound segmentation [66]. In this thesis, we only focus on image segmentation using deep learning.

For medical image segmentation using deep learning, U-Net [49] is the most widely used neural network architecture. U-Net has shown to perform well for different modalities [67], and ultrasound is no exception [66]. U-Net was first proposed in 2015 and up to date the best performing architectures only make minor adjustments to the original. However, the performance between different U-Net realizations can differ drastically, based on the choice of hyperparameters and design details [68]. Furthermore, an improvement for one task or dataset does not necessarily carry over to other tasks or datasets and the lack of proper validation is a problem in current research [69]. This makes it hard to design good baseline models to compare against when doing research.

The **nnU-Net** [51, 70] provides a pipeline that automatically configures the

hyperparameters to a given task. It combines the knowledge gained through years of educated trial-and-error performed by the research community since the conception of U-Net into a single pipeline. The resulting models offer competitive performance that is in many cases the current state of the art. Therefore, this model is the preferred choice for image segmentation in this thesis.

2.6 Generative AI for image synthesis: diffusion models

This section gives a theoretical background on generative AI for image synthesis, which is explored in depth in the article in chapter 5. The section starts by defining key concepts in the theory of generative models and finishes with an in-depth theoretical decomposition of diffusion models.

Generative Models

Generative models aim to learn and replicate the underlying patterns of a dataset by estimating or approximating the true data distribution. It represents the likelihood of observing various data points across the space of all possible values. In the image domain, this probability density function would give a likelihood to each combination of pixel values in an image. An image consisting of random noise would have a low probability, while specific pixel configurations—those forming realistic images relevant to the domain being modeled—would have a higher likelihood. Thus, this distribution captures the patterns, dependencies, and structures inherent in the data and can be thought of as a mathematical model of how real-world data is generated. However, generative models can also learn incorrect patterns and dependencies, leading to artifacts or biases that do not accurately reflect the true data distribution.

The primary objective of a generative model is to capture the key characteristics of the target distribution, enabling it to generate new samples that are statistically similar to the observed data and appear to originate from the same source. There are various methods to achieve this, and not all generative models require an explicit mathematical formulation of the data distribution. Some approaches, such as normalizing flows, explicitly define the probability distribution of the data and allow for direct computation of likelihoods. Others, like generative adversarial networks (GANs), bypass the need for an explicit probability model by learning to generate realistic samples through adversarial training. Common generative modeling techniques include variational autoencoders [71], generative adversarial networks (GANs) [72], normalizing flows [73], and diffusion models [74]. This thesis specifically focuses on diffusion models, which learn the data distribution

implicitly by modeling a stochastic process that transforms noise into structured data.

Diffusion models

The central idea of a diffusion process is to transform noise into an image in an iterative fashion. In order to do this, the model is trained at several steps in a noisy sequence $x_0 ... x_T$, where the original image x_0, is combined with more and more Gaussian noise $x_1, x_2, ...$, until x_T which is approximately pure Gaussian noise. The learning objective of a diffusion model is to approximate the unknown reverse transformation, e.g. learn to go from x_i to x_{i-1}. During training, the model learns to reverse the added noise at random steps, as illustrated in algorithm 1. During inference, the model runs iteratively on the input data. At each step, a portion of the predicted noise at that step is subtracted and another portion of new Gaussian noise is added. Algorithm 2 illustrates the sampling process. The remainder of this section goes into more details on how the equations in the algorithms are derived.

Algorithm 1 Pseudocode of the training procedure for a diffusion model, as proposed by Ho et al. [75]

repeat
  $x_0 \sim q(x_0)$
  $t \sim \text{Uniform}(\{1, \dots, T\})$
  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
  Take gradient descent step on
    $\nabla_\theta \left\| \epsilon_t - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, t\right) \right\|^2$
until converged

Algorithm 2 Pseudocode of the sampling procedure for a diffusion model, as proposed by Ho et al. [75]

$x_T \sim \mathcal{N}(0, \mathbf{I})$
for $t = T, \dots, 1$ **do**
  $z \sim \mathcal{N}(0, \mathbf{I})$ if $t > 1$, else $z = 0$
  $x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\, \epsilon_\theta(x_t, t) \right) + \sigma_t z$
end for
return x_0

In the **forward process**, the transformation from a x_{t-1} to x_t adds Guassian noise and the literature [74–76] uses the following notation for the stochastic forward

diffusion process:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{2.19}$$
$$q(x_{1:t}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$$

Thus, at each step in the forward process, the image from the previous step is scaled by a factor $\sqrt{1-\beta_t}$ and Gaussian noise with variance β_t is added. The hyperparameters $\beta_0 ... \beta_T$, defined in a noise scheduler, control how much noise is added in each step. The second equation expresses the accumulation of Gaussian noise over all steps in the forward process as a sequence of each individual stochastic transition. Ho et al. and Sohl-Dickstein et al. [74, 75, 77] derive a closed form to directly sample x_t, the noisy sample at timestep t.

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \quad \text{with } \bar{\alpha}_t = \prod_{t=1}^{T} \alpha_t \tag{2.20}$$

In the **reverse process**, the goal is to undo the Gaussian noise added at each step in the forward process, reconstructing a clean image from the noisy version. This goal is achieved by training a model to approximate the data distribution of the previous noise-adding step in the diffusion process. Then, during inference, the model generates samples by running iteratively on white noise. The target distribution, the distribution of the noise-adding step, is a Gaussian distribution. Thus, approximating this distribution boils down to estimating the mean and variance. Instead of directly approximating the mean, Ho et al. [75] found that predicting the accumulated noise gives better results. Ho et al. and Sohl-Dickstein et al. [74, 75, 77] show the Gaussian reverse transition $q(x_{t-1}|x_t)$ can be written in closed form when conditioned on the clean image x_0:

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\beta}_t \mathbf{I})$$
$$\tilde{\mu}_t(x_0, x_t) := \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right) \tag{2.21}$$
$$\tilde{\beta}_t := \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$$

where ϵ_t is the Gaussian noise of the image at step t. Thus, $q(x_{t-1}|x_t, x_0)$ can be calculated from x_t and ϵ_t. In practice, ϵ_t is not known during inference and in order to approximate $q(x_{t-1}|x_t, x_0)$, the model learns to estimate the accumulated noise ϵ_t

given the noisy image x_t and time step t. This estimate by the model is denoted as $\epsilon_\theta(x_t, t)$, with θ the model parameters. This can be rewritten using equation 2.20:

$$\epsilon_\theta(x_t, t) = \epsilon_\theta(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, t)$$

which is used in the optimization step in algorithm 1. During inference, equation 2.21 is used during the sampling process in algorithm 2. It can be shown that the optimization problem described above is a reweighted variant of maximizing the log likelihood of the reverse transition distribution under the learned model [74, 75, 77]. The diffusion model of Ho et al, [75] only estimates the mean of the reverse transition distribution and keeps the variance constant. Nichol and Dhariwal [76] further improve upon this model by optimizing the noise scheduler, learning the variance, and optimizing a different variant of the learning objective.

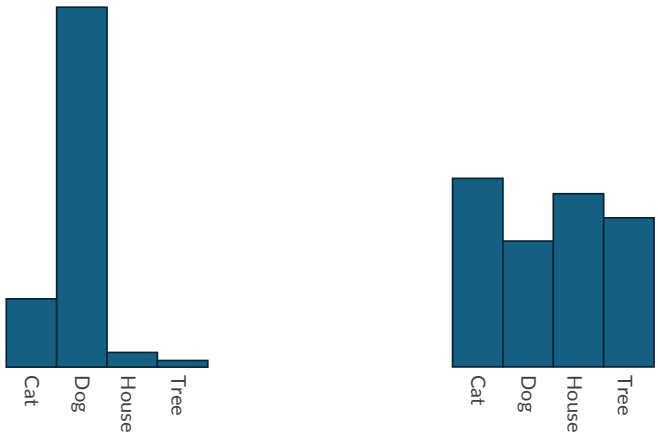
Evaluation metrics for generative models

The **Inception Score (IS)** [78] evaluates the quality and diversity of images generated by a model by using a pretrained Inception v3 network [79] to measure how well the generated images resemble distinct, recognizable classes while maintaining variety. For a generative model that generates high quality images, the softmax output of an inception model run on a generated image should contain a high value for a single class and a low probability for the remaining classes, as in Figure 2.13(a). This means the generated image is clearly recognized as one of the classes by the inception model and is not vague or random. On the other hand, the sum of all these distributions of individual images should be well spread out over all classes, as in Figure 2.13(b). This means the model creates images of all categories and does not always repeat the same class. Thus, the output distributions of the individual samples should be very different from the distribution of the total. The inception score is the exponential of the average Kullback–Leibler (KL) divergence between the distributions of the individual samples and the total. For N generated samples,

$$IS = \exp\left(\frac{1}{N}\sum_{i=1}^{N} D_{\mathrm{KL}}[p_i(y) \| p_{all}(y)]\right) \tag{2.22}$$

where $p_i(y)$ is the predicted class probability distribution for the i-th generated image, and

$$p_{all}(y) = \frac{1}{N}\sum_{i=1}^{N} p_i(y)$$



(a) Example distribution of a single generated image.

(b) Distribution of all generated images.

Figure 2.13: Example distributions of possible softmax outputs of an inception model for generated images. The class labels are only for illustrative purposes. In reality, the distribution is over the classes the inception model is trained on. The most common choice is a model trained to classify the 1000 ImageNet classes. The inception score is the average of the KL divergences between the distributions of single images and the distributions of the total.

The **Fréchet Inception Distance (FID)** [80] is another widely used metric for evaluating the quality of images generated by a model. FID measures the distance between the distributions of real images and generated images in the feature space of a pretrained Inception v3 network [79]. Unlike the Inception Score, which focuses on the diversity and recognizability of generated images alone, FID evaluates how closely the generated images resemble the real images in the feature space. Specifically, FID calculates the Fréchet distance between the multivariate Gaussian distributions of the features extracted from real and generated images:

$$FID = \|\mu_r - \mu_s\|^2 + \mathrm{Tr}(\Sigma_r + \Sigma_s - 2(\Sigma_r\Sigma_s)^{\frac{1}{2}}) \tag{2.23}$$

where μ_r and μ_s are the means and Σ_r and Σ_s are the covariances of the features for the real and synthetic images respectively.

Both the IS and the FID rely on a pretrained classification network. In practice, the most commonly used model is the Inception v3 network trained on ImageNet. However, this can give misleading results when the generative model is trained on a different dataset [81]. Since ultrasound images have characteristic speckle patterns and unique visual attributes, the features and outputs extracted from the pretrained

ImageNet model might not be ideal.

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Towards Robust Cardiac Segmentation using Graph Convolutional Networks

3

Fully automatic cardiac segmentation can be a fast and reproducible method to extract clinical measurements from an echocardiography examination. The U-Net architecture is the current state-of-the-art deep learning architecture for medical segmentation and can segment cardiac structures in real-time with average errors comparable to inter-observer variability. However, this architecture still generates large outliers that are often anatomically incorrect. This work uses the concept of graph convolutional neural networks that predict the contour points of the structures of interest instead of labeling each pixel. We propose a graph architecture that uses two convolutional rings based on cardiac anatomy. While this architecture does not improve performance on classical measures like Dice score and Hausdorff distance, it does eliminate anatomical incorrect segmentations. Additionally, we propose to use the inter-model agreement of the U-Net and the graph network as a predictor of both the input and segmentation quality in real-time. The results show that from the 100 high agreement samples, 93 were in distribution, while from the 100 low agreement samples, only 7 were in distribution. Finally, this work contributes with an ablation study of the graph convolutional architecture on the publicly available CAMUS dataset and an evaluation of clinical measurements on the clinical HUNT4 dataset. Source code is available online: https://github.com/gillesvntnu/GCN_multistructure

3.1 Introduction

Cardiovascular diseases are the most common cause of death, accounting for one in three deaths in the United States [1]. Ultrasound imaging is the standard diagnostic tool for cardiology, as it is safe, inexpensive, real-time, and non-invasive. Accurate segmentation of the cardiac structures from ultrasound images is important for diagnosis, as it enables the extraction of standard clinical measurements such as left ventricular (LV) volume, ejection fraction (EF), and global longitudinal strain (GLS), which all provide important insights into the patient's cardiovascular health. In today's clinics, it is mainly done manually or semi-automatically. Clinical

guidelines recommend that clinical measurements should be repeated over three cardiac cycles [2]. However, this is typically not performed since it takes too much time for the clinician. Automating segmentation would make repeating the measurements on multiple cardiac cycles trivial.

With exponential growth in computing power and access to large amounts of digitized data, deep learning methods have become the most common approach for automatic medical image segmentation, showing average performance comparable to inter-observer variability [3–5]. Deep learning segmentation approaches usually directly predict a label for each pixel in the image. The U-Net architecture [6] is the most commonly used architecture for pixel-wise segmentation. U-Net is a fully convolutional neural network (CNN) that efficiently combines high and low-level features by integrating skip connections in an encoder-decoder structure. The output is a multi-channel segmentation map, with each channel corresponding to a structure.

Whereas the pixel-wise approach of U-Net gives it excellent accuracy on average, it has two fundamental issues:

1. The U-Net has no anatomical constraints, which can result in outliers and anatomical incorrect segmentations, as shown in Fig. 3.1.

2. The U-Net has no way of reporting when it is uncertain on how to segment an image, for instance in out-of-distribution and low image quality cases.

This lack of robustness hinders clinical use. Since the segmentation accuracy (e.g. Dice) of LV segmentation is on average already very high with U-Net methods, the goal of this paper is not to create a new network which improves these segmentation metrics, but instead investigate new methods for dealing with these two fundamental issues of anatomical incorrectness and detection of failing cases.

3.1.1 Robust U-Net-based segmentation

Several works have tried addressing the fundamental outlier problem of U-Net using shape priors. These works encourage the U-Net to produce anatomically valid shapes, either with a specialized loss function expressing anatomical correctness [7], an atlas prior [8], closeness to a learned embedding [9], temporal consistency, [10] or with joint landmark detection [11]. However, Painchaud et al. [12] showed that methods using soft constraints on pixel labeling methods still produce anatomically incorrect outputs. Their work proposed a post-processing approach that refines the output of the U-Net to the closest anatomically valid shape [12] in a large latent space. The architecture consists of an autoencoder trained on ground

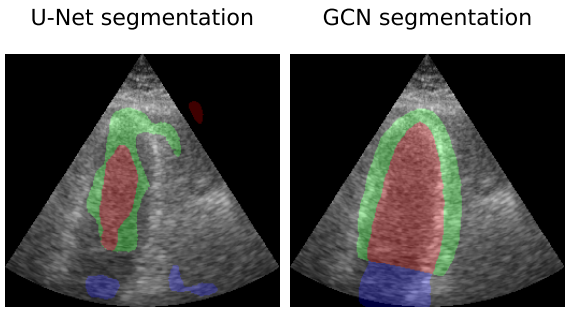


Figure 3.1: Though U-Nets can achieve a high average Dice accuracy on large datasets (> 0.94), they can still produce anatomical incorrect results as shown here to the left, with multiple atria disconnected from the LV and an incoherent myocardium around the LV, none of which are anatomically plausible. The anatomical correct output of the proposed graph convolutional network (GCN) is shown to the right.

truth segmentations and generates a latent space of valid anatomically correct segmentations. During inference, the output of the U-Net-based model is then fed to the encoder producing a latent representation. A nearest neighbor search is performed in the latent space and fed to the decoder to produce an anatomically corrected output. This method shows excellent results but is too slow for real-time applications. However, the work introduces a real-time variant that works as a denoising autoencoder. In this work, we compare our results to this version on the CAMUS dataset [4].

3.1.2 Graph Neural Networks

Graph Convolutional Networks (GCN) address the problem of segmentation in a fundamentally different way. Instead of predicting a pixel-wise segmentation map, these models predict the contour of the segmentation as a graph. The contours are sampled into keypoints which form the nodes of a graph. Previous works have shown the effectiveness of this method for MRI segmentation [13] and X-ray [14]. In our previous work, we demonstrated the feasibility of GCNs for ultrasound segmentation [15] using the EchoNet dataset [16]. However, this work was limited to single-structure segmentation (LV endocardium) on a single view (apical four-chamber) and lacked a thorough evaluation.

3.1.3 Contributions

In this work, we extend our work on the GCN approach for cardiac ultrasound segmentation with a focus on the two fundamental issues highlighted in the introduction; anatomical correctness and detection of failing cases.

1. **Multi-structure segmentation:** We explore methods for extending the graph to segment multiple structures, such as the LV epicardium and left atrium (LA), enabling the segmentation to be used for more measurements such as strain. We propose a clinically motivated design and show that this eliminates anatomically incorrect shapes in the segmentation output of the network. We perform an ablation study of the GCN architecture on the public CAMUS dataset and compare it with U-Net. Additionally, we evaluate the final models on the clinical HUNT4 dataset.

2. **Segmentation quality predictor:** We combine the outputs of the GCN and the U-Net and show that this can be used to detect out-of-distribution and unsuitable low image quality cases, resulting in bad segmentation output.

3. **Open-source framework and demo:** We make our code publicly available. We provide two parts: the full PyTorch framework to reproduce the results (https://github.com/gillesvntnu/GCN_multistructure) and the C++ code to run the real-time demo application described later in this article (https://github.com/gillesvntnu/GCN_UNET_agreement_demo).

3.2 Methods

3.2.1 U-Net baseline models

We compare the GCN with two U-Net architectures. U-Net 1 [4], a fixed architecture optimised for speed, and nnU-Net V2 [17, 18], a self-adapting framework optimised for accuracy. For U-Net 1, we use the same architecture and training procedure as in [4], but with additional augmentations as listed in Table 3.1. These are the same augmentations as in [15]. The nnU-Net is used out of the box using the default configuration, but without the final ensemble step [17] [18]. For both U-Nets, the input images are resized to 256x256 pixels as a pre-processing step.

3.2.2 General structure of the GCN

The general structure of a GCN for cardiac segmentation [15] consists of a CNN encoder pre-trained on ImageNet [19] that acts as a feature extractor and a graph convolutional decoder that reconstructs the keypoints of the contours of the cardiac structures. Fig. 3.2 shows a general overview of the GCN architecture. The encoder transforms a grayscale ultrasound image of $\mathbb{N}^{H \times W}$ pixel values to a vector embedding of $\mathbb{R}^{X}$ values, where X is the feature embedding size of the last layer of the encoder. A dense layer transforms these embeddings to a representation in the keypoint space of size $\mathbb{R}^{n \times C1}$, where n is the number of keypoints and C1 is the number of output channels in the first layer of the GCN decoder. The decoder layers

Table 3.1: Characteristics of the U-Net baselines. U-Net 1 and nnU-Net are the same as in [4] and [17] respectively. The "number of channels" column indicates the number of channels at the first, bottom, and last convolution of the U-Net. Table 3.7 shows the inference time and number of parameters for each network.

Specification	U-Net 1	nnU-Net
Number of channels	32 ↓ 128 ↑ 16	32 ↓ 512 ↑ 32
Lowest resolution	8×8	4×4
Upsampling scheme	2×2 repeats	Deconvolutions
Normalization scheme	None	InstanceNorm
Batch Size	32	49
Learning Rate	1e-3	1e-2
Scheduler	None	Polynomial
Loss function	Dice	Dice & cross-entropy
Augmentations	Rotations, scaling, cropping, brightness adjustments and mirroring	Rotations, scaling, Gaussian noise, Gaussian blur, brightness, contrast, simulation of low resolution, gamma correction and mirroring

perform graph convolutions in the keypoint space and gradually decrease the number of channels until the final output with dimensions $\mathbb{R}^{n\times 2}$, representing the relative pixel coordinates for each keypoint in the image. The size of the intermediate channels of the keypoint embeddings C_i is a tunable parameter.

Every graph convolutional layer consists of a linear layer followed by an activation function. It takes the embeddings of adjacent keypoints as input to update the embedding of each keypoint. In the ring of LV keypoints $p_{1..n}$, the embeddings of keypoints $p_{i-w..i+w}$, are the input to update the embeddings of keypoint p_i, where w is the receptive field of the GCN decoder. The same weights are reused at each keypoint, creating an inductive bias of locality. Section 3.2.4 describes how to expand this architecture to segment multiple structures simultaneously.

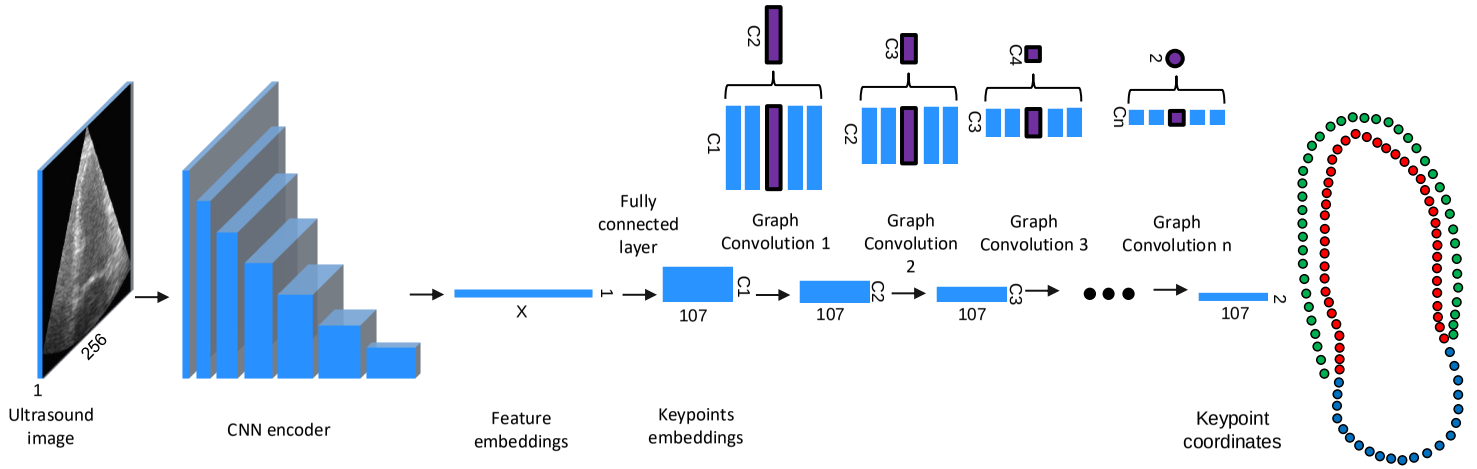


Figure 3.2: The architecture of the GCN. The CNN encoder transforms the input ultrasound image of width W and height H to an embedded vector of size X. A dense layer transforms this embedding to an embedding in keypoint space, with 107 keypoints and $C1$ channels. The decoder consists of a sequence of graph convolutions over these keypoint embeddings. The final outputs are the 2D coordinates of the keypoints in the image.

3.2.3 Keypoint extraction

The keypoints are extracted in a standardized way from the annotations that are available as pixel-wise labels of the left ventricle (LV), left atrium (LA), and myocardium (MYO). The following algorithm extracts anatomical landmarks from the annotations and uniformly samples the contours between these landmarks to fill in the remaining keypoints, as shown in Fig. 3.3. The anatomical landmarks A-G have a unique physical meaning/location, while the rest of the keypoints are sampled uniformly on the contour between these anatomical landmarks as described in the algorithm below:

1. Extract the annulus points, the corner points where the MYO meets the LA. These are points A and B in Fig. 3.3. Connecting these points gives the base line.

2. Extract the base points of the MYO by extending the base line and selecting the furthest point with the MYO label on the base line. These are points C and D in Fig. 3.3

3. Extract the apexes of the LV, MYO, and LA, defined as the furthest points from the base line with the corresponding label. These are points E,F, and G respectively in Fig. 3.3

4. Sample the contour of each structure to complete the keypoint extraction. Sample the endocardium contour line between E and A in Fig. 3.3 for n equidistant points and do the same for the endocardium contour line between E and B, with n a model parameter. The total number of endocardium points is then $2n + 3$, consisting of the sampled points together with A,B, and E. Follow the same procedure for the epicardium, using F,C, and D as corner points. This results in another $2n + 3$ keypoints for the epicardium, as the number of endo- and epicardium keypoints needs to be equal in our architecture. Finally, sample the LA border between G and A for m equidistant points and do the same between G and B, with m another model parameter. The total number of LA keypoints is then $2m + 1$, consisting of the sampled points together with G. We do not add A and B to the LA points because these points already belong to the endocardium keypoints. In this work, we choose $n=20$ and $m=10$, resulting in a total of 107 keypoints. The right side of Fig. 3.3 shows the final set of keypoints.

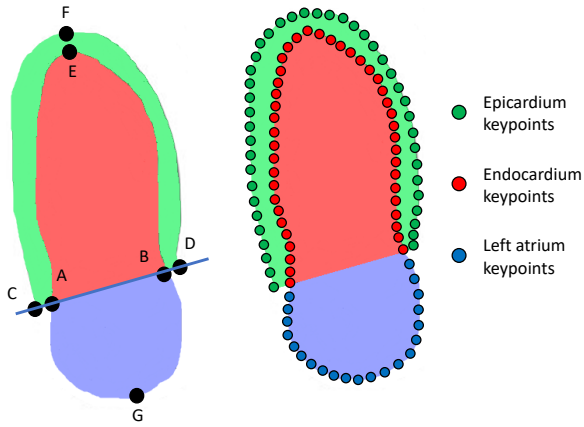
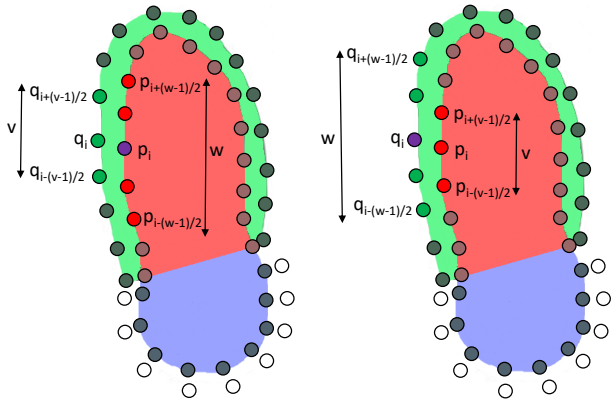


Figure 3.3: Schematic diagram showing the preprocessing to transform pixel labels to keypoints positions. A and B are the base points of the LV. C and D are the base points of the MYO. E, F, and G are the apexes of the LV, MYO, and LA respectively.

3.2.4 Multi-structure segmentation

The graph convolutions of the GCN are performed over the keypoints in the spatial neighborhood in the graph. Since the original approach only included the endocardium, the graph convolutions had to be adapted for the additional structures; the epicardium and the LA.

To include the LA keypoints, the ring of n endocardium keypoints $p_{1..n}$ from the original GCN was extended to a ring of keypoints $p_{1..n+m}$, where m is the number of LA keypoints. The endocardium and LA keypoints together form the keypoints in the inner ring. In our implementation, $n = 43$ and $m = 21$ keypoints are used. For the epicardium keypoints, a second ring of keypoints $q_{1..n}$ was added, which are zero-padded to form a ring of $q_{1..m+n}$ outer keypoints. In the graph convolutional layers of the decoder, the embeddings of inner keypoints $p_{i-(w-1)/2..i+(w-1)/2}$ and outer keypoints $q_{i-(v-1)/2..i+(v-1)/2}$ are the input to produce the embeddings of each inner ring keypoint p_i, where w is the primary receptive field and v is the secondary receptive field of the GCN decoder. The weights are reused at each keypoint, creating an inductive bias of locality. Fig. 3.4(a) visualizes the inner convolution schematically. For each epicardium keypoint q_i, the embeddings of the outer keypoints $q_{i-(w-1)/2..i+(w-1)/2}$ and inner keypoints $p_{i-(v-1)/2..i+(v-1)/2}$ are the input to the convolutional layer. Also, for the epicardium the weights are reused at each keypoint, but the weights of the inner ring are different than the weights of the outer ring. Fig. 3.4(b) visualizes the outer convolution schematically.



(a) Inner ring convolution **(b)** Outer ring convolution

Figure 3.4: Schematic diagram showing the multi-structure convolution. $q_{1..n}$ are zero-padded epicardium keypoints, $p_{1..n}$ are endocardium and left atrium keypoints. The highlighted points are used as input to update the embedding of the purple keypoint, with w the primary receptive field and v the secondary receptive field.For illustrative purposes, the diagram does not show the actual number of keypoints used in this work.

3.2.5 Displacement method data representation

An alternative way to represent the epicardium keypoints is to use the thickness of the MYO at each of the endocardium keypoints. The position of each epicardium keypoint is then defined by the normal distance relative to the corresponding keypoint of the endocardium. The normal direction is calculated using the two neighboring keypoints. Only the final layer of the graph decoder needs to be adjusted to the new output format. The displacement representation prevents the network from producing erroneous segmentations where the epicardium keypoints are inside the endocardium keypoints.

3.2.6 GCN training procedure

The training procedure of the GCN is the same as in [15]. It uses the Adam [20] optimizer using a learning rate of 1e-5. The model is trained for 5000 epochs and the model weights with the best validation accuracy are retained. The training data is resized to 256x256 pixels and augmented by using rotations, scaling, cropping, brightness adjustments, and mirroring. The loss function is the sum of the Euclidean distances of all predicted keypoints plus the mean absolute error of the displacements, if applicable.

3.2.7 Combining U-Net and GCN

GCN - U-Net cascade model

We explore a cascade network of GCN with displacement method followed by U-Net. The idea is that the GCN with displacement method will produce an anatomically correct initial shape, which the U-Net can refine to a more accurate pixel-wise segmentation. In the first stage, the GCN with displacement method is trained as before. After training, the GCN performs inference on the training set, and the resulting keypoints are transformed into pixel-wise segmentation outputs. In the second stage, a U-Net is trained with the concatenation of the grayscale ultrasound image and the segmentation output of the GCN as input. The only change to the U-Net architecture is that the first layer takes an extra input channel. We use the same U-Net architectures as described in section 3.2.1. For U-Net 1, the input segmentation channel is additionally augmented separately with rotation, scaling, and translation to avoid the U-Net from fully relying on the segmentation outputs of the GCN. For nnU-Net, the GCN segmentation serves as an extra input channel and the framework is used out of the box without custom augmentations. Fig. 3.5 shows a graphical representation of the cascade model.

GCN - U-Net ensemble

For estimating ejection fraction using the GCN - U-Net ensemble, both networks run in parallel and construct their segmentations, leading to two different ejection fraction estimations. The average of these two measurements is the output of the ensemble.

3.2.8 Inter-model agreement as a quality predictor

The second goal of this work was to develop a method to detect when the segmentation fails.When the GCN and U-Net generate segmentations in parallel, the agreement of the two models in terms of Dice agreement between the two segmentation outputs contains additional information. The idea is that if the two methods, each with their own data representation, generate similar segmentations, there is a higher probability of the segmentations being correct. On the other hand, when one of the methods makes a large error, for example, due to an out-of-distribution case, it is unlikely the other method will make the same mistake because of their fundamental differences in data representation, as demonstrated in Fig. 3.1. Thus we propose to use the inter-model agreement of these two different segmentation architectures to detect failing segmentation cases.

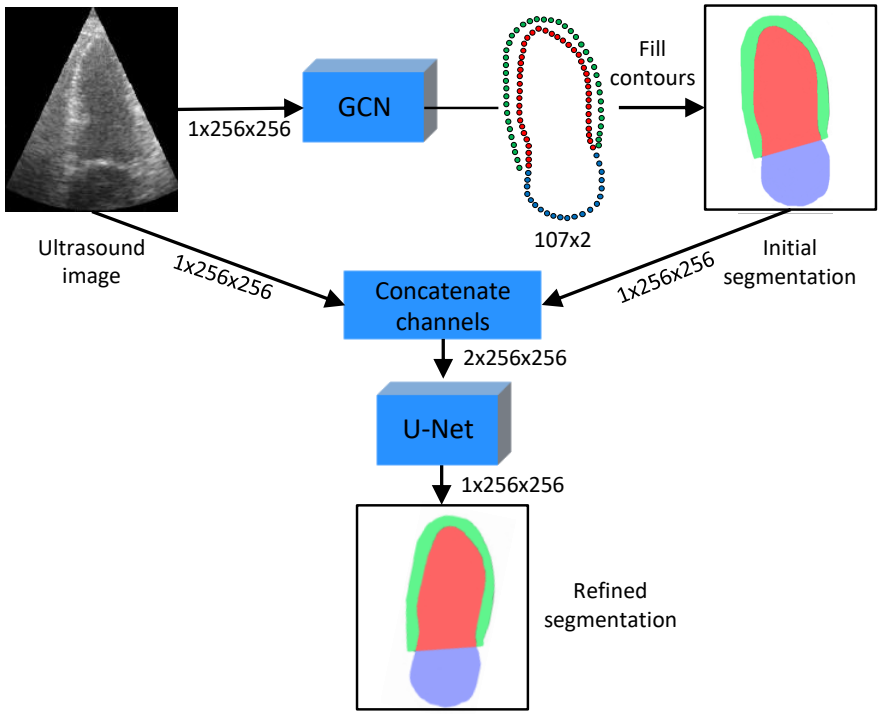


Figure 3.5: Graphical representation of the GCN - U-Net cascade. The cascade model concatenates the initial segmentation produced by the GCN with the original ultrasound input and feeds this to the U-Net to produce the final, refined segmentation. The segmentations shown are only illustrative.

3.3 Experimental setup and Results

3.3.1 Datasets

CAMUS

The CAMUS dataset is a publicly available dataset of 500 patients including apical 2 chamber (A2C) and apical 4 chamber (A4C) views obtained from a GE Vivid E95 ultrasound scanner, equalling 2000 image annotation pairs [4]. The annotations are available as pixel-wise labels of the left ventricle (LV), left atrium (LA), and myocardium (MYO), split into 10 folds for cross-validation.

HUNT4

The *Helse Undersøkelsen i Nord-Trøndelag* ultrasound dataset (HUNT4Echo) is a large-scale clinical dataset of 2462 patient examinations of LV-focused A2C and

A4C recordings used for clinical measurements [5]. Each recording contains 3 cardiac cycles. A fraction of 311 patient exams, the training set, contains single frame segmentation annotations in both ED and ES as pixel-wise labels of the LV, LA, and MYO. For 1913 patient exams, there is a reference value of the biplane LV volumes in end-diastole (ED) and end-systole (ES), obtained manually using the clinically approved EchoPAC software (GE HealthCare).

Differences

Due to differences in annotation conventions and scanning approaches, the HUNT4 and CAMUS datasets can not be used jointly. For instance, the myocardium is consistently annotated to be much thicker in CAMUS than in HUNT4. Although both datasets contain cardiac images of the same clinical views, the images in the HUNT4 dataset are consistently LV-focused, which is not the case for CAMUS. Whereas we consider the HUNT4 annotations to be more clinically correct, we have used and included results for the CAMUS dataset as well in our evaluation since it is publicly available whereas HUNT4 is not.

3.3.2 Ablation study

The ablation study uses the first subgroup of the CAMUS dataset, meaning it tests on the first cross-validation split, validates on the second and uses the remaining eight splits for training. The first experiment varies the decoder while keeping the CNN encoder fixed to MobileNet-v2 [21]. The second experiment varies the encoder.

Table 3.2 summarizes the results of the first part of the ablation study that focuses on varying the complexity of the decoder. The channels are the embedding sizes of the decoder layers, corresponding to $C_{1..n}$ in Fig. 3.2. The secondary receptive field is the number of keypoints used from the second ring, as described in section 3.2.4. The primary receptive field is kept constant at $w = n + m$, the total number of inner keypoints, as in the original GCN. When there are no channels, the decoder is replaced by a single dense layer, converting the feature embeddings from the encoder directly to keypoint coordinates. The difference in overall Dice score between the best and worst performing variation is not statistically significant, $p = 0.77$ and $p = 0.58$ with and without displacement method respectively, using the Wilcoxon signed-rank test [22]. The remainder of this work uses the GCN version with two decoder layers and a secondary receptive field of one.

The second part of the ablation study varies the CNN encoder. Table 3.3 compares the Dice scores of the network with MobileNet-v2 [21] and ResNet-50 [23] as encoder. The difference in overall Dice score is statistically significant ($p < 0.05$) for the version with displacement method and not significant ($p = 0.41$) for the

Table 3.2: Ablation study GCN part one. The difference in Dice score between the best and worst performing variation is not statistically significant with or without displacement method.

Decoder	Channels	Secondary receptive field	Dice LV	Dice MYO	Dice LA
GCN	48,32,32,16,16,8,8,4	11	.910±.07	.813±.15	**.867**±.15
	32,16,8,4	5	.910±.07	.811±.14	.862±.09
	8,4	1	**.914**±.05	**.819**±.13	.861±.09
	None	n.a.	.911±.07	.811±.14	.860±.11
GCN with displacement method	48,32,32,16,16,8,8,4	11	.908±.08	.810±.11	.863±.09
	32,16,8,4	5	.909±.07	.809±.12	.863±.09
	8,4	1	**.911**±.06	**.812**±.11	**.864**±.10
	None	n.a.	.907±.08	.809±.12	.856±.11

Table 3.3: Ablation study GCN part two. The difference in Dice score between the two decoders is statistically significant.

Decoder	CNN encoder	Dice LV	Dice MYO	Dice LA
GCN	MobileNet-v2 [21]	**.914**±.05	.819±.13	.861±.09
	ResNet-50 [23]	.912±.06	**.824**±.11	**.874**±.08
GCN with displacement method	MobileNet-v2 [21]	.911±.07	.802±.16	.862±.09
	ResNet-50 [23]	**.911**±.06	**.812**±.11	**.864**±.10

version without displacement method. While slightly more accurate, the ResNet-50 encoder makes the model slower and an order of magnitude larger, as shown in Table 3.7. Therefore, the remainder of this work uses MobileNet-v2 as encoder.

3.3.3 Anatomical correctness and segmentation accuracy

As the purpose of this work was to create a more anatomical correct segmentation method, an experiment was conducted to measure the number of anatomical incorrect segmentation. For this purpose we use the publicly available CAMUS dataset and the same criteria for anatomical correctness as Painchaud et al. [12]. In addition, we calculate the Dice and Hausdorff distance to demonstrate the trade-off between anatomical correctness and segmentation accuracy. We performed this experiment using our GCN architectures and other state-of-the-art (SOTA) cardiac segmentation methods for comparison, and the results are summarized in Table 3.4. As this study focuses on real-time applications, we compare them with the real-time version of the post-processing method proposed by Painchaud et al., the robust Variational AutoEncoder (rVAE) [12]. Furthermore, we compare to U-Net 1 [4] and

Table 3.4: Anatomical correctness and segmentation accuracy (Dice and Hausdorff) comparison with SOTA on all cross-validation splits of CAMUS.

Method	#Anatomical incorrect*	Dice score†	Hausdorff distance (mm)†
rVAE post processing [12]	5-22‡	.916-.923±n.a.‡	5.7-6.2±n.a.‡
nnU-Net [18]	25	**.949±.03**	**4.56±2.6**
U-Net 1 [4]	68	.943±.04	5.43±4.7
GCN	12	.936±.04	5.20±2.3
GCN displacement	**0**	.932±.05	5.96±2.8
GCN displacement - U-Net1 cascade	32	.943±.03	5.17±2.9
GCN displacement - nnU-Net cascade	**0**	.938±.03	5.15±2.6

*Anatomical correctness was measured with the same criteria as Painchaud et al. [12].
†The Dice score is the average of the LV_{Endo} and LV_{Epi} Dice scores [4]. The Hausdorff distance is the average of the LV_{Endo} and LV_{Epi} [4].
‡Painchaud et al. [12] report several values from different U-Nets. Here, the extreme values are reported.

nnU-Net [18]. To be able to compare to previous work, the Dice score is calculated in the same way as in the work of Painchaud et al. [12] where the myocardium is added to the LV lumen to create an "epicardium" region which results in overall higher Dice scores. The Dice score and Hausdorff distance are calculated for the LV by itself (LV_{endo} in [4]) and the LV and MYO together (LV_{epi} in [4]). The final result is the average of these two values. With $p < 0.05$ using the Wilcoxon signed-rank test [22], the Dice score of nnU-Net is significantly higher than that of the GCN, regardless of the presence of the displacement method.

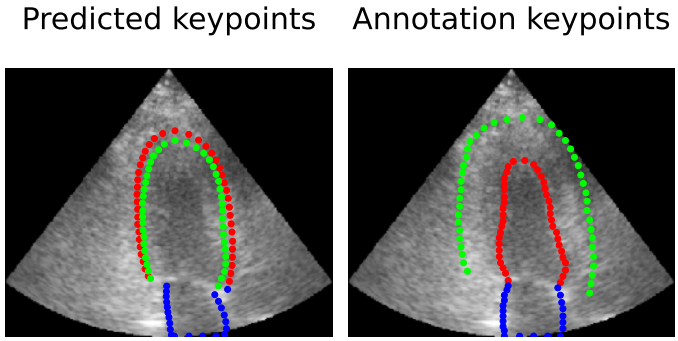
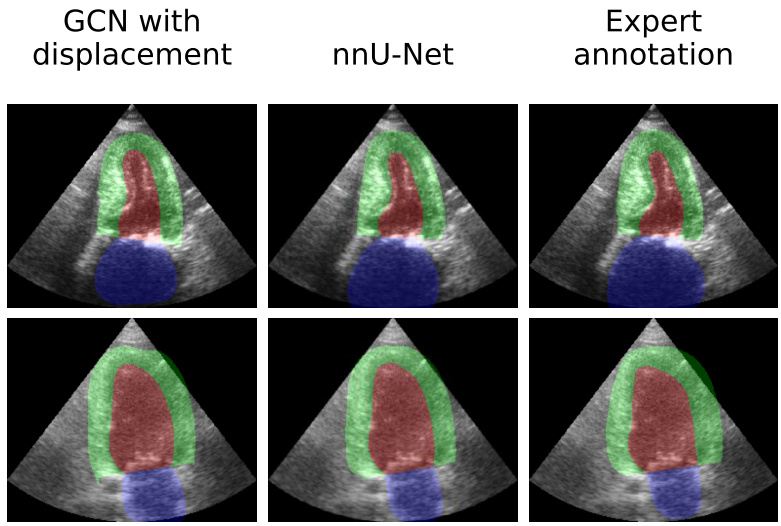
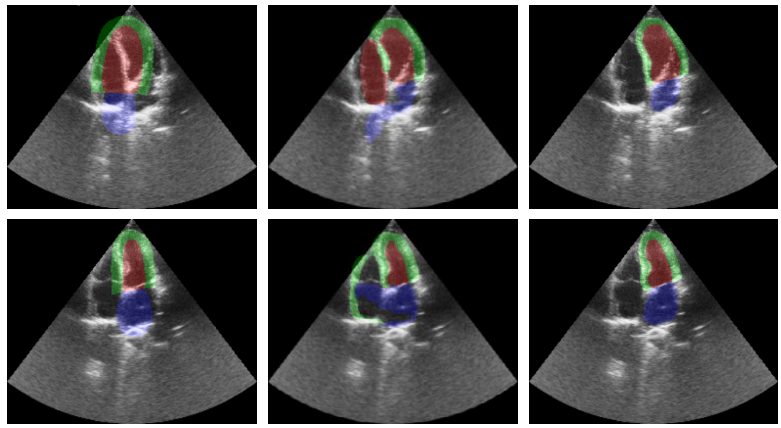


Figure 3.6: Anatomically incorrect case for the GCN without displacement method. The predicted keypoints of the endo- and epicardium are flipped and partly overlapping. The displacement method eliminates these errors.



(a) Median cases



(b) Worst cases

Figure 3.7: Case analysis and comparison of the GCN with displacement method and nnU-Net on CAMUS. The cases are selected based on the overall Dice score between the annotation and the GCN or U-Net segmentations.

The anatomically incorrect cases for the GCN are cases where the model confuses the endo- and epicardium keypoints, as in Fig. 3.6. Using the displacement method avoids this from happening, resulting in no anatomically invalid segmentations. Fig. 3.7 shows the median and worst cases of the GCN with the displacement method and nnU-Net.

3.3.4 Inter-model agreement as quality predictor

The goal of this experiment was to evaluate the proposed inter-model agreement as a segmentation quality predictor and to investigate whether GCN has a better worst-case performance than nnU-Net. The inter-model agreement is measured as the Dice similarity between the segmentation masks of nnU-Net and GCN with displacement method. For this experiment, we use the clinical dataset HUNT4 as it better represents the ultrasound images used in practice. Fig. 3.8 shows a histogram of the inter-model Dice scores on the HUNT4 evaluation set. For the experiment, the samples are split between cases with high and low inter-model agreement. The threshold for low inter-model Dice is 0.8, equalling ∼2% of cases. The threshold of high inter-model dice is 0.9, equalling ∼92% of cases. A total of 200 samples were extracted by combining and randomizing 100 samples below the low threshold and 100 samples above the high threshold. Each sample contains 3 images: the input ultrasound image and the two segmentations produced by the two architectures in random order. A clinician manually classified each sample using the following criteria:

- Overall quality of ultrasound image: 'High', 'Medium', 'Low', or '**Unsuitable** for measurements'. The last category is reserved for images with so low quality that they should not be used for any clinical measurement.
- Correct view?: 'Yes' or 'No'. 'Yes' means the view is A4C or A2C.
- LV-focused?: 'Yes' or 'No'. 'Yes' means the LV is central in the image and the depth is adjusted to the LV.
- Anatomically correct segmentation? 'Yes' or 'No'. We use the same criteria for anatomical correctness as Painchaud et al. [12].
- Correct placement of segmentation? 'Yes' or 'No'. 'Yes' means the segmentation is placed correctly.

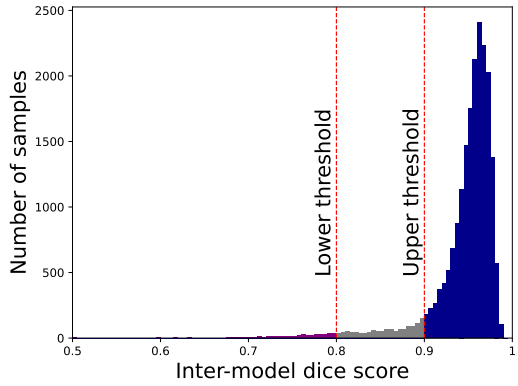


Figure 3.8: Histogram of inter-model Dice scores between nnU-Net and GCN on the HUNT4 evaluation set. The upper and lower threshold determine the set of images that are sampled for the experiment that evaluates inter-model agreement as a quality predictor.

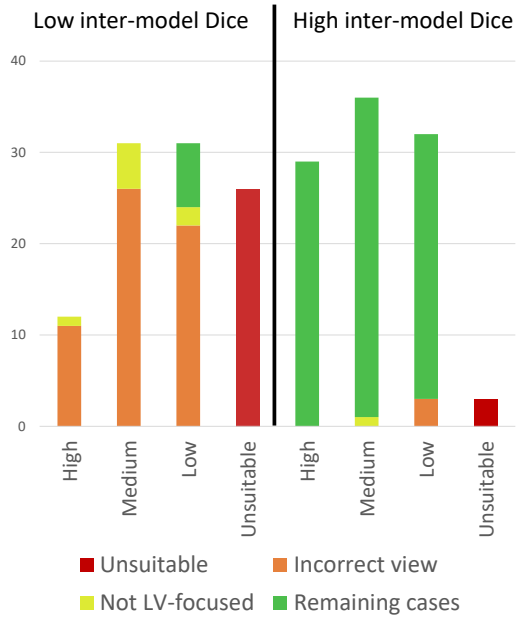


Figure 3.9: Comparison of input quality between samples with a high and a low inter-model agreement. The exact number of cases can be found in Table 3.5. From the 100 high agreement samples, 93 were in distribution, while from the 100 low agreement samples, only 7 were in distribution.

Table 3.5 summarizes the results of the experiment in which the first subtable

(a) shows that when the inter-model agreement is high (> 0.9) the two networks both produce valid segmentations. The second subtable (b) shows that when the inter-model agreement is low (< 0.8), the GCN produces slightly more correct segmentations. If we regard samples that have suitable image quality, the correct view, and are LV-focused as in-distribution samples, we see that from the 100 high agreement samples, 93 were in distribution, while for the 100 low agreement samples, only 7 were in distribution. Furthermore, all 7 in-distribution samples with low agreement had low quality. Fig. 3.9 visualizes these results showing the difference in input quality between samples with high and low inter-model agreement.

3.3.5 Clinical evaluation on HUNT4

We use the clinical dataset HUNT4 to evaluate the accuracy of clinical measurements instead of CAMUS because the CAMUS dataset only contains volume measurements which were measured using the annotations and not using clinically approved software. The HUNT4 dataset also contains considerably more patients. We compare the accuracy of the U-Net and the GCN which both were trained on the HUNT4 training set. The GCN used for this experiment is the GCN with displacement method.

The automatic estimation of EF mimics the procedure for manual LV volume calculations suggested by clinical guidelines [2]. The automatic procedure contains the following steps:

1. Detect ED and ES of each cycle using the timing neural network described by Fiorito et al. [24] for both the A2C and A4C recordings of the same patient. This means the view was labeled manually during acquisition and the timing is estimated using deep learning.
2. Segment the ED and ES frame of each cycle for both the A2C and A4C recordings of the same patient.
3. Combine the *usable* segmentations in the A2C recordings with the *usable* segmentations in the A4C recordings for biplane volume calculations using the modified Simpson method. A segmentation is *unusable* if the procedure described in section 3.2.3 fails to extract the LV apex or base points, which position is required for the modified Simpson method. The EF is then calculated as $EF = \frac{ED\,volume - ES\,volume}{ED\,volume}$. Measurements from all cardiac cycles are combined and averaged.

Our results include 1877 out of 1913 patients, equalling a feasibility of 98.1%. In

Table 3.5: Results of the experiment on using the inter-model agreement as a quality predictor. The GCN uses the displacement method in this experiment.

a Analysis for cases with high inter-model agreement

Image set	**Quality of ultrasound image**	**Number of cases**	**Number of anatomically correct cases**		**Number of cases with correct placement**	
			nnU-Net	GCN	nnU-Net	GCN
All images	High	29	29	29	29	29
	Medium	36	36	36	35	35
	Low	32	31	**32**	30	**31**
	Unsuitable	3	-	-	-	-
Correct view	High	29	29	29	29	29
	Medium	36	35	35	34	34
	Low	29	**27**	26	**26**	25
Correct view and LV-focused	High	29	29	29	29	29
	Medium	35	35	35	34	34
	Low	29	**26**	25	**26**	25

b Analysis for cases with low inter-model agreement

Image set	**Quality of ultrasound image**	**Number of cases**	**Number of anatomically correct cases**		**Number of cases with correct placement**	
			nnU-Net	GCN	nnU-Net	GCN
All images	High	12	12	12	0	0
	Medium	31	26	**29**	0	**2**
	Low	31	25	**30**	2	**4**
	Unsuitable	26	-	-	-	-
Correct view	High	1	1	1	0	0
	Medium	5	2	**5**	0	**2**
	Low	9	7	**9**	2	2
Correct view and LV-focused	High	0	-	-	-	-
	Medium	0	-	-	-	-
	Low	7	7	7	2	2

13 cases the timing network failed. In the remaining 23 cases, either U-Net 1 or nnU-Net failed to produce a *usable* segmentation for all cycles. Because the GCN predicts the LV apex and base points directly as one of the keypoints, it can produce an EF for every recording. However, for a fair comparison, we exclude these cases from our results.

Fig. 5.16 shows the Bland-Altman plots of the automatic methods compared to the manual reference. Table 3.6 shows the Mean Absolute Error (MAE) of EF measurements between the automatic method and manual reference. The ensembles average the EF estimates of the two models. The difference in MAE between nnU-

Table 3.7: Inference time comparison of the proposed architectures and baseline models on an NVIDIA GeForce RTX 3070 Ti Laptop GPU.

Architecture	Inference time in ms	Number of parameters in millions
nnU-Net [18]	5.40±0.008	33.4
U-Net 1 [4]	1.88±0.011	2.0
MobileNet-v2-GCN	**1.35±0.007**	3.3
MobileNet-v2-GCN with displacement method	1.40±0.020	3.3
ResNet-50-GCN	3.37±0.010	25.2
ResNet-50-GCN with displacement method	3.55±0.009	25.2

Net and all other methods is significant with $p < 0.05$ using the Wilcoxon signed-rank test [22].

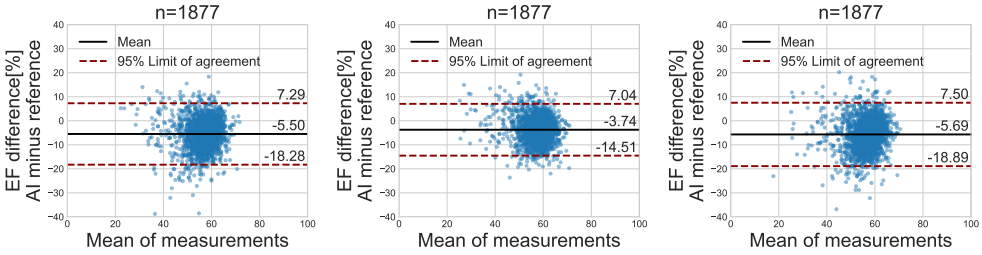
Finally, we use the findings of the previous experiment and filter the ED and ES frames based on inter-model agreement. Fig. 3.11 shows the same Bland-Altman plots when only using the frames where the inter-model Dice between U-Net or nnU-Net and the GCN is above 0.85. In 137 out of 1877 cases, there was no ED or ES frame in all three cycles of the recording with Dice above 0.85, resulting in the patient being excluded. As a result, the filtered results only include 1740 out of 1913 patients, equalling a feasibility of 91%.

Table 3.6: Evaluation of automatic EF estimation on HUNT4. The mean absolute error of nnU-Net is statistically significantly lower than all the other models.

Method	**Mean absolute error**
U-Net 1 [4]	6.79±6.5
nnU-Net [18]	**5.38±5.5**
GCN with displacement	7.07±6.7
U-Net 1 GCN with displacement ensemble	6.66±6.1
nnU-Net U-Net 1 ensemble	5.92±5.7
nnU-Net GCN with displacement ensemble	6.00±5.7

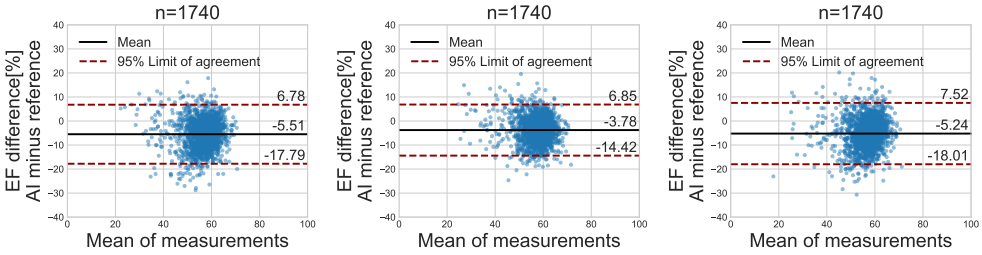
3.3.6 Inference time

Table 3.7 summarizes the inference times of the proposed architectures and baseline models on an NVIDIA GeForce RTX 3070 Ti Laptop GPU. To measure the inference time, the model performs a warmup run on 1000 random dummy inputs. Afterward, the model performs 10 test runs on 100 random inputs. The final inference time is the average runtime of each of the test runs divided by 100. These values do not include the data loading time and pre- and post-processing steps.



(a) U-Net 1 **(b)** nnU-Net **(c)** GCN

Figure 3.10: Bland–Altman plots of the automatic EF measurements compared to the manual reference. All frames with a usable segmentation are used.



(a) U-Net 1 **(b)** nnU-Net **(c)** GCN

Figure 3.11: Bland–Altman plots of the automatic EF measurements compared to the manual reference when only using the frames for which the inter-model Dice agreement is more than 0.85.

3.4 Discussion

3.4.1 Anatomical correctness and segmentation accuracy

In this work, we developed a multi-structure graph convolutional network (GCN) for cardiac ultrasound segmentation. While the GCN method can eliminate anatomical incorrect segmentations, it comes at the cost of slightly reducing the segmentation accuracy compared to U-Net. Whereas the GCN does not outperform U-Net in terms of accuracy for multi-structure segmentation, the keypoint representation has its own advantages. Both architectures have their unique data representation and solve the problem in a fundamentally different way. The pixel-wise approach of U-Net can give the most accurate segmentations, but can also produce anatomically incorrect results. On the other hand, the GCN with displacement method has lower accuracy but does not produce anatomically incorrect segmentations as the cardiac shapes are embedded as a strong bias in the network. However, the GCN can still fail to place the anatomically correct shape correctly in the image. Fig. 3.7(b) shows an example

where both the GCN and U-Net fail to generalize to samples with increased depth. From an abstract point of view, GCN can be seen as a heavily regularised version of pixel-wise architectures. The displacement method regularizes the architecture even further. Finally, it is worth noting that nnU-Net is an order of magnitude larger and slower than the GCN.

3.4.2 Ablation study

The results of the ablation study show that reducing the depth of the decoder changes the performance of the GCN non-significantly, indicating that the pre-trained CNN is the most important contributor. There are two possible explanations for this behavior. It could be that the architecture of the GCN decoder is not efficient for processing data in keypoint embedding space. On the other hand, it could also be that the decoder part is superfluous, as the shapes of the cardiac structures are easy enough to not require a shape encoding as the embedding of the graph convolutional layers.

3.4.3 Combination with U-Net

Combining GCN and U-Net in a cascade gives a mixed result of the characteristics of both architectures. For the GCN - nnU-Net cascade, there are no additional augmentations to the GCN output that serve as an extra input channel to the nnU-Net, as it is used out of the box. As a result, the nnU-Net cascade relies too much on the GCN output and only learns to do minor adjustments to the GCN intermediates. On the other hand, the GCN - U-Net 1 training procedure augments the GCN intermediates with additional rotation, scaling, and translation augmentations specifically for the cascade network, as described in subsection 3.2.7. This results in the U-Net correcting some cases of wrong placement of the GCN but also re-introduces anatomical incorrect outliers. Table 3.4 reflects these findings numerically.

3.4.4 Inter-model agreement

Despite the excellent Dice and Hausdorff metrics of U-Net, this pixel-wise segmentation approach can fail completely in certain cases as shown in Fig. 3.1. In this work, we have proposed using the inter-model agreement as a simple and effective approach to detect these failing cases.

The distinct characteristics of U-Net and GCN make them appealing for joint use to quantify uncertainty. Although averaging their clinical estimates doesn't enhance accuracy, the agreement between the models offers valuable insight for identifying difficult cases and erroneous segmentations of U-Net. The leading approach to uncertainty quantification in deep learning is Bayesian Neural Networks [25], but

their multiple passes during inference hinder real-time use. Deep ensembles are another option [25], but require multiple models to run on each input. Combining U-Net with GCN approximates deep ensembles practically. Specifically, U-Net and GCN are unlikely to make the same mistake since the GCN is designed to avoid the most common error of U-Net i.e. anatomically incorrect segmentations.

The results of the experiments using inter-model agreement indicate a clear distinction in input quality between high and low agreement cases. Fig. 3.9 shows significantly fewer high-quality images and fewer unsuitable cases with high inter-model agreement as compared to the cases with low inter-model agreement ($p < 0.05$). Furthermore, most of the cases with low inter-model agreements were the wrong view. In only 15 out of 74 suitable low agreement cases the view was correct. The majority (55) of these wrong view cases are apical long axis (ALAX) views or views rotated towards ALAX. These images are in the dataset because the clinicians used these views in practice even though they were instructed to use A4C and A2C views to measure EF. One of the reasons is that sometimes it is not possible to get a clear A2C or A4C view so the clinician trades view correctness for image quality. However, the segmentation models are only trained on LV-focused A2C and A4C views, resulting in poor performance. Finally, only 7 out of 100 low agreement cases had suitable quality, the right view, and were LV-focused as compared to 93 out of 100 high agreement cases. This demonstrates the inter-model agreement estimator as an efficient method to detect out-of-distribution cases in the form of very low image quality or wrong views.

3.4.5 Clinical evaluation

For accuracy on EF measurements, the nnU-Net exhibits the narrowest limits of agreement and the fewest outliers in comparison to both GCN and U-Net 1. However, when frames with low inter-model agreement are omitted, the limits of agreement for all three networks improve, as can be observed when comparing Figs. 5.16 and 3.11. This improvement can be attributed to the elimination of outliers stemming from problematic inputs or flawed segmentations. Thus, while the GCN does not improve EF accuracy directly, it can help discard cases where automatic EF is not appropriate, consequently narrowing the limits of agreements for the remaining cases. The negative mean in Figs. 5.16 and 3.11 show that each of the deep learning methods underestimates the reference ejection fraction. Because there are many possible sources of error, both from the deep learning methods and from the reference annotations, the reason for the negative bias is not trivial. The article by Olaisen et al [5] discusses this more in-depth.

3.5 Real-time demo application

To demonstrate the potential of the inter-model agreement to detect out-of-distribution cases and faulty segmentations in real-time, a real-time application [26] was created using the FAST framework [27]. It shows the segmentation output of the GCN and nnU-Net side by side together with a status bar that visualizes the agreement between the models. Fig. 3.12 shows a screenshot of the application in action.The demo shows the application in use while a clinician is operating a GE Vivid E95 scanner. The video demonstrates the effectiveness of the inter-model agreement as a method to detect out-of-distribution and low image quality cases. The video is available at https://doi.org/10.6084/m9.figshare.24230194.

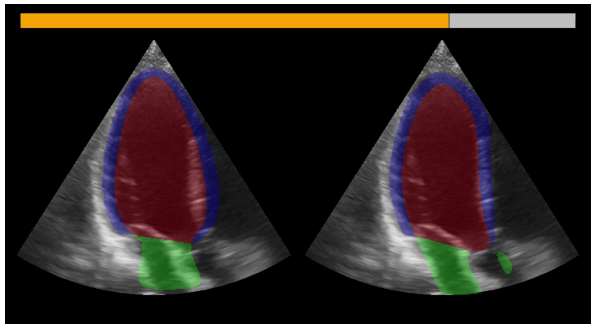


Figure 3.12: Screenshot of the real-time demo application. The GCN and nnU-Net segmentations are shown on the left and right side respectively. The color-coded status bar on top visualizes the agreement between the models.

3.6 Conclusion

This paper proposes a multi-structure graph convolutional network (GCN) to reduce anatomical incorrect results produced by U-Nets in cardiac ultrasound segmentation. We propose the displacement method as a clinically motivated regularization that eliminates anatomically incorrect segmentations for multi-structure segmentation. The ablation study showed that changing the architecture of the GCN decoder does not affect performance significantly. This indicates the encoder is the most important component of the network

Measuring the inter-model agreement between GCN and U-Net yields a simple and highly effective method for quantifying uncertainty and thereby detecting out-of-distribution and failing cases. Whereas the average performance of both U-Net and the GCN is within inter-observer variability, their worst-case behavior is still

drastically worse than human annotators. These outliers occur mostly due to a lack of generalization capabilities, which is still a fundamental problem in cardiac segmentation. Low inter-model agreement can detect out-of-distribution cases and thus can be used as a warning signal to the user using the automatic segmentation tool in practice.

The idea of GCNs is a relatively new concept for doing cardiac segmentation that shows promising initial results towards more robust cardiac segmentation. The GCN can be implemented as a robust independent model or as a supplementary model to validate the segmentation outcomes of pixel-wise techniques such as U-Net.

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Regional quality estimation for echocardiography using deep learning

4

Automatic estimation of cardiac ultrasound image quality can be beneficial for guiding operators and ensuring the accuracy of clinical measurements. Previous work often fails to distinguish the view correctness of the echocardiogram from the image quality. Additionally, previous studies only provide a global image quality value, which limits their practical utility. In this work, we developed and compared three methods to estimate image quality: 1) classic pixel-based metrics like the generalized contrast-to-noise ratio (gCNR) on myocardial segments as region of interest and left ventricle lumen as background, obtained using a U-Net segmentation 2) local image coherence derived from a U-Net model that predicts coherence from B-Mode images 3) a deep convolutional network that predicts the quality of each region directly in an end-to-end fashion. We evaluate each method against manual regional image quality annotations by three experienced cardiologists. The results indicate poor performance of the gCNR metric, with Spearman correlation to the annotations of $\rho = 0.24$. The end-to-end learning model obtains the best result, $\rho = 0.69$, comparable to the inter-observer correlation, $\rho = 0.63$. Finally, the coherence-based method, with $\rho = 0.58$, outperformed the classical metrics and is more generic than the end-to-end approach. The image quality prediction tool is available as an open source Python library at https://github.com/GillesVanDeVyver/arqee.

4.1 Introduction

Image quality is one of the main challenges in ultrasound imaging and can differ significantly between patients and imaging equipment. In echocardiography, many factors influence image quality such as the ultrasound scanner, the patient, and the probe. Several quantitative measurements using the images are performed. However, this requires image quality good enough for the given measurement. Different measurements have different image quality requirements, for instance, left ventricular (LV) volume, ejection fraction (EF), and strain measurements require good image quality in the entire myocardium. On the other hand, mitral

annular plane systolic excursion (MAPSE) only requires good image quality in the annulus. Good image quality should generally provide measurement values with low uncertainty. Estimating image quality can be useful in the following ways:

- To guide operators to achieve as good image quality as possible while scanning.
- To automatically select the best images, recordings, and the best cardiac cycles to use for a given measurement.
- As quality assurance, e.g. to warn the user when an image is not good enough for a measurement, and to automatically approve/disapprove individual myocardial segments based on image quality.
- In data mining projects, to exclude cases with insufficient quality for reliable measurements.

We distinguish between two types of quality of ultrasound images: view quality/correctness and image quality. In this work, we will focus on image quality specifically. For view correctness, previous work has demonstrated that 3D ultrasound can serve as training data to automatically identify the transducer rotation and tilt in relation to the desired standard view and can guide the user to the correct position [1–3]. For image quality, the classic ultrasound signal-processing metrics are the contrast ratio (CR) [4], contrast-to-noise ratio (CNR) [5], and generalized CNR (gCNR) [6]. These three metrics need a region of interest (ROI) and a background region to compare against. More recently, global image coherence (GIC) [7] has been proposed as a general quality metric that does not require the selection of these two regions. The image coherence measures how well the signals of the transducer elements align after delay compensation, with more alignment corresponding to clearer and sharper images. From the above mentioned methods, only the GIC can be used directly and automatically for measuring image quality separately as it does not require selecting an ROI and noise region. However, this approach requires channel data, which is not readily available in practice and does not give regional metrics.

Several automatic methods for measuring ultrasound image quality have been published applicable to cardiac imaging. Abdi et al. [8] used a recurrent neural network to predict the global quality of cardiac cine loops. The criteria for quality assessment take both image quality and view correctness into account. In subsequent studies [9, 10], they used an architecture that performs both view classification

and global quality prediction simultaneously. The image quality metric is a global criterion based on the manual judgment of the clarity of the blood-tissue interface. Labs et al. [11] used a multi-stream neural network architecture where each stream takes in a sequence of frames and predicts a specific quality criterion. The criteria are global and take both view correctness and image quality into account. Karamalis et al. [12] detect attenuated shadow regions with random walks resulting in a pixel-level confidence map. Unlike the other methods above, this method is not based on deep learning. It provides a local, pixel-level metric, but it only measures the visibility of regions and not the quality of their content.

All of the automatic methods mentioned above have the limitation that they only provide a global image quality evaluation and/or do not assess the image quality separate from the view correctness.

4.2 Methods

In this work, we developed and compared three fully automatic methods to asses regional image quality in cardiac ultrasound separate from the view correctness:

- Classical ultrasound image quality metrics, such as contrast ratio and contrast to noise ratio, applied in cardiac regions automatically extracted using deep-learning segmentation.
- Deep-learning predicted ultrasound coherence, which is a measure of how coherent a signal is received by the transducer elements, together with deep-learning segmentation.
- End-to-end prediction of regional image quality.

The rest of this section first presents the datasets used to develop these methods, and then presents each of the three methods.

4.2.1 Datasets

VLCD

The **Very Large Cardiac Channel Data Database (VLCD)** consists of channel data from 33280 frames from 538 recordings of 106 study participants [7]. It contains parasternal short axis (PSAX), parasternal long axis (PLAX), apical long axis (ALAX), apical two-chamber (A2C), and apical four-chamber (A4C) views.

We split the VLCD dataset on the study participant level into train, validation, and test sets, 70%, 15%, and 15% respectively.

HUNT4

The Nord-Trøndelag Health Study dataset (HUNT4Echo) is a clinical dataset including among others PSAX, PLAX, ALAX, A2C, and A4C views. Each recording contains 3 cardiac cycles. We use two subsets of the HUNT4Echo dataset.

- **Segmentation annotation dataset** A fraction of 311 study participant exams, the segmentation annotation set [13], contains single frame segmentation annotations in both ED and ES as pixel-wise labels of the left ventricle (LV), left atrium (LA), and myocardium (MYO) in ALAX, A2C, and A4C views.
- **Regional image quality dataset** For this work, we created an additional dataset of image quality labels. The local image quality labels are manual annotations that asses the image quality of the cardiac regions of interest on a subset of the HUNT4 dataset in ALAX, A2C, and A4C views. Section 4.2.2 describes the annotation process in more detail.

4.2.2 Regional image quality annotation on HUNT4

An annotation tool was developed specifically for this project using the open-source Annotation Web software[1] [14]. The tool was made to enable clinicians to annotate regional image quality as efficiently and accurately as possible. The tool is freely available and can be adapted to other image quality projects. The image quality annotations were performed by three experienced cardiologists using the following protocol:

1. Annotate the end-diastole (ED) and end-systole (ES) frame of each recording, and optionally other frames if the image quality changes significantly during the recording.
2. If the majority of the cardiac regions of interest is out-of-sector, label it as out-of-sector. Otherwise, label the part of the region that is inside the sector according to the definitions in Table 4.1. We ignore the out-of-sector regions in the remainder of this work.

For the first round of annotations, each of the three clinicians annotated the same 10 frames from 5 recordings of 2 study participants. We used this dataset to

[1] https://github.com/smistad/annotationweb

Table 4.1: Definitions of image quality labels for annotating cardiac regions of interest. We only look at the signal quality of the region that is inside the ultrasound sector. If more than 50% is outside the sector, the region is treated as out-of-sector and excluded from analysis.

image quality label	Definition	Quality score
Not visible	Less than 50% of what is inside the sector is visible.	1
Poor	Low signal-to-noise ratio and/or smearing/low resolution, but can still see at least 50% of the myocardium/annulus. Diffuse border between annulus and cavity.	2
Ok	Medium signal-to-noise ratio and/or some smearing/lower resolution, can see at least 70% of the myocardium/annulus.	3
Good	Good signal-to-noise ratio and clear border in at least 80% of the myocardium/annulus. Very little smearing/good resolution. Clear border between annulus and cavity	4
Excellent	Within the best 10% of what you have seen on this system.	5

calculate the inter-observer variability. For the second round of annotations, the three clinicians collectively annotated 458 frames from 158 recordings of 65 study participants. The annotations from the second round form the **regional image quality dataset**. This dataset was split randomly at the study participant level into train, validation, and test sets, allocating 70%, 15%, and 15% of the data to each set respectively.

4.2.3 Regional image quality estimation

Classical ultrasound image quality metrics

For the classical image quality metrics, deep-learning segmentation is used to extract the annulus regions and each of the myocardial segments as regions of interest and the LV as the background region. Appendix C gives more details about the procedure for dividing the segmentation into regions. The four classical ultrasound image quality metrics below were tested. We apply histogram matching [15, 16] to a Gaussian distribution ($\mu = 127, \sigma = 32$) for the B-Mode grayscale images before calculating pixel-based quality metrics.

- **Pixel intensity** is the average pixel intensity value in each region.
- **Contrast Ratio (*CR*)** [4] is defined as

$$CR = \frac{\mu_{\text{segment}}}{\mu_{\text{LV}}}$$

where μ_{segment} is the average intensity in each region and μ_{LV} is the average intensity inside the LV lumen.

- **Contrast to Noise Ratio (*CNR*) [5]** is defined as

$$CNR = \frac{\mu_{\text{segment}} - \mu_{\text{LV}}}{\sqrt{\sigma^2_{\text{segment}} + \sigma^2_{\text{LV}}}}$$

where σ_{segment} is the standard deviation in each region and σ_{LV} is the standard deviation inside the LV lumen.

- **Generalized CNR (*gCNR*)** [6] is defined as the maximum performance that can be expected from a hypothetical pixel classifier based on intensity using a set of optimal thresholds. It is calculated as

$$gCNR = 1 - \frac{1}{2} \sum_{i=0}^{MAX_i} \min\{p_{\text{segment}}(i), p_{\text{LV}}(i)\}$$

where $p_{\text{segment}}(x)$ is the probability density function of the pixel intensities inside the region the *gCNR* is calculated for, $p_{\text{LV}}(x)$ the probability density function of the pixel intensities inside the LV lumen, and MAX_i the maximum possible pixel intensity. Fig. 4.1 shows an example of the probability density functions for one of the regions as ROI and the LV lumen as background.

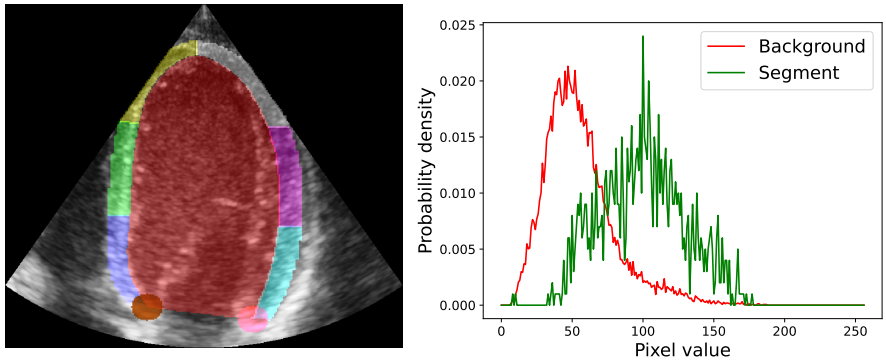
Local, deep-learning predicted image coherence as quality metric

We use the VLCD dataset to calculate the coherence factor (CF) [17] for each pixel in the ultrasound image. This factor is the ratio between the amplitude of the sum of the received signals to the sum of the amplitudes of those signals,

$$CF = \frac{\sum_{n=1}^{N} S_i}{\sum_{n=1}^{N} |S_i|}$$

where S_i is the delayed signal for the i-th transducer element. This is equivalent to taking the coherent sum of the signal and dividing it by the incoherent sum of each signal. Thus, the coherence factor measures of how well the complex signals of all transducer elements align. The remainder of the signal-processing chain is the native processing of the GE HealthCare Vivid E95 system[2] but without the log compression. The result is a **coherence image** with the same dimension as the

[2] Gundersen et al. [18] describe this signal-processing chain in more detail.



(a) Segmentation **(b)** Probability density functions

Figure 4.1: Calculating gCNR for a region of the myocardium. The left side of the figure shows the segmentation with the MYO and annulus points divided into regions. The right side shows the probability density functions of the segment and background pixels used to calculate the gCNR. In this example, we use the mid region on the left side as ROI and the full LV lumen as background. These correspond to the green and red masks respectively in the left part of the figure.

B-Mode image. The final preprocessing step applies gamma normalization with $\gamma = 0.5$ on the coherence images,

$$t_{i,normalized} = t_i^{\gamma}$$

where t_i are the pixels of the target coherence image. The corresponding B-mode images are generated from the channel data using the same, native signal-processing pipeline.

The HUNT4 dataset, like most ultrasound datasets, does not include channel data. Therefore, VLCD was used to train an image-to-image network that takes as input the grayscale B-mode image and predicts the coherence image, which is then used to calculate local image coherence. We use a lightweight U-Net architecture inspired by the U-Net 1 architecture in [19], with characteristics listed in Table 4.2. As coherence is related to image quality, we only apply augmentations that do not influence the quality of the image. Furthermore, the coherence should be invariant to different gain settings, so we additionally augment with brightness adjustments on the B-mode image while keeping the target coherence image unchanged. During training and validation of the coherence prediction model, we sample a random frame from each recording in the train and validation set respectively during each

Table 4.2: Characteristics of the coherence prediction network. The general architecture is a U-Net. The "number of channels" row indicates the number of channels at the first, bottom, and last convolution of the U-Net respectively.

Input size	256×256
Number of channels	$16 \downarrow 128 \uparrow 16$
Number of output channels	1
Lowest resolution	4×4
Upsampling scheme	2×2 repeats
Normalization scheme	BatchNorm
Batch Size	32
Optimizer	Adam
Initial learning Rate	1e-2
Scheduler	None
Loss function	Negative structural similarity index measure (SSIM) [20]
Inter-layer Activation	Gaussian error linear unit (GELU) [21]
Final layer Activation	Sigmoid
Epochs	500
Augmentations	Rotations ($-45° \leq$ angle $\leq 45°$), horizontal mirroring, gamma correction ($0.75 \leq \gamma \leq 1.25$, only on B-mode), scaling ($0.75 \leq$ magnification ≤ 1.25), contrast/brightness adjustments ($0.2 \leq$ gain ≤ 2, $0 \leq$ bias ≤ 128, only on B-mode). Each augmentation is applied individually with 0.5 probability.

epoch. During testing, we use all frames in the test set. The local image coherence quality metric of a region is the average pixel value of all pixels corresponding to the region in the coherence image. This is the same as the pixel intensity metric above but applied to the coherence image instead of the B-Mode image.

End-to-end deep-learning quality prediction

The end-to-end learning approach trains a convolutional neural network on the regional image quality dataset to predict the quality of each region directly. We treat the problem as a regression task where the model predicts a score for each segment. Table 4.1 shows the correspondence between quality scores and annotation labels. As architecture, we use a MobileNetV2 [22] that predicts the image quality labels of all regions simultaneously. Appendix D describes the ablation study conducted to justify this setup.

4.3 Experimental setup

4.3.1 Evaluation of coherence prediction from B-mode

We use the structural similarity index (SSIM) [20], peak signal-to-noise ratio (PSNR)[3], and relative pixel error (RPE) to evaluate the coherence image prediction

[3]The maximum pixel value for coherence images is 1.

network. We define the RPE as

$$RPE = \frac{|t_i - p_i|}{max(t_i, \epsilon)}$$

where t_i are the pixel values of the target coherence image, p_i the pixel values of the predicted coherence image, and $\epsilon = 1e-4$. Section 4.4.1 contains the results of this experiment.

4.3.2 Evaluation of quality metrics

The correlation and accuracy of each quality metric were measured by comparing them to the expert annotations on the test set of the regional image quality dataset. For the classic image quality methods and regional coherence method, linear regression models were used to map the quality metric values to image quality labels. The train and validation set were used together to fit the linear regression model and evaluate it on the test set. Section 4.4.2 contains the results of this experiment.

4.3.3 Comparison to inter-observer variability

We compare the end-to-end, local image coherence, and gCNR-based model to the inter-observer variability on the data obtained in the first round of annotation, explained in subsection 4.2.2. For the local image coherence and gCNR-based models, the same linear model as subsection 4.3.2 was used to map the metrics to quality labels. The inter-observer variability is calculated from the aggregate of the three unique pairwise score errors between each of the three annotators:

$$e_{\text{inter-observer}} = e_{12} \cup e_{23} \cup e_{13}$$

where e_{ij} is the score difference between operator i and j. The error metrics of the automatic methods are calculated from the aggregate of the pairwise score errors between the output of the method and each of the three annotators:

$$e_M = e_{1M} \cup e_{2M} \cup e_{3M}$$

where e_{iM} is the score difference between operator i and method M. Section 4.4.3 contains the results of this experiment.

4.3.4 Relation to variability in clinical measurements

This experiment evaluates whether there is a relation between the predicted quality and the agreement between different methods for clinical measurements. The

Table 4.3: Average metric values of the coherence prediction network on the test set of the VLCD.

SSIM [20]	0.994 ± 0.003
Relative pixel error (RPE)	$5.6\% \pm 0.89\%$
PSNR[3].	35.99 ± 2.00 dB

hypothesis is that with lower image quality the variability, and thus the uncertainty, of the measurements between methods and between experts increases. More specifically, this analysis compares peak global longitudinal strain (GLS) and ejection fraction (EF) measurements obtained either fully automatically with AI tools or manually by using GE HealthCare EchoPAC software on HUNT4 [13, 23]. For AI estimation of GLS and EF, the deep-learning methods proposed by Østvik et al. [24] and Smistad et al. [25] were used respectively. The study participants in HUNT4 used for model development were excluded from the analysis. For GLS, the predicted quality is the average quality of all segments over the full recording. For EF, the predicted quality is the average quality of all segments in the end-diastole (ED) and end-systole (ES) frames of all cycles in the recording. Section 4.4.4 contains the results of this experiment.

4.4 Results

4.4.1 Results of coherence prediction from B-mode

Table 4.3 summarizes the average metric values on the test set. Fig. 4.2 shows an example of the best, median, and worst-case predictions according to the relative pixel error. Fig. 4.3 illustrates how the predicted coherence images are almost independent of the brightness/gain and contrast/dynamic range of the input B-Mode images. The main finding is that the difference between the estimated and ground truth coherence images is small and thus the predicted coherence images can be used to obtain coherence-based quality metrics for B-mode for which the channel data is not available.

4.4.2 Results of quality metrics

Table 4.4 summarizes the results of the evaluation of the quality metrics. Fig. 4.4 shows box plots of the quality metrics per image quality label for the end-to-end, coherence, gCNR, and intensity models. Fig. 4.5 shows examples of B-mode images with varying quality together with labels from the annotators and automatic quality metrics. The main finding is that the end-to-end model performs the best, followed by the local image coherence metric. The classical ultrasound image quality metrics perform poorly.

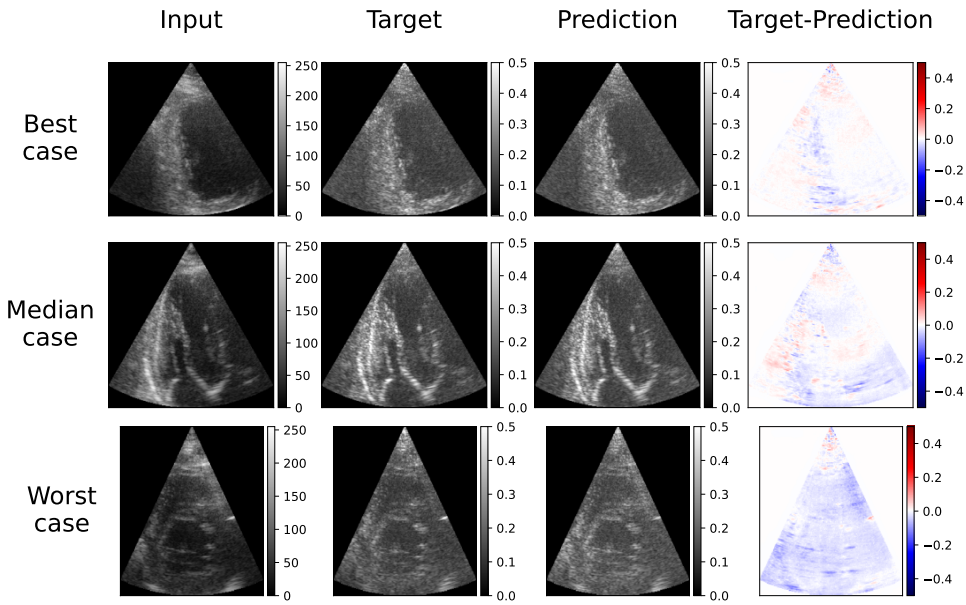


Figure 4.2: Best, median, and worst case of image-to-image coherence prediction task with relative pixel errors of 4.0, 5.4, and 9.1% respectively. The first column shows the input B-mode image. The second column shows the coherence image as predicted by the image-to-image network. The third column shows the ground truth coherence image as calculated from the channel data. Finally, the rightmost column shows the color-coded difference of the target minus the predicted image.

Table 4.4: Comparison of quality metrics on the test set of the regional image quality dataset, i.e. the second round of annotations. Accuracy is defined as the ratio of agreement between the automatic measurement, rounded to the nearest quality category, and the annotation on the segment level.

Quality metric	Spearman correlation	Mean absolute error (MAE)	Accuracy
Pixel intensity	0.30	1.31±0.91	20%
CR	0.28	1.32±0.92	22%
CNR	0.25	1.38±0.96 0.	20%
gCNR	0.20	1.50±0.96	15%
Local image coherence	0.48	1.07±0.80	29%
End-to-end learning	**0.65**	**0.59±0.45**	**48%**

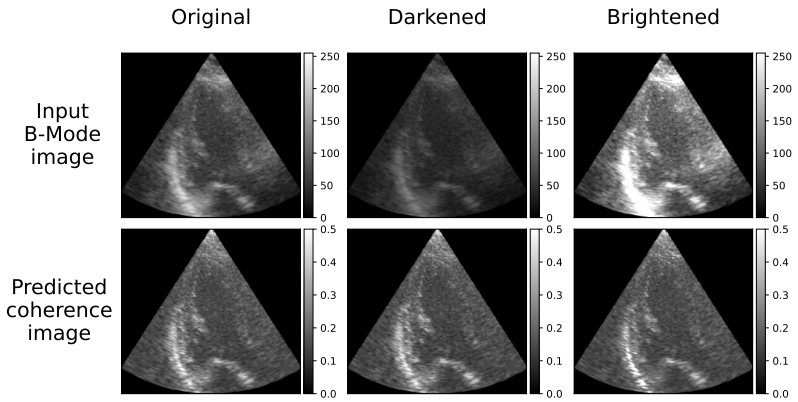


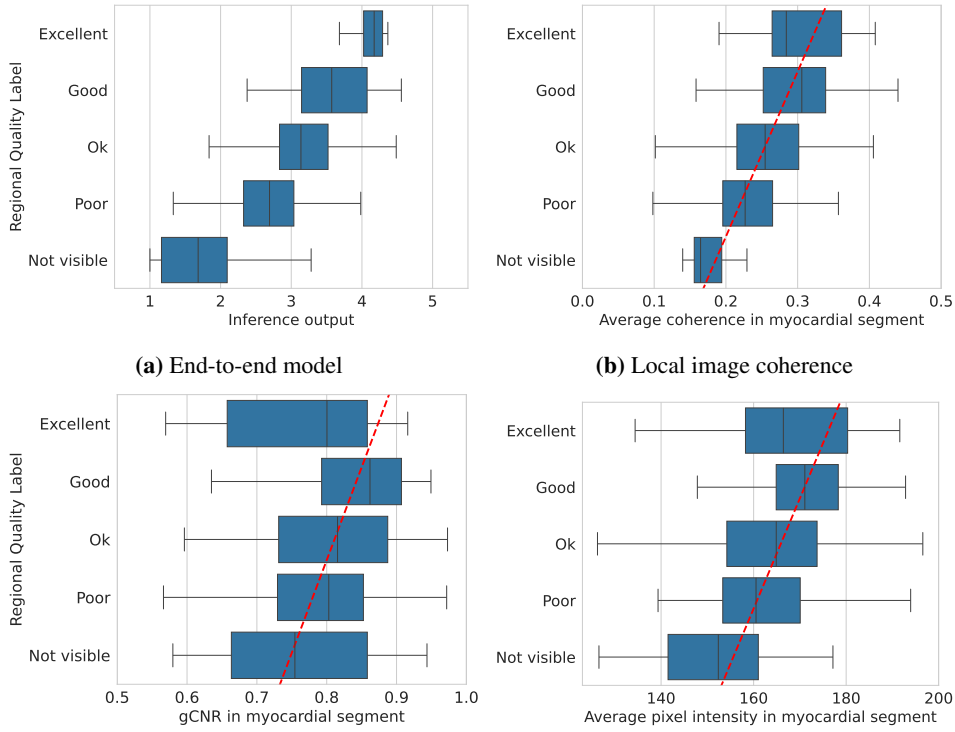
Figure 4.3: Effect of brightness on coherence prediction. The first row shows a B-Mode image from the regional image quality dataset, brightened and darkened with gamma correction (γ = 0.9 and 1.1). The second row shows the predicted coherence images generated by giving the corresponding input from the first row to the network. The predicted coherence is unaffected by the adjustments in brightness, apart from the saturation effect in the brightened image reducing the information in the input, as can be seen in the basal part of the inferolateral wall.

Table 4.5: Comparison of automatic methods to inter-observer variability on the first round of annotations. For this data, there are three annotation values for each quality label. Each of the automatic methods is compared against each of the three annotators and the results are averaged. For the inter-observer variability, the labels are compared pairwise between each of the three annotators.

Method	Spearman correlation	MAE	Accuracy
End-to-end model	**0.69**	**0.59±0.59**	**46%**
Local image coherence	0.58	1.19±1.01	27%
gCNR	0.24	2.23±1.58	13%
Inter-observer	0.63	0.67±0.60	40%

4.4.3 Results of comparison to inter-observer variability

Fig. 4.6 shows the bar plot comparing the automatic methods to the inter-observer variability and Table 4.5 lists the corresponding average metric values. Using the Wilcoxon signed-rank test [26] and a significance level of $p = 0.05$, we find that the difference in Mean absolute error (MAE) between each of the methods is statistically significant. The difference between the inter-observer MAE and the MAE of each of the methods is also statistically significant, i.e. the inter-observer MAE is higher than the MAE of the end-to-end model and lower than the MAE of the other two models.



(a) End-to-end model

(b) Local image coherence

(c) gCNR

(d) Pixel intensity

Figure 4.4: Box plots of quality metrics versus regional quality labels on the test set of the regional image quality dataset. The predictions of the end-to-end model have the strongest correlation to the quality labels. The dotted line represents the linear regression model that maps the quality metrics to quality labels, as described in section 4.3.2. The inference output of the end-to-end model can be used directly without additional linear model.

4.4.4 Results of relation to variability in clinical measurements

Fig. 4.7 shows box plots visualizing the agreement between the measurements obtained automatically and with EchoPAC for each predicted quality category. The standard deviations in these plots represent how well the AI estimates agree with the manual references. The main finding is that the limits of agreement are narrower for higher qualities.

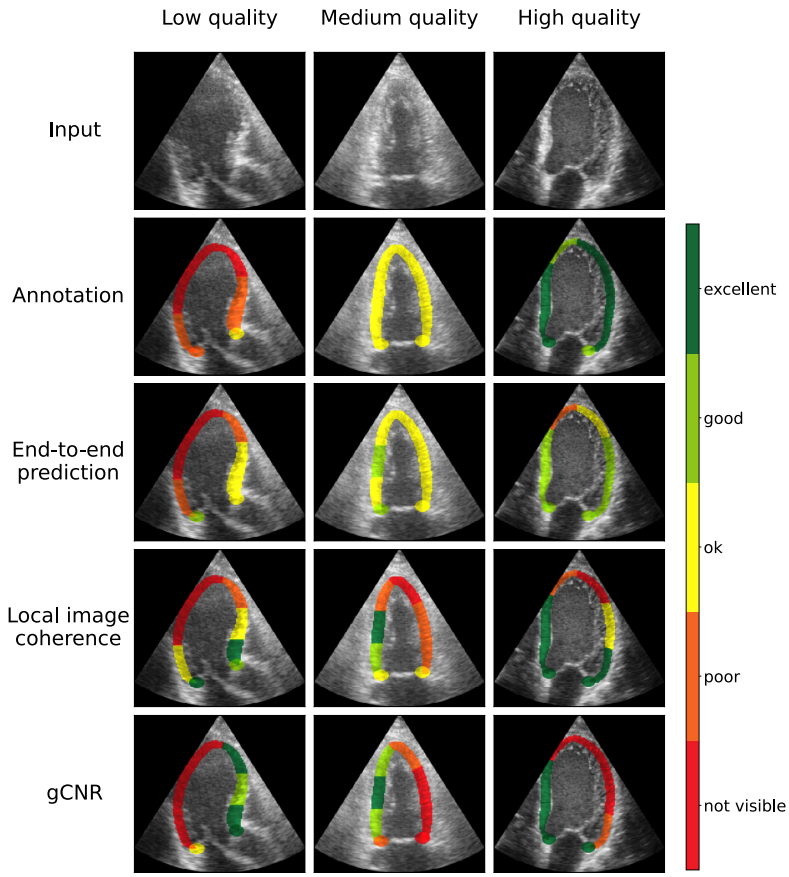


Figure 4.5: Example cases of annotations and automatically predicted regional quality from the test set. The visualization uses the regional quality metrics to color-code the output of divided segmentation output. The end-to-end model predicts the regional qualities directly from the B-mode without using the segmentation output. The local image coherence metric uses the segmentation output to select ROI. The gCNR metric uses the segmentation output to select ROI and background region. The background region, i.e., the LV lumen, is not shown in the image.

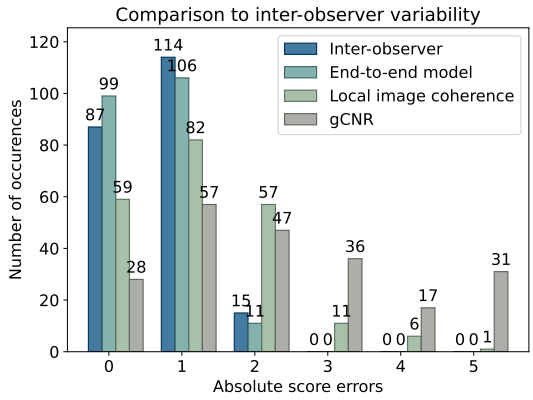
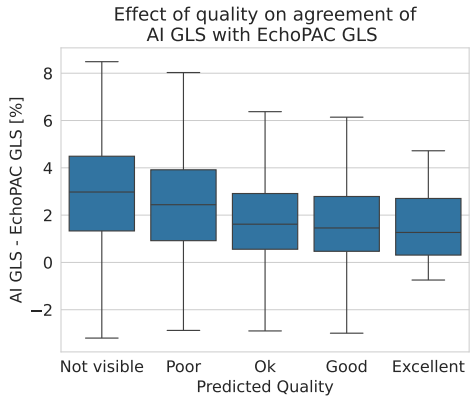
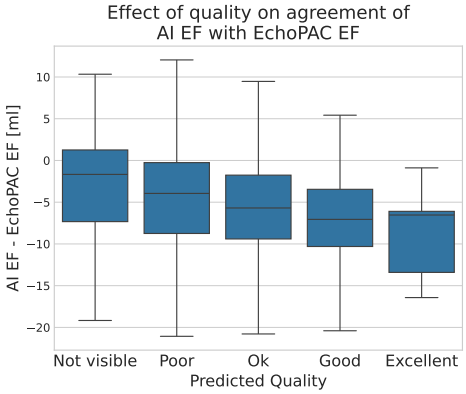


Figure 4.6: Bar plot comparing inter-observer variability to automatic methods. A method with lower variability will have the most occurrences with low score errors. Here we can observe that the variability of the end-to-end model is on par with the inter-observer variability, while the two other methods are not.



(a) AI GLS [24] vs EchoPAC GLS (2D Strain tool) [23].



(b) AI EF [25] vs EchoPAC EF [13].

Figure 4.7: Box plots of the difference between clinical measurement values obtained automatically by AI [24, 25] and reference measurements obtained manually using GE HealthCare EchoPAC on the HUNT4 data [13,23] per image quality category, as predicted by the end-to-end model. The decrease in standard deviation with better image quality indicates a better agreement between the methods for higher image quality. Additionally, there is a noticeable change in bias between different quality categories. We believe this effect is partly caused by physiological differences correlated with image quality and is out of scope for this work.

4.5 Discussion

4.5.1 Challenges and considerations

Assessing image quality based on human perception is inherently a subjective task, even when supported by clear definitions of image quality categories. This creates challenges for training and evaluation as there is no ground truth as the reference labels are a subjective estimation themselves. Therefore, it is not realistic to expect the automatic models to agree with reference labels as well as on tasks with well-defined ground truth labels. This, together with a rather fine scale of image quality categories, explains the low accuracies in Tables 4.4 and 4.5. Fig. 4.6 shows how the end-to-end model has on average less error than the annotators between each other. This indicates the model has learned to produce quality labels that strike a middle ground between the subjective assessments of the annotators.

The end-to-end learning model overestimates low-quality regions and underestimates high-quality regions, as can be seen in Fig. 4.4(a). This means that the model can only explain a limited amount of variability in the image quality labels and is a result of minimizing the mean squared error (MSE) while dealing with subjective, and thus noisy reference values with fixed boundaries. We can eliminate this effect by fitting a linear model on the validation set that maps predicted image quality to image quality labels and applying it when doing inference on the test set. This increases MSE but gives more uniform performance over the image quality labels.

One reason for the weak correlation between the annotations and the pixel-based methods is the rough selection of ROI and background region. Fig. 4.4 shows how the average metrics of the classic pixel-based and coherence metrics increase for each quality label until the *good* label, and then drop again for *excellent*. This is because on the one hand in these high-quality images, the blood speckles can be visible inside the LV lumen, which is used as background region, and on the other hand the myocardium tissue, which is used as ROI, is less blurred resulting in a smaller spread of pixels with high intensity. This can be seen for the anterolateral wall and apex in the rightmost column of Fig. 4.5. One possibility to only select regions belonging to the tissue is to perform automatic pixel selection methods like Otsu thresholding [27] or percentile filters, but in our experiments this reduced the performance even further.

4.5.2 Design choices

The different methods in this study have a trade-off between accuracy and versatility. The default end-to-end network gives the best results but requires specific image quality labels for the task. Next, the coherence-based method is more generic and can potentially be applied more generally without the need for view-specific image quality annotations. Rindal et al. [7] showed that the GIC is not significantly different between apical views, but is higher for apical views than parasternal views. Thus, while a single image-to-image model can learn to predict coherence for different views, the mapping from coherence to image quality should be done for each group of views separately. Another advantage is that coherence can be used to give a global image quality metric without the need for a segmentation model. Finally, the pixel-based methods can be applied automatically in the most general way given a segmentation model to select ROI and background regions but also give the lowest accuracy.

The ablation study of the end-to-end learning model showed that increasing the complexity of the model did not improve the performance. This is a result of the relatively small dataset size and the specific task of the model. For a more general model of image quality prediction with more varied input, e.g. one model for all cardiac views, a larger dataset and more complex model may be required.

4.5.3 Clinical use

Image quality estimation can be the first step towards a method for giving reliability estimates to clinical measurements and quality control of fully automatic methods. Fig. 4.7 shows that the variability in clinical measurements goes down with higher predicted quality. However, image quality is only one source of variability, so a reliability model would also need to include view correctness and other factors that determine whether a given input is difficult to assess.

More direct use cases of the quality prediction model include the automatic selection of the best frame to perform a clinical measurement when multiple options are available, data cleansing in data mining, and automatic disapproval of segments for regional strain analysis. All the methods explored in this work are computationally efficient and can be run in real-time while scanning, and can thus be used as a guidance tool to enable clinicians to acquire images with better image quality.

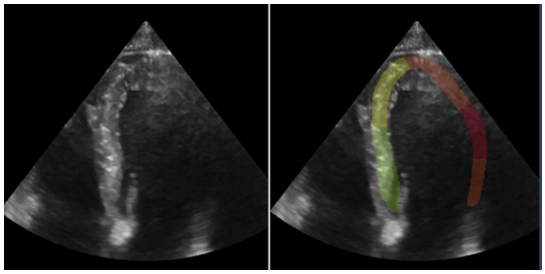


Figure 4.8: Screenshot of the real-time demo application. The left side shows the input B-Mode image. The right side shows the output of the segmentation color-coded by the output of the end-to-end quality network. The color codes are the same as in Fig. 4.5.

4.6 Real-time demo application

To showcase the functionality of the end-to-end, real-time quality network, a real-time application was developed using the FAST framework [28]. The demo is a split-screen application that shows the B-Mode input to the left and the segmentation regionally color-coded by the quality as predicted by the end-to-end network to the right. Fig. 4.8 provides a screenshot of the application in use. We provide a demo video [29] illustrating the application in action while a clinician operates a GE Vivid E95 scanner. The video can be accessed at https://doi.org/10.6084/m9.figshare.26413984.v2.

4.7 Conclusion

In this work, we developed and compared different deep-learning methods for regional image quality estimation in cardiac ultrasound. We show that classic pixel-based methods, such as (g)CNR, together with automatic image segmentation, give low agreement with the quality assessment of cardiologists. We developed a U-Net model to predict the coherence factor for each pixel in the ultrasound image and showed that the resulting coherence image can be used to assess the image quality in a pixel-based way with better performance than the classic measures. The best results, below inter-observer variability, are obtained by using an end-to-end deep-learning model. Finally, we show higher predicted quality is associated with lower limits of agreement between fully automatic and manual methods for the clinical measurements EF and GLS.

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C Extraction of cardiac regions of interest

We use the nnU-Net [30, 31] architecture to segment the cardiac images. The nnU-Net is used out of the box using the default configuration but without the final ensemble step. Instead, we train and validate on a single predefined 80% train, 10% validation, and 10% test split from the HUNT4 segmentation annotation dataset. Table 5.2 summarizes the characteristics of the nnU-Net architecture. This model is described in more detail and compared to other segmentation models in our previous work [32]. We use two nnU-Nets, one for apical two- (A2C) and four-chamber (A4C) views, and one for apical long axis (ALAX) views. The nnU-Net for A2C and A4C views segments the left ventricle (LV), left atrium (LA) and myocardium (MYO). The nnU-Net for ALAX views additionally segments the aorta (AO).

The segmentation of the MYO is divided into eight regions using the following algorithm:

1. Extract the annulus points. For A2C/A4C views, these are the points where the MYO meets the LA. For ALAX views, these are the points where the MYO meets the LA and AO. Points A and B are the annulus points in Fig. 4.9.
2. Extract the apex of the LV, defined as the furthest points from the base points within the lv lumen. This is point C in Fig. 4.9.
3. Divide both the left and right part of the endocardium border, defined as the border between the LV and MYO regions, into three parts with equal length. This gives points D, E, F and G in Fig. 4.9.
4. Find the closest points on the outer MYO border for points C, D, E, F and G. These are points H, I, J, K and L in Fig. 4.9.
5. Fill in the regions by connecting the points via the contour, resulting in the MYO divided into six regions.
6. Draw circles[4] with a radius of 2 millimeters around the annulus points, points A and B in Fig. 4.9. We use these additional two regions to asses the local

[4]Due to the unequal pixel spacing in depth and width, the annulus regions become ovals in the 256x256 segmentation maps. When plotting the images with equal spacing in width and depth, these regions become circles again.

Table 4.6: Characteristics of the nnU-Net [30, 31] used. The "number of channels" row indicates the number of channels at the first, bottom, and last convolution of the U-Net respectively. The number of output channels reflects the number of structures the model segments. For A2C/A4C views, the nnU-Net segments the LV, MYO, and LA. For ALAX views, the nnU-Net additionally segments the AO.

Input size	256×256
Number of channels	$32 \downarrow 512 \uparrow 32$
Number of output channels	3 for A2C/A4C views, 4 for ALAX views
Lowest resolution	4×4
Upsampling scheme	Deconvolutions
Normalization scheme	InstanceNorm
Batch Size	49
Optimizer	Adam
Initial learning Rate	1e-2
Scheduler	Polynomial
Loss function	Dice & cross-entropy
Inter-layer activation	Leaky Relu
Final layer Activation	Softmax
Epochs	500
Augmentations	Rotations, scaling, Gaussian noise, Gaussian blur, brightness, contrast, simulation of low resolution, gamma correction and mirroring. For more details, see [31].

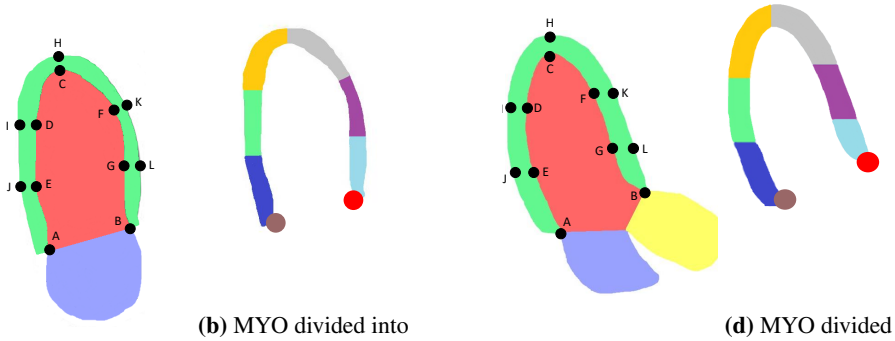
image quality of the annulus points in the image. The result are the eight regions, as in Fig. 4.9.

7. Remove any parts of regions that fall outside of the sector. The apical top regions in Fig.4.1(a), i.e., the yellow and white masks, are examples of this. If more than 50% of all pixels inside the region fall outside the sector, we exclude the region from analysis.

The goal is to automatically quantify the image quality in each of these eight regions. The LV lumen is used as background region.

D Ablation study of the end-to-end learning model

In the ablation study, we evaluated the impact of modifying the architecture of the end-to-end learning model. Three different network architectures were tested: Cardiac View Classification (CVC) network [3], MobileNetV2 [22], and EfficientNet [33]. We tested approaching the problem both as a classification and regression task, with only the final dense layer and loss function being changed accordingly. Additionally, three basic network attention variations were tested using the automatic segmentation output. The default model did not use any attention and predicts each label directly, as in Fig.4.10(a). For the other two versions, the region masks extracted from the segmentation were dilated with a square dilation filter of



(a) Point extraction **(b)** MYO divided into regions **(c)** Point extraction **(d)** MYO divided into regions

Figure 4.9: Division of MYO into regions. A and B represent the MYO base points. C represents the LV apex. D, E, F and G are obtained by dividing the inner LV border into equal parts. H, I, J, K and L are obtained by finding the closest corresponding point on the outer MYO border.

size 50x50 pixels and used as an additional input to the networks which then predict the label of one region at a time. The dilation filter reveals the direct vicinity around each region so the boundary between tissue and background becomes visible. The first variant used this dilated mask as hard attention by blacking out the other parts of the image, as shown in Fig. 4.10(b). For the second variant, the masks were used as soft attention as input to a side branch of the network, as proposed by Eppel [34]. For this version, we created an attention map and added it element-wise to the output of the first layer, corresponding to version 'c' in Eppel [34]. Fig. 4.10(c) shows this configuration.

The ablation study consists of two parts. Both parts used the training configuration listed in Table 4.7 and data from the regional image quality dataset. In the first part, we examined the effect of changing the convolutional backbone and the effect of framing the problem as a classification or regression task. Table 5.4 compares the predictions with the annotations on the test set for the different configurations using the default end-to-end model. In the second part of the ablation study, the backbone was fixed to MobileNetV2 [22] and the problem was framed as a regression task. Next, the different variants shown in Fig. 4.10 were compared. Table 4.9 summarizes the results on the test set. *The default end-to-end model with no attention, the MobileNetV2 [22] architecture, and the problem framed as a regression task gave the best results and were thus used for this paper.*

Table 4.7: Training configuration of the end-to-end learning model

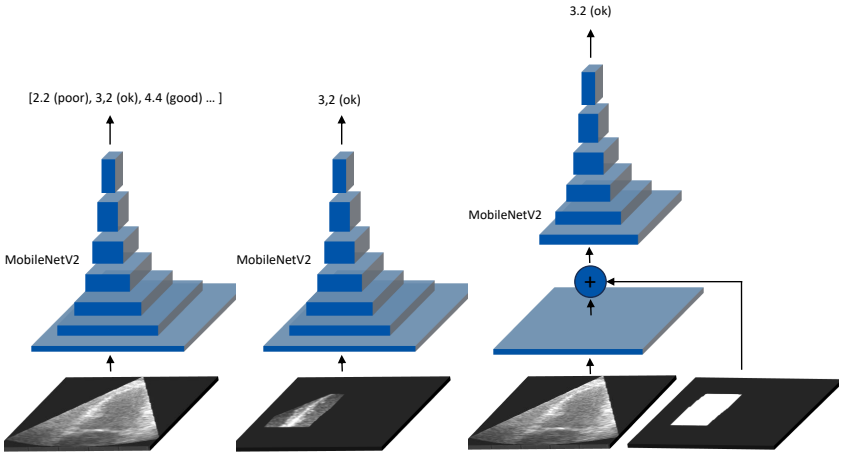
Input size	256x256
Batch size	16
Optimizer	Adam
Initial learning rate	1e-4
Scheduler	None
Loss function regression	Mean squared error (MSE)
Loss function classification	Squared earth mover's distance [35]
Epochs	500
Augmentations	Rotations ($-30° \leq$ angle $\leq 30°$), horizontal mirroring (also mirror labels), gamma correction ($0.9 \leq \gamma \leq 1.1$) and scaling ($0.85 \leq$ magnification ≤ 1.15). Each augmentation is applied individually with 0.5 probability.

Table 4.8: Results on the test set for the first part of the end-to-end learning ablation study. The Cardiac View Classification (CVC) [3] network is trained from scratch. The other networks are pretrained on ImageNet [36]. Each method is compared against annotations from the test set of the regional image quality dataset.

Task	Model	Spearman correlation	Mean absolute error (MAE)	Number of parameters in millions
Regression	CVC [3]	0.62	0.60± 0.45	9.6
	MobileNetV2 [22]	**0.65**	**0.59±0.45**	2.2
	EfficientNet-B0 [33]	0.63	0.61± 0.45	4.0
	EfficientNet-B4 [33]	0.63	0.60±0.44	18
	EfficientNet-B7 [33]	0.63	0.59±0.46	64
Classification	CVC [3]	0.38	0.91± 0.81	9.7
	MobileNetV2 [22]	0.48	0.80±0.71	2.3
	EfficientNet-B0 [33]	0.46	0.80± 0.76	4.1
	EfficientNet-B4 [33]	0.46	0.83±0.73	18
	EfficientNet-B7 [33]	0.37	0.91±0.79	64

Table 4.9: Results on the test set for the second part of the end-to-end learning ablation study using MobileNetV2.

Model variant	Spearman correlation	Mean absolute error (MAE)
Default (no attention)	**0.65**	**0.59±0.45**
Variant 1 (hard attention)	0.47	0.68±0.60
Variant 2 (soft attention)	0.53	0.67±0.57



(a) Default end-to-end model

(b) End-to-end hard attention model

(c) End-to-end soft attention model

Figure 4.10: Different variants of the end-to-end learning model. The default model outputs a list of all quality scores at once, while the other two variants output only a single quality score. The end-to-end soft attention model corresponds to version 'c' in Eppel [34].

Generative augmentations for improved cardiac ultrasound segmentation using diffusion models

5

One of the main challenges in current research on segmentation in cardiac ultrasound is the lack of large and varied labeled datasets and the differences in annotation conventions between datasets. This makes it difficult to design robust segmentation models that generalize well to external datasets. This work utilizes diffusion models to create generative augmentations that can significantly improve diversity of the dataset and thus the generalisability of segmentation models without the need for more annotated data. The augmentations are applied in addition to regular augmentations. A visual test survey shows that experts can not clearly distinguish between real and fully generated images. Using the proposed generative augmentations, segmentation performance improved by 8% in Dice value on an external dataset. Additionally, the limits of agreement for automatic ejection fraction estimation improved by up to 6% of absolute ejection fraction value. These improvements come exclusively from the increased variation of the training data using the generative augmentations, without modifying the underlying machine learning model.

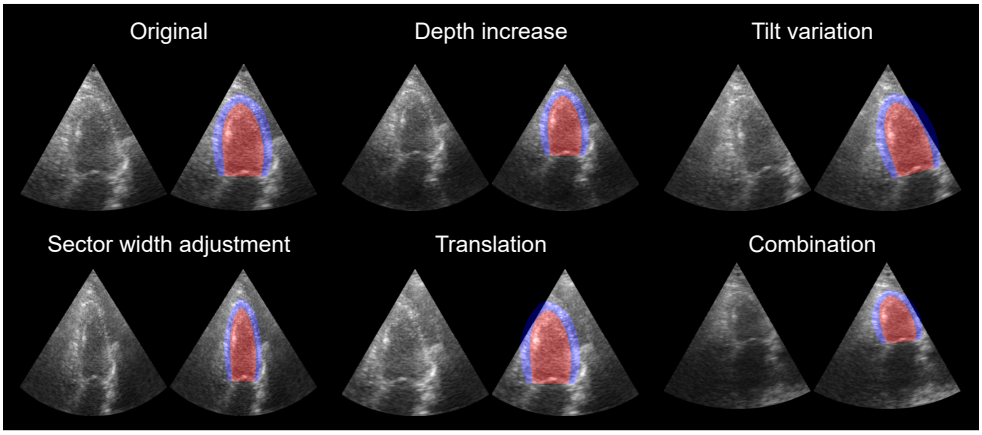


Figure 5.1: Examples of the generative augmentations types used in this work. All the examples are generated from the same original image shown in the top-left corner.

5.1 Introduction

Ischemic heart disease is the leading cause of death worldwide, accounting for 13% of all global fatalities and is the fastest rising cause of death since the beginning of the century [1]. Ultrasound imaging, being cost-effective, safe, and real-time, is the most common technology used for evaluating heart function. Accurately delineating the left ventricle and myocardium directly or indirectly enables the extraction of clinical measurements of heart function, such as ejection fraction (EF) and global longitudinal strain (GLS). However, measurements are labor intensive and even for experienced cardiologists there is high operator-related variability, with a coefficient of variation between 6 and 11% [2]. Automating the process of extracting clinical measures from the recordings reduces inter-observer variability and allows measurements over multiple heart cycles, as recommended in the guidelines [3–6], without additional labor.

Due to data protection regulations and the labor-intensive process of data curation and annotation, only a limited number of open datasets are available, and those that exist are of limited size. Although guidelines for tracing cardiac structures such as the endocardium exist [3], the amount of noise and artifacts in ultrasound, as well as structures such as the trabeculae, make tracing of the true endocardial wall challenging. Thus, if one would ask multiple experts to annotate the endocardial wall, it would result in multiple different annotations. This variability in annotation preference and inter-vendor differences means that combining datasets from different centers should be done with care. This is in stark contrast with the field of computer vision for natural images, where large, labeled datasets are common. The sparsity of annotated data with a consistent annotation protocol creates unique challenges for deep learning networks, particularly in terms of robustness and generalization [7].

Several works have explored using generative AI to improve segmentation in cardiac ultrasound. the idea is to generate images conditioned on segmentation masks to create image-label pairs for training segmentation models. Gilbert et al. [8] and Tiago et al. [9] use generative adversial networks (GANs) for image generation from anatomical models in 2D and 3D respectively. Stojanovski et al. [10] developed a conditional Denoising Diffusion Probabilistic Models (DDPMs) that generates images from segmentation masks. They apply basic transformations to the segmentation masks of an existing labeled dataset and then feed these to the conditional DDPM to generate new images. Jafari et al. [11] and Tiago et al. [12] address domain translation, using CycleGANs and adversarial diffusion models

respectively.

Generative models trained with conditioning on a segmentation mask have the inherent limitation that they can only be trained on labeled data. In cardiac ultrasound, typically only a portion of the data is labeled with pixel-wise segmentation masks as the annotation process is labor intensive.

In this work, we develop a method that augments a labeled cardiac segmentation dataset using an unconditional diffusion model. Our method has two unique advantages. First, since it uses an unconditionally trained DDPM, a dataset can be augmented using a generative model trained on an unlabeled dataset or a dataset with different annotation conventions. Second, since our model does only alter the surroundings of the segmentation masks, the most crucial parts of the image remain untouched. Thus, fine annotation subtleties and details in the original image-label pair are not affected by the generative model. This distinguishes our approach from the work of Kupyn and Rupprecht [13] who repaint the semantic object itself.

The proposed method is most effective for improving the performance of a segmentation model trained on a dataset with limited variation in terms of acquisition and positioning of the left ventricle (LV) in the image. More specifically, this work uses the HUNT4Echo [14] dataset which has annotations of high quality which are time-consuming to obtain. The recordings in this dataset are LV-focused: the recordings were obtained following the clinical guidelines and maximize the area of the ventricle in the image [3]. Thus, the dataset has limited variety in terms depth, sector width, and positioning of the ventricle in the image. This leads to poor performance of segmentation models trained on this dataset when tested on an external dataset with more variation like the public CAMUS [15] dataset. In this work, we explore whether generative augmentations can be used to enrich this limited but high-quality dataset so that the resulting models can generalize better to datasets with more variation.

The contributions of this paper are:

- A method for creating realistic augmentations of cardiac ultrasound images using DDPMs.
- A blinded expert evaluation of the realism of fully generated images.
- An ablation study that evaluates the effect of the proposed augmentations on

the segmentation accuracy.

- A clinical evaluation of the augmentations effect on automatic segmentation-based ejection fraction measurements.

5.2 Datasets

5.2.1 HUNT4

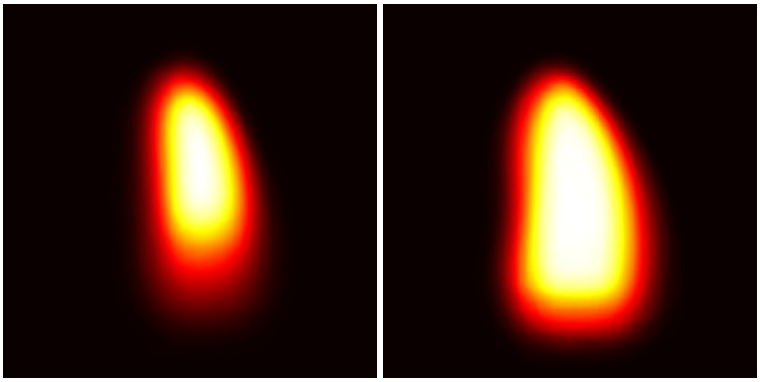
The HUNT4Echo dataset [14], part of the Helse Undersøkelsen i Nord-Trøndelag study, is an ultrasound dataset of 2,211 volunteers of LV-focused apical 2-chamber (A2C) and apical 4-chamber (A4C) views, acquired using a GE Vivid E95 scanner. Each recording includes three cardiac cycles.

The **model development set** is a subset that includes single-frame segmentation annotations for both end-diastole (ED) and end-systole (ES), providing pixel-level labels for the left ventricle (LV), left atrium (LA), and myocardium (MYO). The model development set consists of 1058 annotated ED and ES frames of 529 recordings from 311 patients.

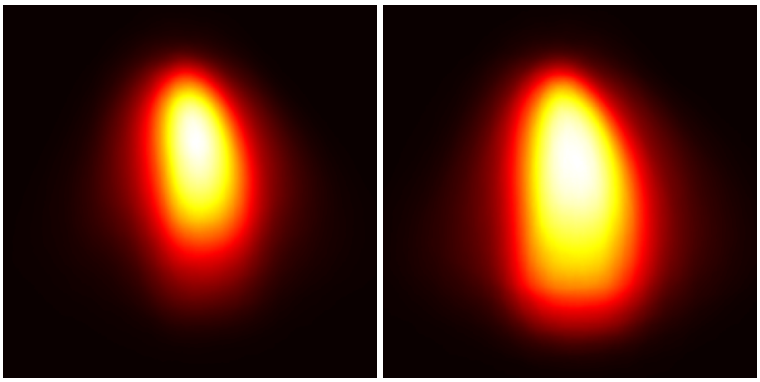
The **ejection fraction set** is a subset disjunct with the model development set of 1900 patients with reference biplane LV volumes in ED and ES. The volumes were obtained following current clinical guidelines by manual tracing and using Simpson's method of discs in the clinically approved EchoPAC software from GE HealthCare on the HUNT4 recordings.

5.2.2 CAMUS

The CAMUS dataset [15] is a publicly available dataset containing single cycle recordings from 500 patients, acquired using a GE Vivid E95 scanner (GE Vingmed Ultrasound AS, Norway). The dataset contains one A2C and one A4C recording for each patient and annotations for both the ED and ES frame in each recording, resulting in a total of 2000 image-annotation pairs. Like the HUNT4 dataset, the annotations are pixel-level LV, LA, MYO. The training, validation and test set contain 400, 50 and 50 patients respectively.



(a) CAMUS **(b)** HUNT4 model development set



(c) CAMUS with generative augmentations **(d)** HUNT4 model development set with generative augmentations

Figure 5.2: Heatmaps of pixels belonging to the LV after resizing to 256×256. This illustrates the difference in scan depth variation and LV positioning in the two datasets, before and after generative augmentations.

5.2.3 Differences and considerations

The HUNT4 model development set and the CAMUS dataset should be combined with care due to differences in annotation conventions. For example, the myocardium is consistently annotated as significantly thicker in CAMUS compared to HUNT4. Fig. 5.6 illustrates this. The segmentation models in this work only segment the LV and MYO labels and the experiments only evaluate on the LV. We elaborate on this choice in the Discussion.

The recordings in the HUNT4 study follow the clinical guidelines for quantitative measurements and maximize the area of LV in the image by adjusting the depth [3].

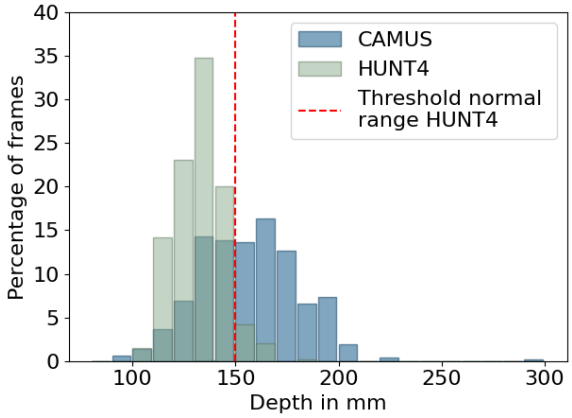


Figure 5.3: Distribution of imaging depths in CAMUS and HUNT4.

This is not the case for CAMUS, resulting in less standardized images. The top part of Fig. 5.2 shows the heatmaps of pixels labeled as LV in both datasets, illustrating that the LV occupies a larger portion of the image in HUNT4 compared to CAMUS. This is a consequence of the distribution of acquisition depth and width, shown in Figs. 5.3 and 5.4. The CAMUS dataset also has a larger variety in terms of EF, as shown in Fig. 5.5.

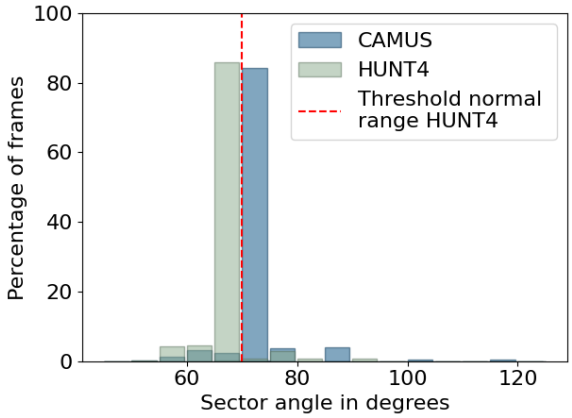


Figure 5.4: Distribution of sector angles in CAMUS and HUNT4.

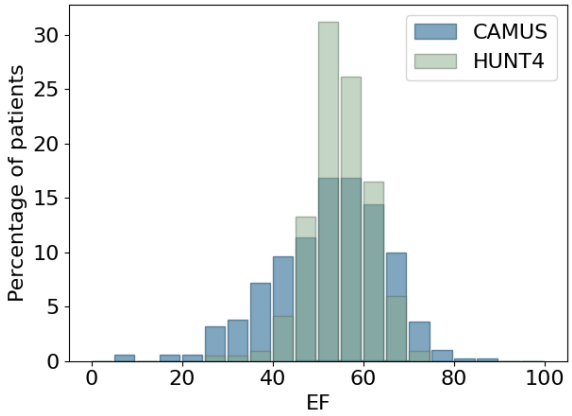


Figure 5.5: Distribution of EF in CAMUS and HUNT4.

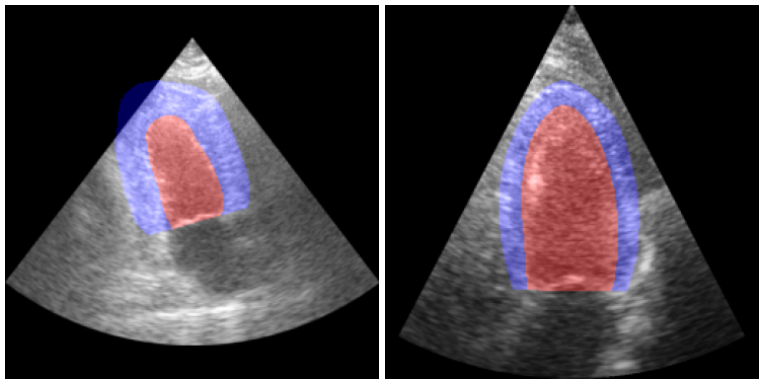
5.3 Methods

5.3.1 Denoising Diffusion Probabilistic Models and RePaint

Denoising Diffusion Probabilistic Models (DDPMs) are a type of generative model that learns to approximate a data distribution by reversing a gradual, multi-step noise addition process. They were first introduced by Sohl-Dickstein et al. [16], and subsequently improved upon by Ho et al. [17], and Nichol and Dhariwal [18]. The latter showed that diffusion models can outperform GAN based models for image synthesis [19]. Diffusion models are also appealing as they do not suffer from the training difficulties often encountered with GANs [20].

The learning methodology of the DDPM has been progressively refined. The DDPM of Ho et al. estimates the noise of the Gaussian target distributions and uses it to calculate the latent of the previous step in the reverse process of the diffusion model. The variance is kept constant with a hyperparameter [17]. Nichol and Dhariwal introduced several improvements to the method of Ho et al. These most important improvements are estimating the variance in addition to the mean, improving the noise scheduler, and using importance sampling during training [18]. This work uses the DDPM described by Nichol and Dhariwal [19].

RePaint [21] is a method that use a DDPM to replace specific regions of an image, as controlled by a given mask. During inference, the method takes as input an image and a mask. After each step of the reverse diffusion process, the part of



(a) CAMUS **(b)** HUNT4

Figure 5.6: Example frames from the CAMUS and HUNT4 datasets. The CAMUS and HUNT4 dataset contain the same cardiac views, but the frames in HUNT4 are consistently LV-focused, while those in CAMUS are not. The annotations conventions are also different in both datasets, which can be seen clearly in the thickness of the annotated myocardium (blue).

the generated image that falls outside of the mask is replaced by the corresponding parts of the input image, with appropriate noise added for that step in the diffusion process. This ensures that only the image regions inside the mask are synthesized, while those outside are left unchanged, and thus allows the unconditionally trained DDPM to be guided by the input image during inference.

5.3.2 Training of the diffusion model

The goal of the proposed method is to enrich the variety of annotated datasets, so the training dataset of the DDPM should be diverse enough. The CAMUS dataset is mostly diverse enough for this purpose. However, since the majority of sector angles in the acquisitions is around 75°, a DDPM trained on the CAMUS dataset struggles to generate images with a different sector angle. Therefore, we apply a preprocessing augmentation step that randomly narrows the sector angle by up to 20 degrees, removing pixels along the peripheral scan lines, and then stretches the cut sector back to 256×256 pixels, as illustrated in Fig. 5.7. Additionally, to preserve data variety while reducing the training dataset size, only every Nth frame is used, where N is a random number between 8 to 12 frames. This reduces the training set size while preserving the variation since consecutive frames are often very similar due to the high acquisition framerate.

The generative model in this work is the RePaint model as described by Lugmayr

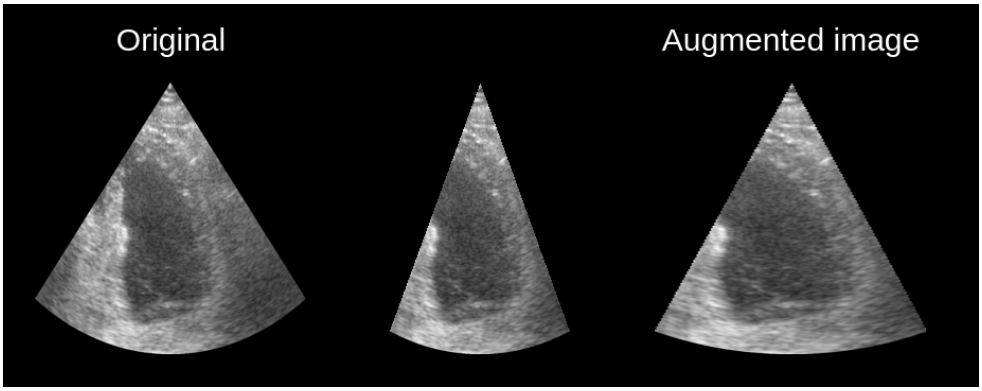


Figure 5.7: Sector width preprocessing augmentation on CAMUS performed before training of the DDPM. This was done to handle the lack of sector width variation in the CAMUS dataset. The sector angle is first reduced and then the sector is stretched back to 256 by 256 pixels. This step increases the variation in sector widths in the RePaint training set. These augmentations are applied solely during the training of the diffusion model and are not part of the generative augmentations described below.

et al [21], where the training and sampling process of the DDPM is replaced by the improved denoising diffusion probabilistic model as described by Nichol and Dhariwal [18]. Table 5.1 summarizes the technical details.

5.3.3 Generative augmentations

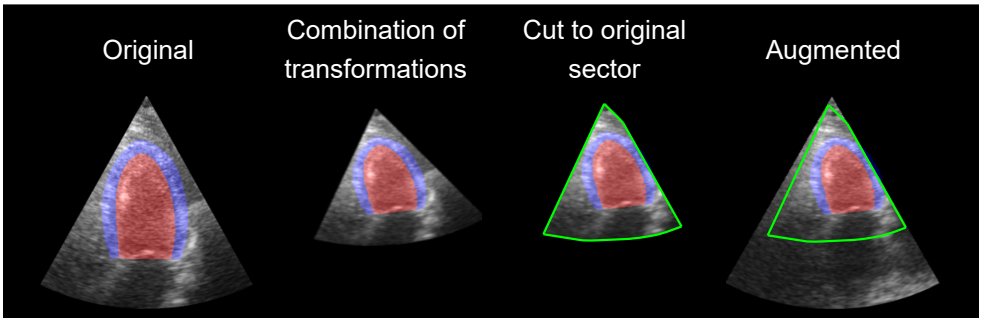


Figure 5.8: Process of creating generative augmentations. First, the frame is transformed with the transformation described in section 5.3.3. Then, the pixels outside the original sector are turned black. Finally, the generative model repaints all black pixels re-creating a complete sector in the process. The green contour delineates the part that is kept from the original image.

To apply generative augmentations to a cardiac ultrasound image, the image is first transformed using random depth, tilt, width and translation transformations

Table 5.1: Key characteristics of the U-Net and its training setup in the diffusion model. The "number of channels" row refers to the number of channels at the first, bottom, and final convolution layers of the U-Net architecture. The "Residual blocks" row refers to the number of blocks per spatial resolution level. For more details, see Nichol and Dhariwal [18, 19].

Number of parameters	44.1 million
Input size	256×256
Number of channels	$64\downarrow\ 256\uparrow\ 64$
Lowest resolution	8×8
Upsampling scheme	Nearest neighbor interpolation
Downsampling scheme	Average pooling
Normalization scheme	GroupNorm
Batch Size	64
Optimizer	Adam
Learning rate	1e-4
Learning rate scheduler	None
Activation	SiLU
Residual blocks	3
Training steps	500k
Self-attention	At resolutions 8 and 16, 4 heads
Diffusion steps	4000
Noise scheduler	Cosine [18]
Learn variance	Yes
Loss	Mean squared error, corresponding to the L_{simple} learning objective in [19].

as described below and illustrated in Fig. 5.1. Then the trained diffusion model is applied to synthesize pixels outside of original input image using the RePaint method. Fig. 5.8 shows this process. Each image is transformed and augmented five times. The augmented training dataset then contains the original image and the five augmented samples.

- **Depth increase**: the depth of the original image is increases randomly by $\lambda = [0, 150]$ pixels by adding black pixels at the bottom of the image and then resizing to 256×256.
- **Tilt variation**: the original image is rotated with a random angle around the sector tip by θ degrees, with $-30° < \theta < 30°$.
- **Sector width adjustment**: the width of the original image is multiplied by a factor λ. If $\lambda > 0$, the image gets stretched out horizontally and cropped back to 256×256. If $\lambda < 0$, the image gets squeezed into the center. Here, $0.5 < \lambda < 1.5$. This augmentation is similar to the work of Gazda et al. [22]. Resizing to 256×256 distorts the image.
- **Translation**: the original image is shifted by a vector with a random angle and length λ, with $0 < \lambda < 50$ pixels. The result is cropped back to 256×256 pixels.

- **Combination**: all of the above augmentations are applied with a 50% chance.

5.4 Experimental setup and results

5.4.1 Evaluation of generated images

The ImageNet Fréchet inception distance (FID) [23] and inception score (IS) [24] of the diffusion model are 23.87 and 1.47 respectively. However, these metrics can give misleading results for generative models that are not trained on ImageNet [25–27]. To qualitatively assess the performance of the model, Fig. 5.9 shows random samples generated together with the most similar cases from the CAMUS dataset identified automatically using the structural similarity index measure (SSIM) [28]. This shows the model does not simply memorize cases from the training set, and produces realistic and varied samples.

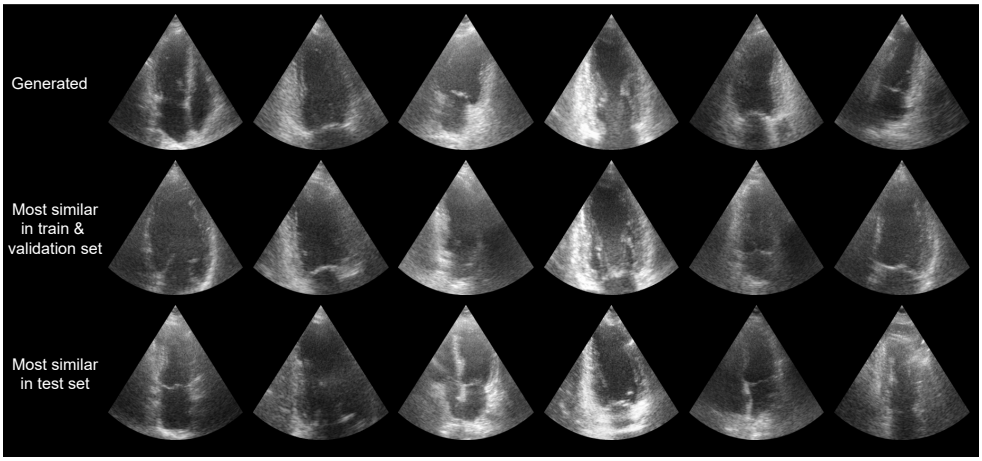


Figure 5.9: Generated samples, together with most similar cases in the train and validation set and the test set of the CAMUS dataset, based on SSIM [28].

To evaluate the realism of the generated images, a survey was conducted with three groups of human evaluators. The first consisted of three senior cardiologists certified by the EACVI in transthoracic echocardiography, each with over 15 years of experience and more than 10,000 examinations. The second group included four clinical researchers. The last group consisted of three engineers specializing in cardiac ultrasound. Each participant was asked to distinguish real from synthetic images. The real images were sampled randomly from the CAMUS dataset. The synthetic images were generated by the DDPM trained on the CAMUS dataset. The

participants were given 50 pairs of images and were told one of the images in each pair was synthetic. The participants than had to select the synthetic image and give an explanation for their selection. Additionally, 5 of the 50 pairs contained two real images without the knowledge of the participants.

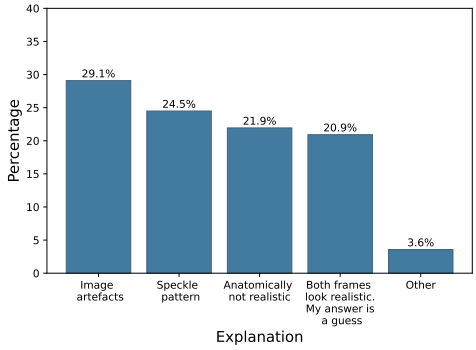
On the 45 pairs with one real and one synthetic image, participants correctly identified the synthetic image 56.4% of the time. When broken down by group, cardiologists achieved an accuracy of 63.7%, while clinical researchers and engineers both identified the correct frame 53.3% of the time. Fig. 5.10 shows the explanations given when the participants correctly identified the synthetic frame, when they were wrong, and when both frames were real in the 5 cases mentioned above.

Using a binomial test with a significance level of 5%, the accuracy of the cardiologists was found to be statistically significantly higher than random guessing ($P = 0.09\%$). However, the engineers and clinical researchers in the survey did not show statistically significant higher accuracy compared to random guessing ($P = 24.6\%$).

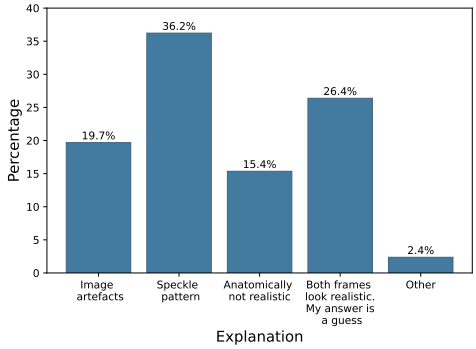
5.4.2 Segmentation ablation study

The goal of the ablation study is to evaluate how different types of generative augmentations, described in subsection 5.3.3, improve segmentation performance. We use the nnU-Net framework [29, 30] as the segmentation model, applying its default configuration but skipping cross-validation. Instead, a single model is trained on the dataset splits defined in section 5.2. Table 5.2 lists the key characteristics of the nnU-Net used.

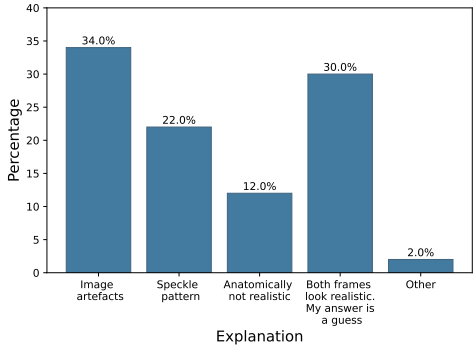
For each type of augmentation, we create a training and validation set that combines all the original images frames with five randomly augmented images generated from each original image. Additionally, we use a *black baseline* augmentation that applies the same transformations as the combination augmentation but does not repaint the missing parts, leaving those areas black. The resulting sets are then used to train separate models using the nnU-Net framework. We evaluate each model's performance on the original test sets of both CAMUS and HUNT4 datasets. Table 5.4 shows the results of the ablation study on the CAMUS dataset, using Dice score and Hausdorff distance as metrics. The bottom part of Fig. 5.2 shows the heatmaps of pixels belonging to the LV after applying the combination of all generative augmentations. Comparing these to the original illustrates that the



(a) Explanations given when selecting the synthetic image as synthetic.



(b) Explanations given when selecting the real image as synthetic.



(c) Explanations given when both images were real.

Figure 5.10: Explanations given for selecting one of the images as synthetic in the visual test.

Table 5.2: Key characteristics of the nnU-Net used for segmentation [29, 30]. The "number of channels" row refers to the number of channels at the first, bottom, and final convolution layers of the U-Net architecture. The "Residual blocks" row refers to the number of blocks per spatial resolution level. The augmentations listed here are performed on top of the proposed generative augmentations. For more details, see Isensee et al. [30].

Number of parameters	33.4 million
Input size	256×256
Number of channels	$32 \downarrow 512 \uparrow 32$
Lowest resolution	4×4
Upsampling scheme	Transposed convolutions
Downsampling scheme	Strided convolutions
Normalization scheme	InstanceNorm
Batch Size	49
Optimizer	Adam
Initial learning rate	1e-2
Learning rate scheduler	Polynomial annealing
Loss function	Dice & cross-entropy
Inter-layer activation	Leaky ReLU
Final layer activation	Softmax
Residual blocks	2
Epochs	500
Deep supervision	At resolutions 128, 64, 32, and 16
Augmentations	Rotations, scaling, Gaussian noise, Gaussian blur, brightness, contrast, simulation of low resolution, gamma correction, and mirroring.

generative augmentations increase the variety of LV location in the image.

The increase in segmentation accuracy of the HUNT4 model on CAMUS originate mostly from an improvement in segmentation accuracy for samples outside the HUNT4 image distribution. Table 5.5 lists the segmentation results for the HUNT4 models on different subsets of CAMUS. The subsets are based on depth and sector angle cutoff values visualized in Figs. 5.3 and 5.4.

5.4.3 Clinical evaluation on HUNT4

This experiment compares the ejection fraction (EF) obtained automatically via segmentation to EF obtained using EchoPAC software (GE HealthCare) for segmentation models trained with different augmented datasets. The automated estimation of EF is the same procedure as in previous works [31, 32] and follows the steps outlined by the clinical guidelines for manual EF estimation [3].

1. Use the timing network proposed by Fiorito et al. [33] to detect the ED and ES frames of each cardiac cycle for both the A2C and A4C recordings of the same patient. Thus, the view is manually labeled during acquisition, while the

Table 5.3: Segmentation results of the ablation study on the test set of the **CAMUS** dataset. The Dice score and Hausdorff distance are only for the LV lumen label. We elaborate on this choice in the Discussion.

Training dataset	Augmentation	Dice score	Hausdorff distance in mm
HUNT4	None	0.802 ± 0.15	29.03 ± 26.01
	Depth increase	0.887 ± 0.05	**7.49 ± 3.25**
	Tilt variation	0.829 ± 0.14	17.31 ± 20.98
	Sector width	0.847 ± 0.11	21.36 ± 23.84
	Translation	0.840 ± 0.12	16.55 ± 19.71
	Combination	**0.887 ± 0.05**	8.17 ± 5.32
	Black baseline	0.810 ± 0.15	26.90 ± 25.07
CAMUS	None	0.943 ± 0.03	4.46 ± 2.52
	Depth increase	0.945 ± 0.03	**4.27 ± 2.34**
	Tilt variation	0.945 ± 0.03	4.30 ± 2.43
	Sector width variation	**0.946 ± 0.03**	4.34 ± 2.41
	Translation	0.944 ± 0.03	4.44 ± 2.43
	Combination	0.944 ± 0.03	4.37 ± 2.43
	Black baseline	0.934 ± 0.03	5.39 ± 2.85

Table 5.4: Segmentation results of the ablation study on the test set of the **HUNT4** model development set. The Dice score and Hausdorff distance are only for the LV lumen label. We elaborate on this choice in the Discussion.

Training dataset	Augmentation	Dice score	Hausdorff distance in mm
HUNT4	None	0.952 ± 0.02	3.34 ± 1.21
	Depth increase	0.954 ± 0.02	3.24 ± 0.99
	Tilt variation	0.954 ± 0.02	3.38 ± 1.06
	Sector width variation	0.953 ± 0.02	**3.23 ± 1.00**
	Translation	0.954 ± 0.02	3.32 ± 0.97
	Combination	**0.954 ± 0.02**	3.31 ± 0.99
	Black baseline	0.947 ± 0.02	4.14 ± 1.85
CAMUS	None	0.886 ± 0.04	6.70 ± 1.81
	Depth increase	0.891 ± 0.04	6.55 ± 1.84
	Tilt variation	0.887 ± 0.04	6.69 ± 1.91
	Sector width variation	0.892 ± 0.04	**6.54 ± 1.78**
	Translation	0.890 ± 0.04	6.55 ± 1.83
	Combination	**0.892 ± 0.04**	6.59 ± 1.82
	Black baseline	0.875 ± 0.04	7.71 ± 2.11

Table 5.5: Segmentation results on different CAMUS subsets for a segmentation model trained on HUNT4 without generative augmentations and with the combination of all generative augmentations.

Training dataset	CAMUS Test subset	Dice score	Hausdorff distance in mm
HUNT4 without generative augmentations	Depth < 150 mm ($n = 1088$)	0.855 ± 0.11	14.48 ± 16.61
	Depth ≥ 150 mm ($n = 912$)	0.729 ± 0.18	45.83 ± 30.19
	Sector angle < 70° ($n = 146$)	0.869 ± 0.10	12.47 ± 16.47
	Sector angle ≥ 70° ($n = 1854$)	0.792 ± 0.16	30.06 ± 28.80
HUNT4 with generative augmentations	Depth < 150 mm ($n = 1088$)	**0.893 ± 0.05**	**7.45 ± 3.80**
	Depth ≥ 150 mm ($n = 912$)	**0.886 ± 0.07**	**9.34 ± 8.37**
	Sector angle < 70° ($n = 146$)	**0.893 ± 0.05**	**7.11 ± 3.10**
	Sector angle ≥ 70° ($n = 1854$)	**0.890 ± 0.07**	**8.40 ± 6.56**

timing is obtained through deep learning.

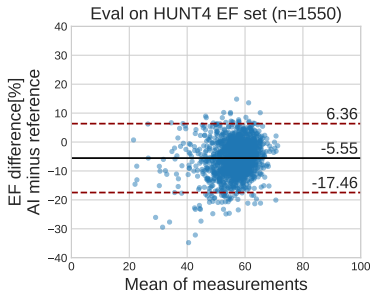
2. Use the segmentation network to segment all ED and ES frames.
3. Use the modified Simpson method to calculate the LV volume in ED and ES using the A2C and A4C frames. Each A2C frame is combined with each A4C frame for each cardiac cycle and the results are averaged.
4. Calculate $EF = \frac{ED\,volume - ES\,volume}{ED\,volume}$

When the segmentation fails for each cycle, the algorithm can not extract an EF value and the exam is omitted from analysis [32]. For a fair comparison between the models trained with and without generative augmentations, we only include exams for which both versions manage to extract an EF value.

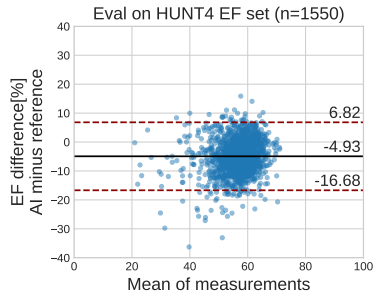
Similar to the segmentation results, the performance gains of the HUNT4 model originate mostly from an improvement in segmentation accuracy for frames outside the normal range. Fig. 5.11 shows the Bland-Altman plots comparing the automatic EF measurements with the manual reference for segmentation models trained without generative augmentations and with the combination of all generative augmentations. Appendix E contains additional analysis of automatic EF and also evaluates automatic on CAMUS.

5.5 Real time demo

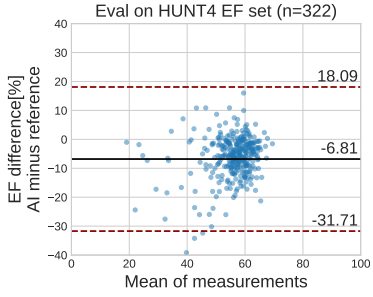
To have a clear comparison between the model trained with and without generative augmentations, a real-time application was created using the FAST framework [34]. It shows the segmentation output of the segmentation model trained in the usual



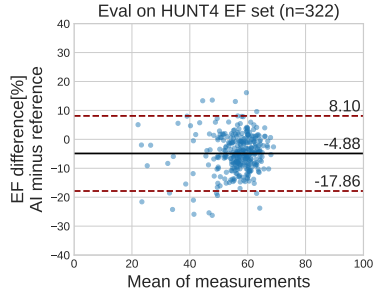
(a) Trained on **HUNT4 without** generative augmentations, tested **in normal range**.



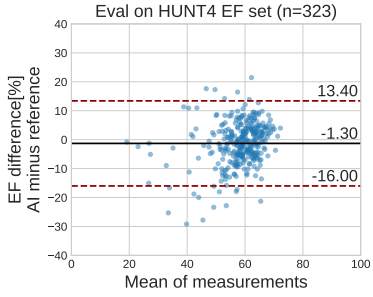
(b) Trained on **HUNT4 with** generative augmentations, tested **in normal range**.



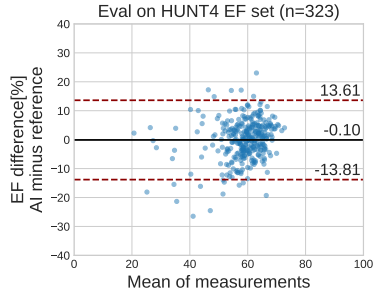
(c) Trained on **HUNT4 without** generative augmentations, tested **outside normal range**.



(d) Trained on **HUNT4 with** generative augmentations, tested **outside normal range**.



(e) Trained on **CAMUS without** generative augmentations, tested **outside normal range**.



(f) Trained on **CAMUS with** generative augmentations, tested **outside normal range**.

Figure 5.11: Bland–Altman plots comparing the manual reference with automatic EF measurements obtained via segmentation trained with and without generative augmentations. The exams outside the normal range are the exams where at least one frame used in the calculation is outside the normal range of HUNT4 (depth > 150mm or sector angle > 70°).

way and the model trained with the combination of all generative augmentations side by side in real-time on a GE Vivid E95 scanner. Fig. 5.12 shows a screenshot of the application. The video is available at https://doi.org/10.6084/m9.figshare.28219919.

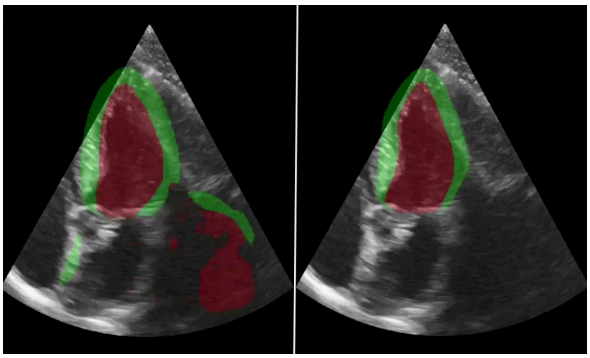


Figure 5.12: Screenshot of the real-time demo application. The left side shows the segmentation of the segmentation model trained in the usual way. The right side shows the segmentation of the same model trained with the combination of all generative augmentations.

5.6 Discussion

The results of the visual assessment shows that the DDPM can generate highly realistic ultrasound images that are hard to distinguish, even for experienced users. Although the senior cardiologists could distinguish synthetic images better than random guessing, their accuracy was still low.

Generative AI can hallucinate, and the DDPM in this work is no exception. Fig. 5.13 shows an example of a problematic hallucination. In this case, the augmentation would add noise to the training data. For segmentation augmentations, the most crucial parts are the parts with the reference segmentation masks, which are original and real. The remaining background region is of less importance and in most cases it is not detrimental if the generated surroundings are not perfectly accurate, although there are exceptions as the example in Fig. 5.13. Still, the comparison to the baseline augmentations where the surrounding pixels remain black shows that it is still important that the surroundings look realistic.

Our study does not include the LA because it is often not fully visible in the original scan. This is especially true for the LV-focused HUNT4 images. When using data

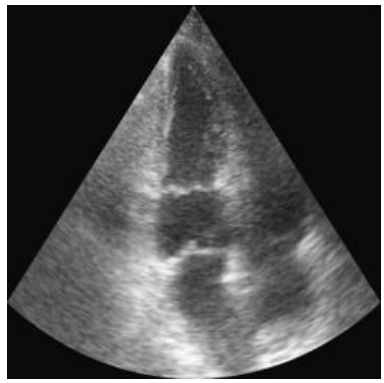


Figure 5.13: A hallucination by the RePaint model, generating a second tricuspid valve.

augmentation, particularly depth augmentation, the RePaint model can generate parts of the LA that were missing in the original scan. This creates problems because the original labels only correspond to the visible parts of the LA in the original image. Therefore, we restrict ourselves to the LV and MYO in this work.

The segmentation models in this work segment the LV and MYO, but the experiments only evaluate on the LV. The experiments do not evaluate on the MYO because there is a notable difference between the annotation conventions between HUNT4 and CAMUS. The annotations in the CAMUS dataset consistently label the MYO notably thicker than the HUNT4. Fig. 5.6 shows an example of this. There are also differences in annotations conventions for the LV lumen, but these are less marked than for the MYO.

The clinical evaluation on HUNT4 showed the generative augmentations lead to narrower limits of agreement agreement with the reference in terms of EF. The reduction is limits of agreement originates from a reduction in failed segmentation outliers. Fig. 5.14 shows visualizations of segmentation outputs for study participants where the segmentation models with and without augmentations lead to the largest differences in automatic EF. In the training set of the HUNT4 development set, the LV stretches over most of the sector. Without the generative augmentations, the model overfits on this phenomena and fails to segment the LV correctly when it only takes up a relatively small area. This explains why the depth augmentation are the most successful augmentation in the ablation study.

The clinical evaluation shows that the bias changes depending on which dataset the

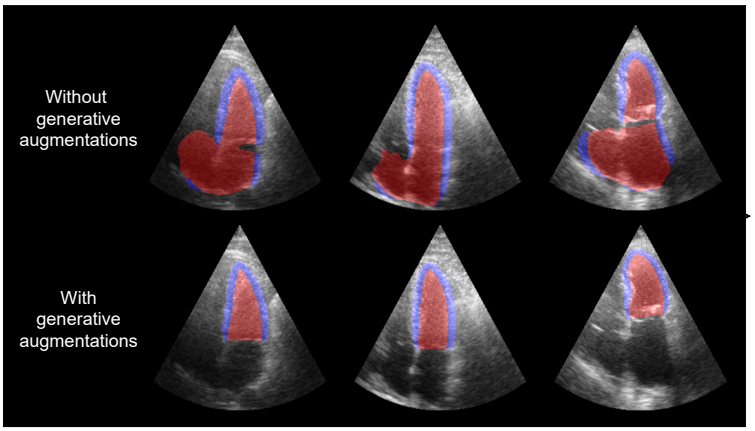


Figure 5.14: Segmentation results for study participants with the largest difference in automatic EF for models with and without generative augmentations. The model trained without generative augmentations fails to correctly segment the LV for frames with increased depth.

segmentation model was trained on. This bias can be corrected for by measuring its magnitude on a subset of the target domain and adjusting accordingly on new, unseen data. The reason for the change of bias can be the data distribution in the training set, the annotating conventions, or the methodology of the tools used for manual measurement [4]. An in debt analysis of the bias is out of scope for this work.

Both the ablation study and the clinical evaluation on HUNT4 show that the model trained on HUNT4 benefits the most from the generative augmentations. The HUNT4 dataset is more standardized, contains recordings from mostly healthy volunteers and contains less variation then the CAMUS dataset. Thus, the increase in variation from the generative augmentations mostly benefits this dataset. However, also the segmentation model trained and tested on CAMUS shows small but statistically significant ($p < 0.05$) improvement using the Wilcoxon signed-rank test [35].

The proposed generative augmentations improve the variation in terms of acquisition and positioning and sizing of the LV in the image, but do not diversify in terms of shape of the heart itself. In practice, these two types of variation might be correlated, as more diversity in patients would naturally leads to more variation in scan sectors.

The study only looks into augmentations for segmentation on cardiac images, but there is nothing preventing similar generative augmentation methods to be applied to other tasks or other imaging modalities. Of course, the type of generative augmentations in this work are tailored to cardiac ultrasound, but the concept of generative augmentations itself is flexible. Any segmentation task for which enriching the positioning of the reference masks can not be achieved realistically with regular augmentations could use the proposed approach.

5.7 Conclusion

This work explores using generative augmentations for cardiac ultrasound segmentation. Our results show that generative models can generate highly realistic images indistinguishable from real images by experts in the field. We show how these generative models can be used to improve the performance of segmentation models through generative augmentations.

The proposed generative augmentations are most useful for datasets with limited variation in terns of acquisition and positioning of the left ventricle in the image. This is relevant in the medical domain, as datasets are often limited according to the acquisition protocol and preference of the personnel at a given center.

Although this work only studies segmentation for cardiac ultrasound, the concept of generative augmentations could be generalized to other tasks or imaging modalities.

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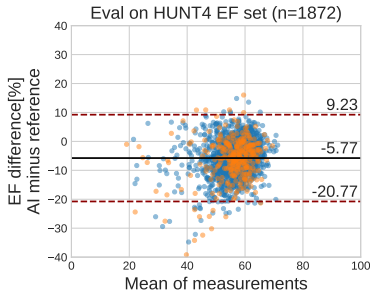
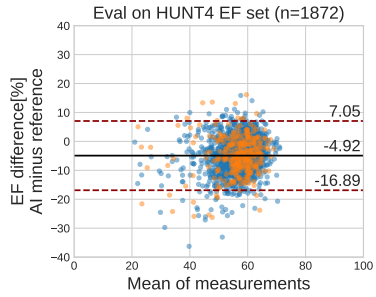
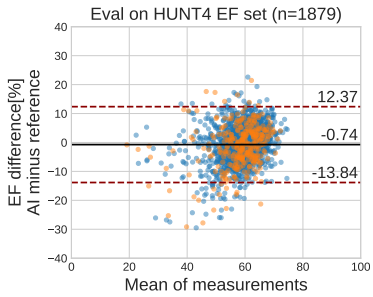
E Extensive evaluation of automatic EF



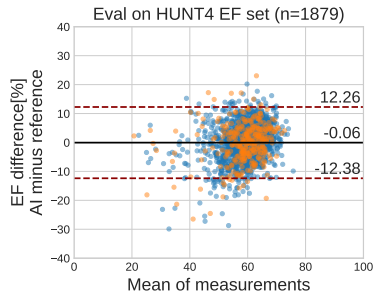
(a) Trained on **HUNT4 without** generative augmentations



(b) Trained on **HUNT4 with** generative augmentations

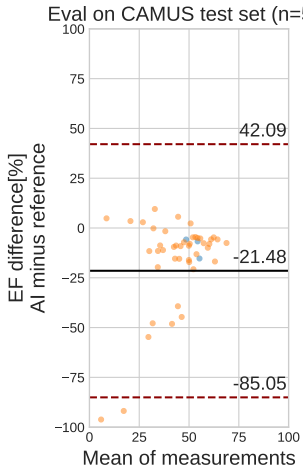


(c) Trained on **CAMUS without** generative augmentations

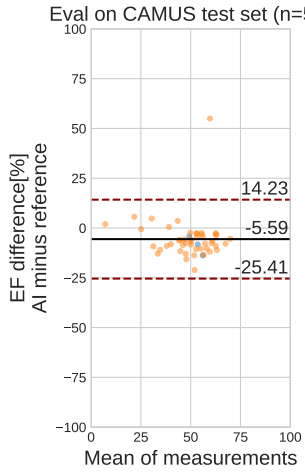


(d) Trained on **CAMUS with** generative augmentations

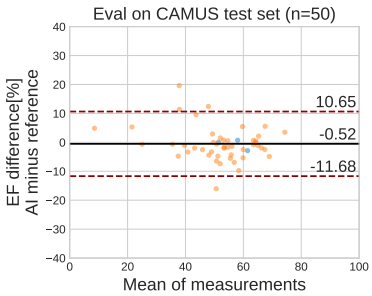
Figure 5.15: Evaluation of automatic EF on the **HUNT4 EF set** obtained via segmentation models trained with and without generative augmentations. The orange dots represent exams where at least one frame used in the calculation is outside the normal range for HUNT4 (depth $>$ 150mm or sector angle $>$ 70°). The reference EF values are obtained using EchoPAC software (GE HealthCare).



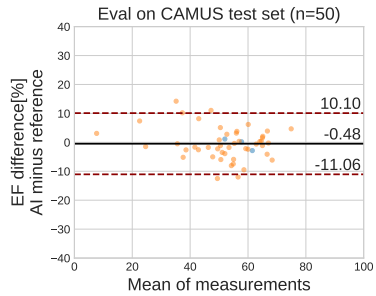
(a) Trained on **HUNT4 without** generative augmentations



(b) Trained on **HUNT4 with** generative augmentations



(c) Trained on **CAMUS without** generative augmentations



(d) Trained on **CAMUS with** generative augmentations

Figure 5.16: Evaluation of automatic EF on **CAMUS** obtained via segmentation models trained with and without generative augmentations. The orange dots represent exams where at least one frame used in the calculation is outside the normal range for HUNT4 (depth > 150mm or sector angle $> 70°$). The reference EF values are obtained using the automatic EF algorithm [31, 32] with the reference segmentation masks.